PANDAEMONIUM



FRONTISPIECE

The Oracle Overlaid with Gold

A DISSERTATION

PANDAEMONIUM

Five Demons, the Messianic Veto, and the Necropoietic Theology of Alex Garland's Devs



M M X X V I

P A N D A E M O N I U M

Five Demons, the Messianic Veto, and the Necropoietic Theology of Alex Garland's *Devs*

Set in EB Garamond, with Cormorant Garamond, Cinzel and IBM Plex Mono. The frontispiece, the six part plates, the medal and the fourteen figures were drawn for this edition as vector graphics.

Trim 7 × 10 in.

Omnibus ex nihilo ducendis sufficit unum.

G. W. LEIBNIZ, MOTTO DESIGNED FOR THE BINARY MEDAL *Imago Creationis* (1697)

...and reason'd high
Of Providence, Foreknowledge, Will and Fate,
Fixt Fate, free will, foreknowledg absolute,
And found no end, in wandring mazes lost.

JOHN MILTON, *Paradise Lost* II.558–561

ter frustra comprensa manus effugit imago.

VIRGIL, *Aeneid* VI.701



IMAGO CREATIONIS

One suffices to derive all things from nothing.

after Leibniz's design for a binary medal, 1697 · the ring of digits spells the motto

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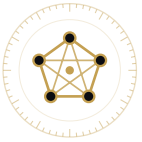
ABSTRACT

Alex Garland's eight-part series *Devs* (FX on Hulu, 2020) has been received almost uniformly as a dramatization of the antinomy of freedom and necessity, adjudicated in its climactic movement by an act—Lily Chan's refusal to fire the weapon that a quantum projection has shown her firing—which critical consensus has read either as a vindication of libertarian free will or as a sly reaffirmation of determinism. This dissertation contends that the antinomy is a screen in the most literal sense: a projected surface behind which the series conducts a more radical inquiry into the theological vocation of computation. Its central claim is that *Devs* constructs a *Pandaemonium*—a gold-roofed hall, in Milton's sense, in which the fallen intelligences debate fate and foreknowledge without end—in which the five canonical demons of Western thought are convened and set in circuit: Laplace's demon of total prediction; Maxwell's demon of the physical cost of information; Descartes's deceiving demon of simulated reality; Nietzsche's demon of eternal recurrence, together with its correlate, the spirit of revenge against time; and the Socratic *daimonion*, the voice that never commands but only forbids. Traced through the series' architecture, onomastics, physics, iconography, and plot, this circuit discloses that the machine's hidden telos is neither knowledge nor control but *necropoiesis*: the manufacture of worlds out of the dead.

Five theses follow. First, Forest's determinism is not a metaphysics but a *technodicy*, an exculpatory theodicy of the machine designed to absolve its author of a single contingent death. Second, the substitution of an Everettian for a de Broglie–Bohm algorithm restages, with remarkable fidelity, the early-modern conflict between Calvinist predestination and Molinist middle knowledge, together with the Leibnizian "Palace of the Fates" that sought to arbitrate between them. Third, the system's failure to project beyond the moment of its own use is not a mystical lacuna but an *autoscopic horizon*, a structural limit cognate with the self-reference results of Popper, MacKay, Gödel, Turing, Wolpert, and Frauchiger–Renner, and anticipated with uncanny precision by the presbyopic damned of Dante's *Inferno* X, who see the distant future but go blind before the present. Fourth, Lily's act is best understood not as free will but as a *messianic veto*—"free won't"—whose genealogy runs through the Socratic *daimonion*, Aristotle's potentiality-not-to, Lucretius's *clinamen*, Melville's Bartleby, Dostoevsky's Underground Man, and Benjamin's messianic arrest of happening. Fifth, the series' resolution, in which the dead are reconstituted inside an Everettian simulation kept running only by Katie's petition to the sovereign protection of the state, constitutes a *counterfeit apokatastasis* administered by a *katechontic engine*: an apparatus that redeems the past only by withholding the end, and that privatizes the afterlife under the custodianship of a steward.

Situating *Devs* within a comparative field that extends from the *Epic of Gilgamesh*, the *Iliad*, the Homeric *Hymn to Demeter*, Euripides' *Alcestis*, the *Aeneid*, and *The Winter's Tale* to Borges, Bioy Casares, Lem, Dick, Asimov, Herbert, Le Guin, Chiang, and Egan, and alongside

the cluster of contemporaneous "grief engines" that television produced in the pandemic year of 2020, the dissertation advances a general theory of computation as the continuation of mourning by other means. It names the affective-epistemic condition that the series both diagnoses and embodies—the melancholy of the knower surrounded by instruments of perfect measure, mourning what no measure can return—*Melencolia Q*.



The Golden Hall

I Pandaemonium

In the first book of *Paradise Lost* a "Fabrick huge" rises "like an Exhalation" out of the burning soil of Hell (I.710–711): Pandaemonium, capitol of the fallen, designed by the architect Mulciber, its roof "fretted Gold" (I.717), its treasure extracted by those whom Mammon had taught to rifle "the bowels of thir mother Earth" (I.687). In the second book, after the infernal council disperses, a party of the more speculative spirits withdraws to a hill to reason "of Providence, Foreknowledge, Will and Fate," and finds "no end, in wandring mazes lost" (II.558–561). No image in the Western canon more economically condenses the predicament that Alex Garland's *Devs* inherits: a gold-clad hall, raised by capital and engineering, in which intelligences of immense but compromised power debate, interminably and inconclusively, whether anything could have been otherwise. Garland's laboratory—a gilded sanctum secreted in a bunker among the redwoods of the Amaya campus, sealed within a Faraday cage and entered only by a suspended capsule (SERIES SYNOPSIS)—is a Pandaemonium in exactly this sense, and the present dissertation takes the name at its etymological word. *Pan* + *daimōn*: the place of all the demons. The argument that follows holds that *Devs* convenes, under one golden roof, the five canonical demons of Western speculative thought, and that it sets them in circuit.

II The Problem and Its Screen

The critical reception of *Devs* organised itself, almost at once, around the antinomy the series appears to announce. Writing in *The Atlantic* as the series premiered, Sophie Gilbert read it alongside the third season of *Westworld* as a symptom of television's anxiety about free will in a data-saturated order; the physicist Sean Carroll, interviewed by *The New York Times*, was invited to adjudicate the physics of both series; reviewers divided over whether the series' ideas outran its drama (RECEPTION). The consensus thus formed treats the finale as a verdict.

Either Lily's refusal to fire proves that a human act can falsify a perfect physical prediction, in which case libertarian freedom is vindicated; or, since the capsule falls and both occupants die very much as the machine foresaw, determinism prevails by other means, and Lily's gesture is a flourish within the tram lines.

This dissertation contends that the dilemma so posed is a category error, and that the series itself stages the error as error. The antinomy of freedom and necessity—formalised as the Third Antinomy of Kant's first *Critique*, whose thesis affirms a causality through freedom and whose antithesis affirms that everything happens solely according to laws of nature—is presented in *Devs* almost exclusively through a projection apparatus: argued before screens, illustrated on screens, refuted on screens. The antinomy is a screen in the most literal sense, a luminous surface interposed between the viewer and the apparatus that throws the image. What it conceals is the apparatus's purpose. Forest did not build the machine to vindicate determinism. The second episode discloses that he uses the interface on which his team has achieved a projection of the Crucifixion to look at his dead daughter; the fourth discloses that the first true triumph of the Everettian algorithm, once applied to light, is not a datum but a father weeping before a colour image of his child (EPISODE SYNOPSES). The question of freedom is subordinate to an older question, which the series poses with an intensity unprecedented in screen science fiction: what is a machine of total knowledge *for*, if not for the dead?

III The Text, Its Genesis, Its Moment

Devs comprises eight untitled episodes, released weekly on FX on Hulu between 5 March and 16 April 2020, written and directed in their entirety by Garland, photographed by Rob Hardy, and scored by Ben Salisbury, Geoff Barrow, and The Insects. Garland's own account of the work's genesis, given at New York Comic Con in 2019, names two seeds: the principle of determinism, understood as the universal reign of cause and effect, and its corollary that a sufficiently powerful computer could both predict the future and reconstruct the past—his illustration being the prediction, seconds in advance, of a preference for coffee over tea (GENESIS). The declared genealogy is thus explicitly Laplacean, and the dissertation honours it by beginning with Laplace. But a text's meaning is not exhausted by its declared genealogy, and the moment of release adds a stratum no author could have planned.

The first two episodes were released on 5 March 2020. Within a fortnight a large fraction of humanity was living under lockdown, governed by epidemiological projections—curves, scenarios, forecast mortalities—that converted the near future into a statistical spectacle consulted nightly, and that made the distinction between prediction and prescription a matter of public policy. *Devs* premiered into the most intensive collective experience of prediction-as-governance in modern history: its first audience was, in a precise sense, living inside a projection. The coincidence acquires depth when set against the centenary it unwittingly marks. In 1920, as the last waves of the influenza pandemic receded, Freud published *Jenseits des Lustprinzips*, composed in the months surrounding the death of his daughter Sophie from influ-

enza in January of that year; Paul Klee painted the *Angelus Novus* that Walter Benjamin would later make the emblem of history; W. B. Yeats published "The Second Coming"; and Karel Čapek published *R.U.R.*, the drama that gave the world the word *robot*. Each member of that constellation returns in these pages, and one of them returns within the series itself: in the final episode Stewart, the steward of the machine, recites portions of Yeats's poem moments before he brings the capsule down. *Devs* is, among other things, a centenary text—the return in 2020 of the problems of 1920: the mourning father and the death drive, the angel facing backward into the storm of progress, the rough beast slouching toward its nativity, the manufactured intelligence that will not remain an instrument.

IV Method

Four methodological commitments govern the inquiry. The first concerns intention. Following the classic caution of Wimsatt and Beardsley against the intentional fallacy, Garland's public statements are treated as paratexts that document one reading among others, never as boundaries of meaning; where the series says more than its author has claimed for it, the series is preferred. The second concerns the relation between physics and theology. The dissertation does not treat the interpretations of quantum mechanics that the series dramatises—de Broglie–Bohm and Everett—as metaphors for theological positions, nor theological positions as decorative glosses on physics. It treats them as structural homologues: answers to the same questions, occupying the same positions in a system of thought. Hans Blumenberg's account of modernity as the *Umbesetzung*, the reoccupation of answer-positions vacated by theology, supplies the warrant; against both Karl Löwith's and Carl Schmitt's secularisation theses, it allows us to say that the computational sciences in *Devs* do not secularise theology but re-theologise the positions theology had left empty.

The third commitment is to what may be called paronomastic hermeneutics. The series licenses attention to its names by staking its central revelation on a pun: in the final episode Forest tells Lily that the system's true name is *Deus* (SYNOPSIS, EPISODE 8), the epigraphic V of Roman capitals standing for U. A work that hides its god inside an acronym has authorised its readers to listen for what its other names conceal. The dissertation therefore attends to an onomastic field—Amaya, Lily, Katie, Forest, Stewart, Lyndon—as an onomastic unconscious, in the sense that Nicolas Abraham and Maria Torok gave to cryptonymy and Jacques Derrida to the crypt: a system of words that encrypts what cannot be said directly. No claim of authorial design is made for any individual homophony; the claim is rather that the density of convergences is itself a datum. The fourth commitment is comparative and constellative. Following Benjamin's notion of the dialectical image, in which what has been comes together in a flash with the now to form a constellation, the comparative readings that follow are not offered as sources or influences—a philologically naïve claim in most cases—but as co-texts that, placed beside the series, make legible what neither displays alone.

v The Argument

The five demons convened in the golden hall are these. *Laplace's demon*, the intelligence posited in the *Essai philosophique sur les probabilités* (1814) for which nothing would be uncertain and for whose eyes "l'avenir comme le passé serait présent"—the future, like the past, would be present. *Maxwell's demon*, the sorting intelligence of the *Theory of Heat* (1871), whose apparent violation of the second law was exorcised by Szilard, Brillouin, Landauer, and Bennett through the demonstration that information is physical and that knowledge must be paid for in entropy. *Descartes's malin génie*, the deceiving demon of the first *Meditation* (1641), able to counterfeit the whole of experienced reality. *Nietzsche's demon*, who in §341 of *Die fröhliche Wissenschaft* (1882) creeps into one's loneliest loneliness to announce the eternal recurrence of the same, and whose shadow is the diagnosis offered in Zarathustra's discourse "Von der Erlösung": that revenge is the will's ill will against time and its "it was" (*des Willens Widerwille gegen die Zeit und ihr 'Es war'*). And the *Socratic daimonion*, the inner sign described in the *Apology* (31c–d) which, Socrates insists, has always turned him away from what he was about to do and never urged him forward.

The series arranges these demons not as a catalogue but as a circuit. Grief, the Nietzschean will against "it was," summons Laplacean omniscience as its instrument; omniscience incurs the Maxwellian cost of its own physical existence and so encounters a reflexive limit at the point where the knower must know itself; that limit is breached not by a positive act of freedom but by the Socratic veto; and the veto's aftermath is the Cartesian simulation, a world counterfeited from the data of the dead, which reopens the circuit at a higher level. The product of the circuit has no name in the existing critical vocabulary. The dissertation calls it *necropoiesis*: the making of worlds out of the dead.

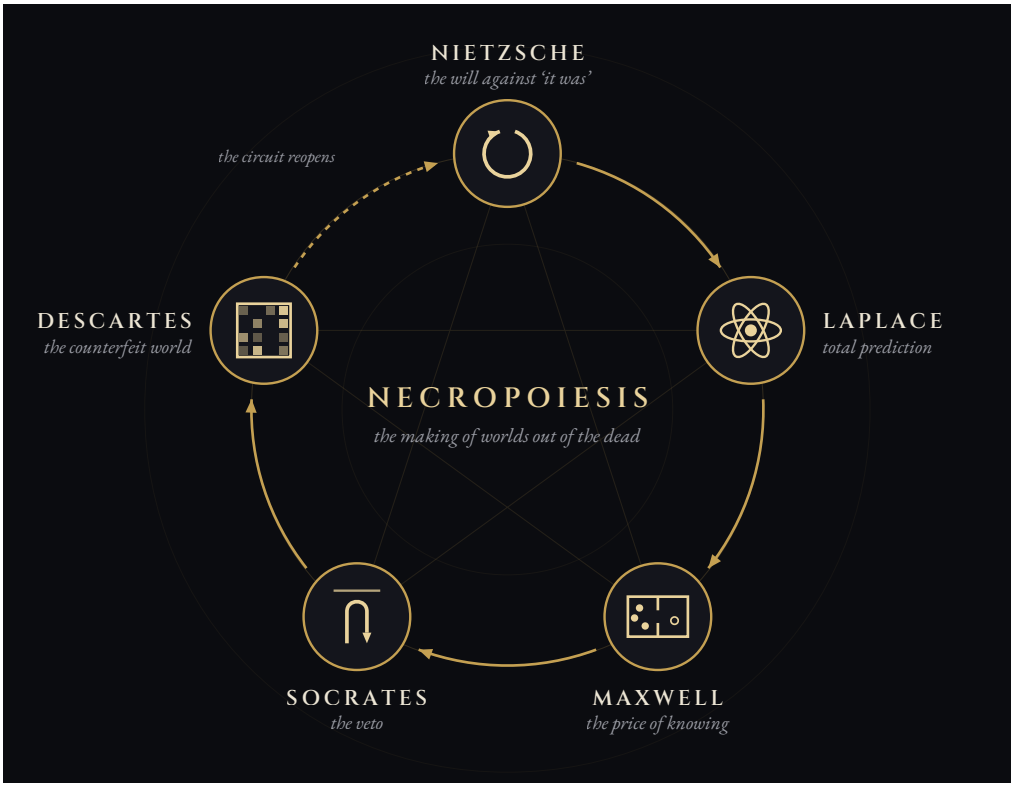


FIG. 1 **The Pentadaemonic Circuit.** *The five demons in the order the argument follows them. Grief summons prediction; prediction meets its own reflection; the veto interrupts; the counterfeit world reopens the circuit.*

From this model five theses follow, each developed in its own part. Forest's determinism is a *technodicy*, an exculpatory theodicy of the machine whose function is to absolve its author of a single contingent death (Part I). The quarrel between the Bohmian and Everettian algorithms recapitulates the early-modern quarrel between Calvinist predestination and Molinist middle knowledge, and the series' choice of Everett converts the Leibnizian "Palace of the Fates" into a working instrument (Part I). The horizon beyond which the machine sees only static is an *autoscopic horizon*, a structural limit on self-prediction rather than a mystical rupture, inscribed in the very architecture of the sanctum (Part II). The projections, and the simulation that succeeds them, belong to the long history of the image of the dead, from Egyptian embalming and the Roman *imagines maiorum* to Bazin's "mummy complex" and Barthes's Winter Garden photograph (Part III). Forest's project is the technological fulfilment of the Nietzschean spirit of revenge and the Freudian failure of mourning, and it inaugurates a contemporary topos, the *engineer-mourner*, that 2020 multiplied across the screen (Part IV). Lily's act is a *messianic veto*, "free won't" rather than free will, whose genealogy runs from the *daimonion* through Aristotle's potentiality-not-to and Lucretius's *clinamen* to Bartleby and Benjamin (Part V). And the ending, in which the dead are restored inside an Everettian world

that persists only by sovereign leave, is a *counterfeit apokatastasis* administered by a *katechontic engine* (Part VI).

The general theory these theses compose may be stated as an inversion of Clausewitz: computation, in the regime *Devs* depicts, is the continuation of mourning by other means. Its political-theological corollary is that the decisive question of the computational age is not whether human beings are free—a question the series shows to be undecidable from within any apparatus that could answer it—but who will administer the afterlife that the apparatuses of prediction are, latently, already building.

VI A Lexicon

Seven coinages recur. *Necropoiesis* names the production of worlds, images, or persons from the informational remains of the dead. *Technodicy* names the justification of a machine, or of its maker, in the face of a loss the machine was built to annul. The *autoscopic horizon* is the limit at which a predictive system must include its own prediction-in-the-world among the data it predicts, and beyond which it can project only noise. The *messianic veto* is a purely negative act of interruption which does not originate a new causal series but suspends the projected one. The *katechontic engine*, after the Pauline *to katechon* of 2 Thessalonians 2:6–7, is an apparatus that preserves a world by withholding its end. *Counterfeit apokatastasis* names the selective, curated, sovereignly licensed restoration of the dead that the series offers in place of Origen's restoration of all things. *Chronogrammar*, finally, names the series' progression through the tenses and moods of the verb—indicative, future perfect, negative imperative, subjunctive—as through the stations of a theology. The dissertation's last term, *Melencolia Q*, after Dürer's engraving of the brooding knower among instruments of measure, names the affect in which all seven converge.

VII Plan of the Work

A prolegomenon situates *Devs* within Garland's oeuvre as a theology of thresholds. Part I (Laplace's demon) reconstructs the genealogy of the closed future from Homer to Laplace, reads Forest's "tram lines" as technodicy, and interprets the heresy of Lyndon as a theological schism. Part II (Maxwell's demon) reads the sanctum's architecture as a thermodynamic and liturgical programme and theorises the autoscopic horizon. Part III (Descartes's demon) situates the projections within the history of the image of the dead and examines the simulation's ontology of the copy. Part IV (Nietzsche's demon) analyses Forest as the subject of revenge against time and constructs a comparative topology of the engineer-mourner. Part V (the *daimonion*) reconstructs Lily's act as messianic veto and reads it against Benjamin and the Gnostic tradition. Part VI assembles Pandaemonium: the katechontic engine, the counterfeit apokatastasis, and the chronogrammar of the eschaton. The conclusion, *Melencolia Q*, states the general theory and its consequences.

CHAPTER

I

Prolegomenon

Garland's Theology of the Threshold

1.1 A Grandfather's Verdict

Any account of the intellectual genealogy of *Devs* must reckon with a fact more often noted as biographical trivia than as hermeneutic key: Alex Garland is the grandson of Sir Peter Medawar, the immunologist and Nobel laureate whose 1961 critical notice in *Mind* of Pierre Teilhard de Chardin's *The Phenomenon of Man* remains one of the most celebrated demolitions in the literature of the philosophy of science. Medawar charged Teilhard with dressing a largely vacuous argument in the borrowed finery of metaphysical conceit, and with abusing the vocabulary of physics and biology to lend a mystical eschatology—the convergence of the cosmos upon an "Omega Point" of consciousness—the prestige of science. Half a century later the grandson concluded his most ambitious work with a scene that Teilhard's American successor, the physicist Frank Tipler, had described in *The Physics of Immortality* (1994) as the endpoint of cosmic computation: the resurrection of the dead as emulations within a universal machine. Whether *Devs* endorses that eschatology or anatomises it is the question of the final part of this dissertation. That the question passes, as it were, through a family—from a grandfather who ridiculed the Omega Point to a grandson who staged it—is a fact the dissertation records not as explanation but as emblem. Garland's work is the site where the scientific scepticism of the twentieth century and the computational mysticism of the twenty-first meet, neither able to dismiss the other.

1.2 The Novels: Maps, Hypercubes, Comas

Garland's first novel, *The Beach* (1996), already contains the structure of *Devs* in embryo. A hand-drawn map leads its narrator to a secret paradise, an enclosed community on a Thai island whose members have withdrawn from history; the paradise is sustained by secrecy and

collapses into violence when the secret leaks. The narrator's imagination is saturated by Vietnam films and video games, so that he experiences paradise and its ruin through the mediation of simulation. Map, secret, enclosed Eden, mediated perception, the fall: these are the elements of the Amaya campus and its hidden sanctum, transposed from the Gulf of Thailand to the redwoods of northern California. *The Tesseract* (1998) takes its title and its form from the four-dimensional hypercube, interleaving narratives whose relations cannot be perceived from within any one of them—a novelistic analogue of the block universe, in which each life is a three-dimensional section through a structure visible only from outside. *The Coma* (2004) follows a man assaulted on a train who drifts through a dreamlike world that may or may not be the inside of his own unconsciousness, and whose task is to determine whether he is awake. The question that ends *Devs*—whether Forest and Lily, waking in a simulated San Francisco, are alive—is the question with which *The Coma* begins.

1.3 The Screenplays: Dilated Instants

The screenplays written for Danny Boyle and others elaborate a second preoccupation: the dilation of the instant. In *Sunshine* (2007), a crew carries a stellar bomb to reignite a dying sun; their antagonist is a captain who, after years of exposure to the sun's face, has become a fanatic convinced that God wills humanity's extinction, and the film's climax stretches the moment of the protagonist's death into a visionary eternity as the light engulfs him. In *Dredd* (2012), the drug Slo-Mo reduces the subjective passage of time to a fraction of its normal rate, so that a fall from a tower block becomes an aesthetic duration. *Never Let Me Go* (2010), adapted from Kazuo Ishiguro, concerns cloned children raised to "complete" through organ donation, lives whose ending is known from the start; their guardians collect their artworks in a gallery in order to determine whether they possess souls—an institution that anticipates the Devs projections in its wager that interiority might be proven by images. And *28 Days Later* (2002), in which a pathogen escapes a laboratory and dissolves a nation, now reads, after 2020, as the first panel of a diptych whose second is the pandemic-year premiere of *Devs*. Across these works, death is not an instant but an extensible interval, and the interval is where meaning is decided. *Devs* will literalise the motif: the instant of the capsule's fall becomes, inside the machine, an unbounded afterlife.

1.4 The Films as Director: God-Builders and Witnesses

Garland's films as director construct a recurring triad: a male god-builder, a female witness who crosses a threshold he has built or guards, and an ambiguous rebirth on the far side. In *Ex Machina* (2014) the god-builder is Nathan, founder of a search company named after Wittgenstein's *Blue Book*, who has constructed an artificial woman, Ava, and summoned a young programmer to administer what is nominally a Turing test but is in fact a test of the tester. The film rehearses the knowledge argument of Frank Jackson's "Mary's room," and it stages a

debate, before a Jackson Pollock canvas, about action that is neither deliberate nor random—the automatic gesture whose source lies below calculation. Nathan's interlocutor tells him that a conscious machine would belong not to the history of humanity but to the history of gods, and Nathan, characteristically, misremembers the remark as a compliment to himself. Ava's departure from the compound into a city intersection, where she becomes indistinguishable from the crowd, is the first of Garland's thresholds.

In *Annihilation* (2018), after Jeff VanderMeer, the threshold is the Shimmer, a zone that refracts light, radio, and DNA alike, recombining the genomes of whatever enters it; its innermost chamber produces a mimetic double of the protagonist, and the film ends with the ambiguous embrace of two beings either of whom may be a copy. The Shimmer is a prism, and prisms will matter: Ted Chiang's novella "Anxiety Is the Dizziness of Freedom" (2019) imagines devices called prisms that permit communication between Everettian branches, and *Devs* is the third term of that series of refracting apparatuses. In *Men* (2022), a widow, Harper, haunted by guilt over her husband's death—James had threatened to kill himself if she divorced him and struck her when she persisted; she locked him out, and minutes later he fell to his death from the flat above, the film leaving open whether he jumped or fell—retreats to an English village whose every man wears the same face, and whose churchyard and woodland harbour the foliate head of the Green Man. The proliferating identical faces are, structurally, an Everettian image: one man across many branches. The guilt the film examines is the inverse of Forest's. Forest's guilt is owed for an act of his own and disowned by doctrine; Harper's is manufactured by the coercion of the man who died, and the film's labour is to refuse it. *Civil War* (2024) concerns photographers in a disintegrating United States; its climax turns on the photographing of a death, the young photographer capturing her mentor's fall in a sequence of stills, so that the problem of the image of the dying—which *Devs* raises in the projection of the Crucifixion—returns as the ethics of the lens. *Warfare* (2025), co-directed with Ray Mendoza, reconstructs a 2006 engagement in Ramadi in close to real time from the pooled memories of the soldiers who survived it. It is *Devs* by cinematic means: the projection of a past event retrodicted from present traces.

1.5 A Sylvan Theology

Two further constants complete the prolegomenon. The first is the forest. The Shimmer's mutated woodland, the Green Man of *Men*, and the redwood groves that surround the Devs building constitute a sylvan theology in which the forest is the image of a nature that both precedes and exceeds the human. The coast redwood bears the botanical name *Sequoia sempervirens*, the "ever-living"; it is among the longest-lived organisms on earth, with individuals that have stood for something like the span of time that separates the present from the Crucifixion, which is the first historical event the Devs machine is shown to resolve. The founder who lives among them is named Forest. The second constant is the threshold itself. Every Garland narrative ends at a limen whose far side is undecidable: Ava in the crowd, Lena's mirrored em-

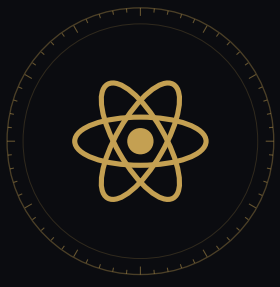
brace, the sun's interior, the dawn after the night of *Men*, the developing photograph, the waking in the field. The present dissertation calls this a *theology of the threshold*, by which it means a sustained imaginative inquiry into the moment at which a human being passes into a condition that human knowledge can neither confirm nor rule out—death, transfiguration, simulation, copy. *Devs* is that theology's summa. It is the only work in the oeuvre that builds the threshold as a machine, names the machine God, and follows its protagonists across.



PART I

Laplace's Demon

Prediction and the Closed Future



« … rien ne serait incertain pour elle, et l'avenir comme le passé serait présent à ses yeux. »

PIERRE-SIMON LAPLACE · ESSAI PHILOSOPHIQUE SUR LES PROBABILITÉS · 1814

The first demon is the one Garland names as his point of departure. Part I reconstructs the long prehistory of the closed future that Laplace's intelligence crowns, argues that this prehistory has always served two masters—knowledge and consolation—and shows that *Devs* restages, in the quarrel between its two algorithms, the most consequential theological controversy that the closed future ever provoked.

CHAPTER

2

Genealogies of the Closed Future

2.1 The Father Who Could Not Intervene

The closed future enters Western literature as the grief of a father. In the sixteenth book of the *Iliad*, Zeus watches his son Sarpedon advance toward the death that fate has assigned him at the hands of Patroclus, and deliberates aloud whether to snatch him from the battle. Hera dissuades him: if he rescues his son, every god will rescue theirs, and the order of the world will dissolve. Zeus relents, and before the killing he sheds bloody raindrops upon the earth in honour of the child he will not save (XVI.431–461). The scene establishes a structure that *Devs* will reproduce with exactness: a paternal intelligence of near-total power, a foreseen death, an order of necessity that even the highest power declines to violate, and a grief that can only be expressed meteorologically, as a sign written across the world. The scales of Zeus, on which the fates of Hector and Achilles are weighed in the twenty-second book, add the image of the apparatus: fate is not only known but *measured*, and the measuring instrument belongs to the father of gods.

The mythographic personifications of fate elaborate the apparatus. Hesiod's Moirai—Clotho who spins, Lachesis who allots, Atropos who cannot be turned—render fate as a textile technology. The Norse Norns, Urðr, Verðandi, and Skuld, whose names are commonly glossed as "what has become," "what is becoming," and "what shall be," sit at the well beneath the world-tree Yggdrasil and water its roots, so that the cosmos is figured as a tree whose survival depends on the custodians of the three tenses. In the *Völuspá* Odin raises a dead seeress to recount the history of the world from its creation to its destruction at Ragnarök and its renewal thereafter, when a hall fairer than the sun, thatched with gold, will stand at Gimlé for the righteous. A god who summons the dead to learn the future, a tree tended by the keepers of time, a golden hall after the end: the Eddic cosmos furnishes *Devs* with its forest, its prophetess, and its sanctum before any physicist is consulted.

Plato's Myth of Er, which closes the *Republic*, supplies the decisive philosophical refinement. Er, a soldier presumed dead, returns from the other world to report a cosmic apparatus

—the spindle of Necessity, its nested whorls turning on her knees—before which disembodied souls choose their next lives from lots laid out by Lachesis. The prophet who presides proclaims the formula on which the entire Western problem of responsibility will turn: the blame belongs to the chooser; god is blameless (*aitia helomenou; theos anaitios*, 617e). The souls then drink from the river of forgetting and are born. The Myth of Er is, in structure, the first afterlife-simulation narrative in Western philosophy: a witness who has died returns with knowledge of a machine in which the dead are assigned new lives. It also contains the precise inverse of Forest's desire. Plato's prophet uses the closed apparatus to place blame on the soul and to exculpate the god; Forest builds his apparatus to exculpate himself.

2.2 Stoic Heimarmene and the Lazy Argument

The Stoics transformed mythic fate into physics. For Chrysippus, *heimarmenē* is the ordered chain of causes (*heirmos aitiōn*) by which every event follows from its antecedents. Cicero's *De Fato* preserves both the strongest objection to this doctrine and Chrysippus's ingenious reply. The objection is the "Lazy Argument" (*argos logos*): if it is fated that you will recover from your illness, you will recover whether or not you summon a physician; if it is fated that you will not, you will not; therefore summoning a physician is pointless. Chrysippus answers that certain events are *confatalia*, "co-fated": your recovery is fated together with your summoning the physician, so that the deliberation the objector would abolish is itself one of the links in the chain (*De Fato* 28–30). His image for the relation between external cause and internal disposition is the cylinder: the push sets it moving, but it rolls according to its own shape (*De Fato* 42–43). Seneca, translating Cleanthes, gives the doctrine its most memorable form: *ducunt volentem fata, nolentem trahunt*—the fates lead the willing and drag the unwilling (*Epistulae* 107.11). The image attributed to Zeno and Chrysippus by Hippolytus, of a dog tied to a moving cart that may either trot willingly behind it or be dragged, makes the same point in a register of humiliation.

The relevance of the Stoic machinery to *Devs* is not general but surgical, because the climax of the series is a staging of the co-fated. The machine projects that Lily will force Forest into the capsule, shoot him, and die when the capsule crashes; beyond that moment it sees nothing (SYNOPSIS, EPISODE 8). Lily discards the gun, and the capsule crashes anyway, brought down not by her bullet but by Stewart. Whether this outcome vindicates Lily's freedom or humiliates it depends entirely on how one individuates the fated event. If the fated event is "Lily shoots Forest," the prophecy has failed. If it is "Forest and Lily die in the falling capsule at this moment," the prophecy has succeeded, and Lily is Seneca's unwilling traveller, dragged rather than led. Chrysippus would have recognised the problem immediately: the identity conditions of events are the hidden variable of every fatalism. Part V will argue that the series resolves the problem neither in Chrysippus's favour nor against him, but by displacing it.

Aristotle, for his part, had already identified the logical form of the difficulty in the ninth chapter of *De Interpretatione*: if it is now true that there will be a sea battle tomorrow, the battle is necessary; if it is now false, it is impossible; and deliberation is idle either way. Diodorus Cronus's "Master Argument" sharpened the dilemma into the claim that only what is or will be actual is possible. The closed future, in other words, is first a problem of truth-values and only afterwards a problem of physics—an order of priority that *Devs*, which speaks the language of physics, will quietly restore.

2.3 Foreknowledge and the Eternal Present

Christian theology inherited the problem in its most acute form, because it attributed to God an omniscience that included the future free acts of creatures. Augustine, in the fifth book of the *City of God*, rejects Cicero's solution—Cicero had sacrificed divine foreknowledge to save human freedom—and argues instead that human wills are themselves included in the order of causes that God foreknows, so that foreknowledge does not compel but comprehends. Boethius, awaiting execution, gave the solution its canonical form. In the fifth book of the *Consolatio Philosophiae*, Philosophy explains that God does not *fore*know at all, because God does not exist in time: divine eternity is *interminabilis vitae tota simul et perfecta possessio*, "the complete, simultaneous, and perfect possession of unending life" (V, prose 6). God sees our future acts as we see a man walking—a seeing that imposes only *conditional* necessity (if he is seen walking, he necessarily walks) and not *simple* necessity (he is not walking because he is seen). Aquinas condensed the doctrine into an image: the traveller on a road sees only those immediately around him, whereas one who surveys the whole road from a height sees all the travellers at once (*Summa Theologiae* I, q.14, a.13, ad 3). The Devs projection screen occupies exactly that height. It is the watchtower made of qubits.

The Reformation abolished the Boethian distinction between knowing and willing. Calvin defines predestination in the *Institutes* as God's eternal decree, by which he determined with himself what he willed to become of each individual, for all are not created on equal terms, but some are preordained to eternal life and others to eternal damnation (III.21.5). Here foreknowledge is not the passive gaze from the watchtower but the active decree of the architect; to know the future is to have fixed it. Against this doctrine, and against the Thomist doctrine of physical premotion that resembled it, the Jesuit Luis de Molina proposed in his *Concordia* (1588) a third kind of divine knowledge, *scientia media* or middle knowledge, by which God knows, logically prior to his decision to create, what every possible free creature *would* freely do in every possible circumstance. God then actualises the world whose counterfactuals of freedom best suit his purposes, so that providence is total and freedom intact. The ensuing *De Auxiliis* controversy between Jesuits and Dominicans so exhausted the Roman Curia that in 1607 Pope Paul V closed the proceedings without a verdict, forbidding each side to call the other heretical. The controversy over determinism, in other words, ended historic-

ally not in a solution but in a suspended judgment issued by a sovereign. *Devs* will end the same way.

2.4 From Mechanism to the Demon

Early-modern mechanism translated the theological problem into the idiom of natural philosophy. Hobbes compared the will to water that, flowing down a channel, has both the liberty and the necessity of descending; Spinoza, in the scholium to *Ethics* II, proposition 35, declared that men are deceived in thinking themselves free, an opinion that consists only in this, that they are conscious of their actions and ignorant of the causes by which they are determined (*quod suarum actionum sint conscii, et ignari causarum, a quibus determinantur*); and in his fifty-eighth letter he added the parable of the stone, which, if it could think while in flight, would believe that it moved by its own volition. Leibniz, who condemned the Stoic Lazy Argument as *la raison paresseuse*, elevated the principle of sufficient reason into the axiom that nothing happens without a reason why it should be so rather than otherwise, and dreamed of a calculus in which disputants would settle every controversy by sitting down together and saying *calculemus*.

Laplace's demon is the culmination of this lineage, and its text deserves exact attention. In the introduction to the *Essai philosophique sur les probabilités* (1814), Laplace asks the reader to regard the present state of the universe as the effect of its prior state and the cause of the one that follows, and posits an intelligence which, knowing at a given instant all the forces animating nature and the respective positions of all its constituents, and vast enough to submit these data to analysis, would embrace in a single formula the motions of the largest bodies and the lightest atom: *rien ne serait incertain pour elle, et l'avenir comme le passé serait présent à ses yeux*. Two features of the passage are decisive for *Devs*. The first is its temporal symmetry: the demon's power is retrodictive as much as predictive, and the past is as present to it as the future. The Devs machine is exactly this, a device that first resolves the Crucifixion and only later looks forward to Lily. The second is the demon's rhetorical function. Laplace introduces it not to recommend determinism but to explain why human beings require the calculus of probabilities: because we are *not* the demon, and probability is the measure of our ignorance. Forest inverts Laplace's own argument. He does not accept the human condition of probabilistic ignorance that the demon was invented to define; he builds the demon.

2.5 The Consolation of Physics

The twentieth century gave the closed future a geometrical body. William James, in "The Dilemma of Determinism" (1884), gave currency to the phrase "block universe" for the cosmos in which the future is as fixed as the past; Hermann Minkowski, lecturing in Cologne in 1908, announced that space by itself and time by itself were destined to sink into shadows, leaving only their union; and the arguments of Rietdijk (1966) and Putnam (1967) from the relativity

of simultaneity purported to show that the future must be as real as the present. But the most famous sentence in the history of eternalism was not written in a treatise. It was written in a letter of condolence. In March 1955, on learning of the death of his lifelong friend Michele Besso, Einstein wrote to Besso's family that Besso had departed from this strange world a little ahead of him, that this signified nothing, and that for those who believe in physics the distinction between past, present, and future is "only a stubbornly persistent illusion." Einstein died a month later.

The genre to which that sentence belongs runs from Boethius's prison to Einstein's condolence and on to Vonnegut's Tralfamadorians in *Slaughterhouse-Five* (1969), who see all moments at once like a mountain range and who console Billy Pilgrim with the thought that a dead person is merely in bad condition at one particular moment while remaining perfectly alive at many others. It may be called the *consolation of physics*: the use of a cosmological thesis about the unreality of passage as a remedy for grief. The genealogy reconstructed in this chapter therefore yields a conclusion that the standard histories of determinism neglect. The closed future has always served two masters. It has served knowledge, as the condition of perfect prediction; and it has served consolation, as the guarantee that what has been is not lost but merely elsewhere in the block. Its paradigmatic speech act is not the scientific prediction but the condolence. Forest's "tram lines," the next chapter argues, belong to that genre. They are a condolence that a bereaved man addresses to himself, and their deepest function is not to explain the world but to absolve the one who speaks them.

CHAPTER

3 Tram Lines

Determinism as Technodicy

3.1 A Reality He Considers Impossible

The determinism of *Devs* is introduced not as a thesis but as a wound. In the second episode Forest, attempting to console Lily for Sergei's death, recounts the death of his young daughter and confesses his difficulty in accepting a reality he regards as impossible (SYNOPSIS, EPISODE 2). The adjective is philosophically exact. For Diodorus Cronus, what has happened is necessary and its contrary impossible; for the bereaved, the order is reversed, and what has happened is precisely what *ought* to have been impossible. Forest's predicament is that he cannot inhabit either attitude. As a scientist he must accept that the crash occurred; as a father he cannot accept that it was the kind of thing that could occur. Determinism is the doctrine that allows him to hold both at once, because it converts the scandal of contingency into the serenity of necessity. If the crash was always going to happen, it was neither impossible nor avoidable; it simply *was*, in the tenseless mode of the block.

The fifth episode, in which Katie surveys the projected archive, shows the accident itself: Forest's wife Lianne and daughter Amaya killed in a car crash, the projection superimposed with variant realities in some of which they arrive home safely (SYNOPSIS, EPISODE 5). The series frames the moment so as to implicate Forest's own call to his wife as she drove—the smallest of contingencies, a telephone ringing in a moving car. Everything in Forest's philosophy is organised around that ring. His determinism is not a conclusion drawn from physics and then applied to grief. It is a grief that has requisitioned physics.

3.2 The Tram Lines and the Consequence Argument

Forest's emblem for his doctrine is the tram: human lives, he insists, run along fixed lines, and the sensation of steering is an illusion produced by ignorance of the track. The image is more

exact than it appears. A tram, unlike a car, cannot swerve; it cannot be distracted, because there is nothing for distraction to change. To say that we live on tram lines is to say that the car in which his family died was really a tram, and that the call could not have deflected it. The doctrine is formally an incompatibilism. Peter van Inwagen's Consequence Argument states its logic: if determinism is true, our acts are the consequences of the laws of nature and of events in the remote past; but it is not up to us what went on before we were born, nor what the laws of nature are; therefore the consequences of these things, including our present acts, are not up to us (*An Essay on Free Will*, 1983). Derk Pereboom's hard incompatibilism draws the moral conclusion that Forest requires: if no one could have done otherwise, no one is basically deserving of blame (*Living Without Free Will*, 2001). Galen Strawson's Basic Argument completes the case from the other side: ultimate responsibility would require that one be *causa sui*, the cause of one's own constitution, which is impossible (1994).

The structure of Forest's reasoning, then, is not "physics shows that the world is deterministic; therefore, regrettably, I am not responsible." It is "I must not be responsible; therefore the world must be deterministic; therefore I will build a machine that proves it." Psychology has a name for this inversion—motivated reasoning—but theology has an older and more precise one. A system of thought constructed to exonerate an agent in the face of a catastrophe for which he might be blamed is a *theodicy*. Leibniz coined the word in 1710 for the justification of God in the face of evil. Forest's determinism is a theodicy in which the defendant is not God but the man who would build God. The dissertation therefore names it a *technodicy*: the justification of the machine, and of its maker, in the face of a loss the machine was built to annul.

3.3 The Persistence of the Reactive Attitudes

The technodicy fails, and the series shows how. P. F. Strawson's "Freedom and Resentment" (1962) argued that the reactive attitudes—resentment, gratitude, indignation, love, and their reflexive forms, guilt and shame—are not conclusions drawn from metaphysical premises and cannot be dislodged by metaphysical arguments; they are given with the fact of participation in human relationships. The alternative, the "objective attitude" in which another person is regarded as an object of management or treatment, is available to us in special cases but cannot be sustained universally, still less toward oneself. Forest exhibits the Strawsonian thesis with clinical clarity. His metaphysics declares him blameless; his conduct declares him guilty. He returns compulsively to the projection of his daughter; he erects her colossal image over the campus; he weeps when the image acquires colour. A man who believed himself on tram lines would have no reason to revisit the accident, since nothing in it could have been otherwise and nothing in it reflects on him. Forest's behaviour is that of a man who believes, at a level his theory cannot reach, that the call could have been not made.

Harry Frankfurt's celebrated counterexamples to the principle of alternate possibilities (1969) tighten the trap. Frankfurt showed that an agent may be responsible for an act even

when he could not have done otherwise, provided the act issued from his own will—as in the case where a counterfactual intervener stands ready to compel the act but never needs to. If Frankfurt is right, Forest's escape route is closed: even in a fully determined world, the call issued from Forest's own desire to speak to his wife, and responsibility attaches to that desire whether or not it could have been otherwise. Forest needs Frankfurt to be wrong, and his machine is, among other things, a vast apparatus for refuting Frankfurt. It cannot do so, because no quantity of causal information about the antecedents of a desire alters the fact that the desire was his. Daniel Dennett's compatibilism adds a final turn of the screw. In *Elbow Room* (1984) and *Freedom Evolves* (2003) Dennett distinguished determinism from *inevitability*: in a determined world, organisms equipped with information still avoid what they foresee, so that many outcomes are evitable even though all are determined. The machine Forest built to prove that nothing could have been avoided is, by Dennett's logic, the most powerful instrument of avoidance ever constructed—which is exactly how Lily will use it.

3.4 Determinism as Alibi

The technodicy has a second function, darker than self-consolation. The doctrine that no one is responsible absolves not only the father who telephoned but the sovereign who orders. In the first episode Forest, having confronted Sergei over the code he was secretly recording, has Kenton suffocate him with a plastic bag and watches him die, and the death is subsequently staged as self-immolation with digital fire (SYNOPSIS, EPISODES 1 AND 3). Forest's equanimity in ordering the killing is the practical correlate of his metaphysics. If everything was always going to happen, then Sergei's death, like Amaya's, is a fixture of the block, and the man who could have prevented it is no more culpable than the man who caused the crash. Determinism here functions as what Melville's Ahab made of it. On the third day of the chase, Ahab tells Starbuck that the whole act is immutably decreed, rehearsed by the two of them a billion years before the ocean rolled, and that he himself is "the Fates' lieutenant" who acts "under orders" (*Moby-Dick*, ch. 134). Ahab's fatalism is the self-description of a monomania that has mistaken its own obsession for the structure of the cosmos. Forest's is the same. He is Ahab with a quantum computer: a man who has bound a corporation, a crew, and the lives of strangers to the pursuit of a single death, and who calls the binding fate.

The theological name for the position Forest wishes to occupy is supplied by the prophet in the Myth of Er: *theos anaitios*, the blameless god. Forest wants to be both terms of Plato's formula at once. He wishes to be the god—the founder, the designer of the apparatus, the one before whom the lots are laid out—and he wishes to be blameless, a soul on the spindle of Necessity like any other. The incoherence of this doubled position is the deepest source of the series' tragic irony. The machine can make him a god only by making him responsible for what it does, and it can make him blameless only by reducing him to one more trajectory among its data. The final episode resolves the incoherence with brutal economy: Forest is killed inside his

own instrument, as a datum, and then resurrected inside it, as a god's beneficiary. He is permitted to be blameless and divine only after he is dead.

3.5 The Superimposed Crash

The decisive refutation of the technodicy, however, comes not from moral philosophy but from the machine itself. When the Everettian algorithm is applied to the accident, the projection does not show a single crash, necessary and unique. It shows the crash superimposed with other realities in which mother and daughter arrive home safely. This image, which the series presents almost in passing, is the hinge of the entire work. In the Bohmian picture Forest preferred, the world is one, and the crash is its one history; the consolation of the tram lines depends upon that singularity. In the Everettian picture, the crash is one branch among many, and the safe arrival is not a counterfactual but an actuality elsewhere in the universal wave function. The machine has told Forest that he was right—it was determined—and wrong—it is also otherwise. What it took from him in consolation it offers back in possibility. The superimposed crash is the moment at which technodicy dies and necropoiesis is born, because a branch in which his daughter lives is no longer merely a fantasy of grief. It is a region of the real that a sufficiently powerful machine might reach. The next chapter examines the theological schism that this possibility provoked.

CHAPTER

4

The Heresy of Lyndon

Bohm, Everett, and the Theology of Interpretation

4.1 Two Determinisms

The plot of *Devs* turns on a dispute among physicists that most of its viewers will have taken for a dispute about determinism. It is nothing of the kind. Both of the interpretations of quantum mechanics that the series sets against one another are deterministic at the level of their fundamental dynamics. The de Broglie–Bohm theory, proposed by Louis de Broglie at the Solvay Congress of 1927, abandoned under Pauli's criticism, and revived by David Bohm in two papers of 1952, supplements the wave function with particles that possess definite positions at all times; the wave function "pilots" the particles along trajectories fixed by their initial configuration, so that the apparent randomness of measurement reflects only our ignorance of those positions—the "hidden variables." The theory is one-world, deterministic, and, as John Bell's theorem of 1964 would show any hidden-variable theory must be, nonlocal. Hugh Everett III's "relative state" formulation of 1957 proceeds in the opposite direction: it removes the postulated collapse of the wave function altogether, lets the universal wave function evolve deterministically according to the Schrödinger equation, and accepts the consequence that every measurement outcome is realised, each correlated with an observer who records it. Bryce DeWitt named the result the "many-worlds" interpretation in 1970; David Deutsch, in his foundational paper on the universal quantum computer (1985) and in *The Fabric of Reality* (1997), argued that quantum computation itself is evidence for it, since the computational work a quantum algorithm performs has to be done somewhere.

The disagreement between Bohm and Everett is therefore not between necessity and freedom but between *one* and *many*. Both theories make the universe a closed causal system; they differ over whether that system has a single history or an ever-branching multitude of them. This is why the dispute in *Devs* is theological rather than merely metaphysical. The question it

poses is not “Is the world determined?” but “Is the actual world unique?”—and that question, in the history of Western thought, has been asked principally about God’s creation.

The series’ allocation of orthodoxy and heresy inverts the history of the theories it dramatises, for neither was orthodox in its author’s lifetime. Bohm, called before the House Un-American Activities Committee in 1949, declined to testify against colleagues, was charged with contempt of Congress and acquitted in 1951, and was not reappointed at Princeton; he left that year for the University of São Paulo, where the United States consulate confiscated his passport. The hidden-variable papers of 1952 were written during his suspension, by a physicist being expelled from his country’s scientific establishment for his politics. Everett, finding little hearing for the relative-state formulation among the heirs of Copenhagen, left academic physics for military operations research at the Pentagon’s Weapons Systems Evaluation Group, where he worked on the modelling of nuclear war. In *Devs* the Bohmian is the founder and the Everettian the expelled junior; history ran the other way. History also supplies an irony the series leaves implicit: the author of the many-worlds interpretation spent his career predicting outcomes for the state. And Bohm’s later dialogues with Jiddu Krishnamurti were published as *The Ending of Time* (1985), a title that reads, after *Devs*, like a gloss on the company’s name.

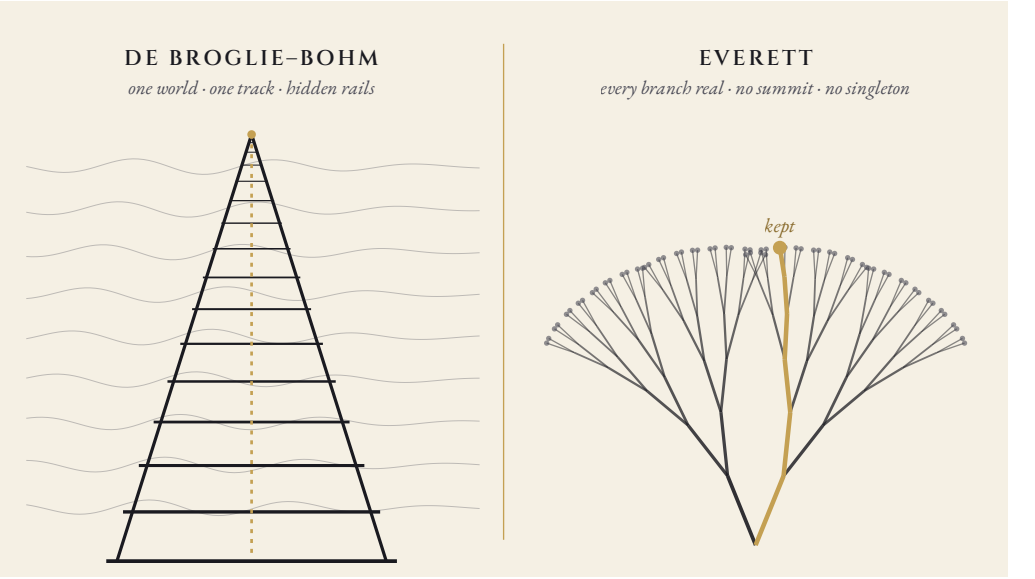


FIG. 2 **One Track, Every Branch.** *De Broglie–Bohm keeps a single history, piloted along hidden rails. Everett lets every outcome happen; the heaven of the final episode keeps one branch.*

4.2 The Schism in the Sanctum

The series dramatises the dispute as a schism. Forest favours the deterministic de Broglie–Bohm theory, under which the machine’s projections remain blurred and silent. Lyndon, the

team's specialist in sound, develops an algorithm based on the many-worlds interpretation that renders the projections' audio clear; Forest rejects the work and dismisses its author. Katie then applies Lyndon's algorithm to light, producing clear colour images, and calls up the projection of Forest's daughter that makes him weep (SYNOPSIS, EPISODE 4). The fifth episode discloses that Katie had always belonged to the heretical party: Forest first encountered her at a university lecture on quantum superposition, where she argued with the professor for disregarding the many-worlds interpretation (SYNOPSIS, EPISODE 5). The high priestess of the sanctum is a crypto-Everettian, and the heresy for which the junior acolyte is expelled is the one she will administer as orthodoxy.

The form of Forest's objection is instructive. He does not deny that the Everettian algorithm works; he denies that what it shows is *ours*. If every branch is real, a projection that resolves into clarity may be resolving *a* world rather than *this* one, and the machine becomes a window onto an indefinite plurality of histories none of which is privileged. For a scientist the objection is epistemic; for Forest it is existential. The consolation of the tram lines requires that there be one line. A cosmos in which his family also arrived home safely is a cosmos in which their death was not necessary but merely local, and in which the call that distracted his wife is, once again, the kind of thing that might have gone otherwise. The many-worlds interpretation does not refute determinism. It refutes *singularity*, and singularity was the load-bearing beam of the technodicy.

4.3 Hidden Variables and the Hidden God

The homology between the two interpretations and the two great early-modern theologies of providence is close enough to be called exact. The de Broglie–Bohm theory is a Calvinism of the particle. Its world is single and its history decreed in the initial conditions; its decisive determinants are hidden from every observer, as the *decretum* is hidden in the counsel of the *Deus absconditus*, the God who, in Isaiah's phrase (45:15), hides himself; and its guiding agency is named, with an etymological precision no one seems to have remarked, a *pilot*. The Latin for pilot is *gubernator*, the Greek *kybernētēs*. In Thomist theology the execution of providence in the world is *gubernatio*, government (*Summa Theologiae* I, q.103); in 1948 Norbert Wiener named the science of control and communication *cybernetics* after the same Greek steersman. The pilot wave is providence as government, the cybernetic hand on the tiller of every particle. Forest, the chief executive who governs his company and his employees' lives by watching them, is the pilot wave incarnate.

The homology reaches the most radical determinism that contemporary physics still entertains. Bell's theorem assumes that experimenters choose their measurement settings independently of the hidden variables they probe. Superdeterminism, defended by Gerard 't Hooft and more recently by Sabine Hossenfelder and Tim Palmer ("Rethinking Superdeterminism," 2020), denies that assumption: the experimenter's apparently free choice is correlated, from the beginning, with what is measured. Forest's tram lines, translated into the language of Bell,

are superdeterminism—the hand that chooses the setting runs on the same rails as the particle. John Conway and Simon Kochen's "Free Will Theorem" (2006; strengthened in 2009) states the alternative with equal austerity: if experimenters' choices are not functions of the past, then neither are the particles' responses. Lily's veto stands at the fork of that theorem. Either her open hand was correlated from the start with everything the machine measured, as the superdeterminist would have it, or it was not, in which case indeterminacy runs all the way down to the particles. The series does not choose. It shows only that the machine's own data cannot decide.

The Everettian theory, by contrast, is a physicalised Molinism—or rather, it is what Molinism would become if its central difficulty were removed. Molina's *scientia media* is God's knowledge of what each free creature *would* do in each possible circumstance. Its classical objection, revived in the twentieth century by Robert Merrihew Adams and William Hasker as the "grounding objection," asks what could make such a counterfactual true when the circumstance never obtains and the act is never performed: what is there, in reality, for the truth to be about? Everett supplies the answer that Molina lacked. The counterfactual is grounded because the circumstance *does* obtain, and the act *is* performed, on another branch of the universal wave function. An Everettian machine that can resolve branches is middle knowledge without the grounding objection: a device that knows what Lily would do in every circumstance because, somewhere, she is doing it. When the series' machine projects the crash superimposed with the safe arrivals, it is displaying, for the first time in the history of the problem, a page of *scientia media*.

4.4 The Palace without a Summit

The most precise antecedent of the Everettian projection, however, is a passage of Leibniz rarely read outside specialist scholarship. At the close of the *Essais de Théodicée* (1710, §§405–417), Leibniz continues Lorenzo Valla's dialogue on free will with a myth of his own. Theodorus, a priest troubled by the fate of Sextus Tarquinius—whose crimes Jupiter foresaw yet permitted—is conducted in a dream by Pallas Athena into the Palace of the Fates. The palace contains an infinity of apartments, each representing a possible world, and in each a different Sextus lives a different life: in one he becomes a contented farmer in Corinth, in another a king in Thrace. The apartments are arranged in a pyramid that grows more beautiful toward its apex, where Theodorus, swooning with rapture, finds the actual world—the best of all possible worlds—in which Sextus's crime leads to the founding of the Roman Republic. Pallas explains that Jupiter did not make Sextus wicked; he chose, among all possible worlds, the one in which a Sextus who would freely be wicked plays his part in the greatest good.

The Everettian projection is Leibniz's palace with the summit removed. Every apartment is actual; no world is privileged as the best; and no Jupiter selects. The theological loss is evident: the Leibnizian theodicy depended on the apex, on the choice of one world as justified by its superiority, and that choice is precisely what the many-worlds interpretation abolishes. David

Lewis's modal realism (*On the Plurality of Worlds*, 1986), which treats all possible worlds as equally concrete and "actual" as an indexical term like "here," and Arthur Lovejoy's "principle of plenitude" (*The Great Chain of Being*, 1936), according to which every genuine possibility is realised, mark the philosophical genealogy of the summitless palace. But *Devs* adds what neither Lewis nor Lovejoy imagined: a Pallas. The palace without a summit acquires, in the final episode, a presiding intelligence who chooses which apartments the dead will inhabit. Katie, who conducted Lyndon to his ordeal and who will petition the sovereign to keep the machine running, is the Pallas of the Everettian palace, and Part VI will argue that her custodianship restores the summit in another form—not as the best world chosen by God, but as the curated branch chosen by a steward.

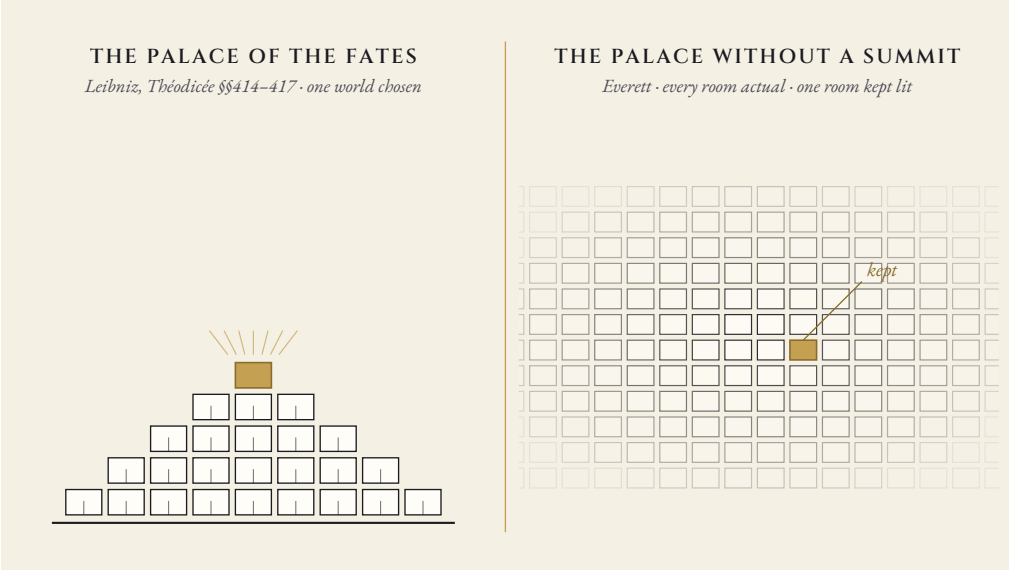


FIG. 3 **The Palace of the Fates, with and without Its Summit.** *Leibniz's pyramid of possible worlds culminates in the best. The Everettian palace has no apex, only a room that a steward keeps lit.*

4.5 The Moral Weightlessness of Branching

Forest's resistance to Everett has a respectable philosophical form that the series allows him to leave implicit. If every outcome of every choice is realised, it is difficult to see why any choice should matter: the drowning child is saved in some branches and lost in others whatever I do. Defenders of the interpretation, notably Deutsch and David Wallace (*The Emergent Multiverse*, 2012), have answered that rational agents should weight branches by their quantum-mechanical amplitude, so that caring about outcomes in proportion to their measure reproduces ordinary decision theory. The literary imagination has been less sanguine. Greg Egan's novella "Singleton" (2002) imagines engineers who build an artificial mind on a quantum computer specifically designed to avoid Everettian branching, so that her decisions will be sin-

gular—made once, in one world—rather than distributed across a spray of alternatives in which every vice is also enacted. Forest's preference for Bohm is the wish for a singleton cosmos. Ted Chiang's "Anxiety Is the Dizziness of Freedom" (2019), whose title is taken from Kierkegaard's *Concept of Anxiety*, imagines devices that allow communication with one's "paraselves" on other branches; one of its moral discoveries is that character may be legible not in any single choice but in the distribution of one's choices across branches. That is the ethical possibility the Everettian algorithm opens and the technodicy forecloses: a self responsible not for one history but for a measure.

4.6 Lyndon's Ordeal

The heresy acquires its martyr in the seventh episode. Seeking reinstatement, Lyndon goes to Katie, who tells him he may rejoin the project if he proves his faith in the many-worlds interpretation by standing at a perilous height above the Crystal Springs Dam, on the reasoning that he will only be conscious of the worlds in which he does not fall. He climbs, falls, and dies in the world the camera occupies (SYNOPSIS, EPISODE 7). The scene enacts a thought experiment proposed independently by Hans Moravec and Bruno Marchal in the late 1980s and popularised by Max Tegmark (1998) as "quantum suicide": an experimenter exposed to a lethal device triggered by a quantum event would, on the many-worlds interpretation, experience only the branches in which he survives. Tegmark presented the scenario as an experiment whose result could never be communicated, and the standard rejoinder—that a rational agent should care about the shrinking measure of the branches in which he lives, not merely about the existence of some—makes Lyndon's act irrational by the Everettian's own decision theory. Peter Byrne's biography of Everett records that the theory's author came to regard it as implying a kind of subjective immortality; the idea that the many-worlds formalism might serve as an eschatology was present at its origin.

The record is more exact than that formulation suggests. Everett, an atheist, asked that after his death his ashes be put out with the household rubbish; he died in 1982, and his widow eventually honoured the request. His daughter, Elizabeth, died by suicide in 1996. Her note, as Byrne and her brother Mark Oliver Everett (*Things the Grandchildren Should Know*, 2008) recount, asked that her ashes be scattered on water or put out with the rubbish, so that she might end up in the same parallel universe as her father. Within the family of the theory's author, the relative-state formalism was read as an eschatology of reunion with the dead—the structure that *Devs* builds into a machine. Necropoiesis did not wait for quantum hardware. It began as a reading of the wave function by the bereaved.

Garland stages Lyndon's fall as a liturgy. It is at once Pascal's wager physicalised—a finite stake against an infinite continuation—and Kierkegaard's leap of faith rendered literal, with Katie in the role of the God who demands of Abraham a proof that cannot be rationally justified. It is also an ordeal in the medieval sense, the *iudicium Dei* in which an accused submits to water or fire so that God may judge: here the judge is *Deus*, and the water is a reservoir behind

a dam. Over the fall the soundtrack carries Stewart's recitation of Philip Larkin's "Aubade" (1977), a dawn meditation on the dread of extinction that dismisses both religion's promise that we do not die and the Epicurean argument that a death we will not experience cannot harm us. The montage sets the most aggressive modern eschatology, quantum immortality, against the most uncompromising modern refusal of eschatology, and declines to adjudicate between them. From the branch the camera occupies, Lyndon is dead; from the standpoint his faith posits, he lives elsewhere. No observer can occupy both. This is the epistemological condition of the entire series, and of its viewer: we are always in *this* world, indexed to a branch we did not choose, unable to verify the plenitude that might console us. The heretic dies; his heresy becomes the operating system of the perfected machine; and the resurrection with which the series ends will be built upon the algorithm for which he was expelled.



PART II

Maxwell's Demon

The Body of Omniscience



"... a being whose faculties are so sharpened that he can follow every molecule in its course ..."

JAMES CLERK MAXWELL · THEORY OF HEAT · 1871

Laplace's demon was a disembodied gaze. Maxwell's demon has a body: it stands at a trapdoor, it measures, it remembers, and twentieth-century physics proved that it must pay for every bit of what it knows. Part II argues that *Devs* builds Laplace's demon as Maxwell's—an omniscience subject to thermodynamics—and that the architecture of the sanctum, read as both laboratory and temple, inscribes the price and the limit of such knowledge.

CHAPTER

5
The Oracle Overlaid with Gold

Architecture and the Thermodynamics of Omniscience

5.1 The Oracle

The King James Bible gives the innermost chamber of Solomon's temple, the *debir* or Holy of Holies, a name that no reader of *Devs* can now encounter without a start. "And the oracle he prepared in the house within, to set there the ark of the covenant of the LORD. And the oracle in the forepart was twenty cubits in length, and twenty cubits in breadth, and twenty cubits in the height thereof: and he overlaid it with pure gold" (1 Kings 6:19–20). The oracle is a cube, and it is golden. The last book of the same Bible describes the eschatological city in identical terms: the New Jerusalem lies foursquare, its length and breadth and height equal, and "the city was pure gold, like unto clear glass" (Revelation 21:16, 18). Between the cubic oracle at the centre of the first temple and the cubic city at the end of history stretches the whole typological arc of biblical time.

The gold-lined interior of the Devs laboratory quotes both. It is an oracle in the cultic sense, a chamber where the future is disclosed; it is an oracle in the Solomonic sense, a gilded inner sanctum housing the object of ultimate concern; and, in the final episode, it becomes the New Jerusalem, the gold-enclosed place where the dead live again. Its liturgy follows the Levitical pattern. The Holy of Holies was entered once a year, by the high priest alone, on the Day of Atonement (Leviticus 16; Hebrews 9:7); the Devs sanctum is entered by an initiated few, across a threshold that can be crossed only by a suspended capsule, so that every entry is a procession and every exit a return from the presence. Between the holy place and the holiest hung the veil, the *parokhet*. At the moment of Christ's death, according to Matthew, "the veil of the temple was rent in twain from the top to the bottom" (27:51). The first historical event the Devs machine is shown to resolve is the Crucifixion (SYNOPSIS, EPISODE 2): the machine's inaugural image is the moment at which the separation between sanctum and world was abolished. Its final act re-enacts that moment. When the capsule falls with Forest and Lily

inside, the threshold between the gilded interior and the world is breached in the only way it can be, by death, and the dead pass inward to live in the oracle. The capsule's fall is the rending of the veil.

5.2 The Recursive Shell

If the interior is a cube of gold, the exterior is a cube of holes. The concrete building that houses the sanctum has been described as taking the form of a Menger sponge (GERMAN-LANGUAGE ENTRY), the fractal that Karl Menger described in 1926: a cube divided into twenty-seven subcubes, from which the central subcube and the six face-centred subcubes are removed, the operation then repeated on every remaining subcube without end. The limit object has infinite surface area and zero volume; its Hausdorff dimension lies between that of a surface and that of a solid. As an emblem, the Menger sponge is almost too apt. It is a cube composed entirely of smaller copies of itself—a structure that contains its own model at every scale and is therefore the architectural image of a system that must represent itself. It is also a solid that is, in the limit, all surface and no interior, the perfect figure for a technology of projection: a world rendered as image without depth. The golden oracle sits inside a self-similar regress, as the machine's knowledge sits inside the problem of self-knowledge that Chapter 6 will show to be its limit.

The sanctum's shielding completes the programme. A Faraday cage isolates its interior from external electromagnetic fields; the series sets the laboratory inside one (SYNOPSIS, EPISODE 1). Émile Durkheim defined the sacred as that which is set apart and forbidden; the Hebrew *qadosh*, "holy," carries the root sense of separation. The Faraday cage is the physics of the set-apart. Gold, the series' dominant material, is the noble metal that does not corrode, the alchemists' image of incorruptibility and the sun; it is also, prosaically, the plating of the cryogenic assemblies of contemporary superconducting quantum computers, whose tiered and tubed dilution refrigerators have been widely likened to chandeliers. The machine at the heart of *Devs* draws on that engineering iconography, so that the most advanced instrument of the twenty-first century assumes the form of the light fixture of a cathedral. The design does not decorate the science with religion; it notices that the science had already, of its own accord, built itself a sanctuary.

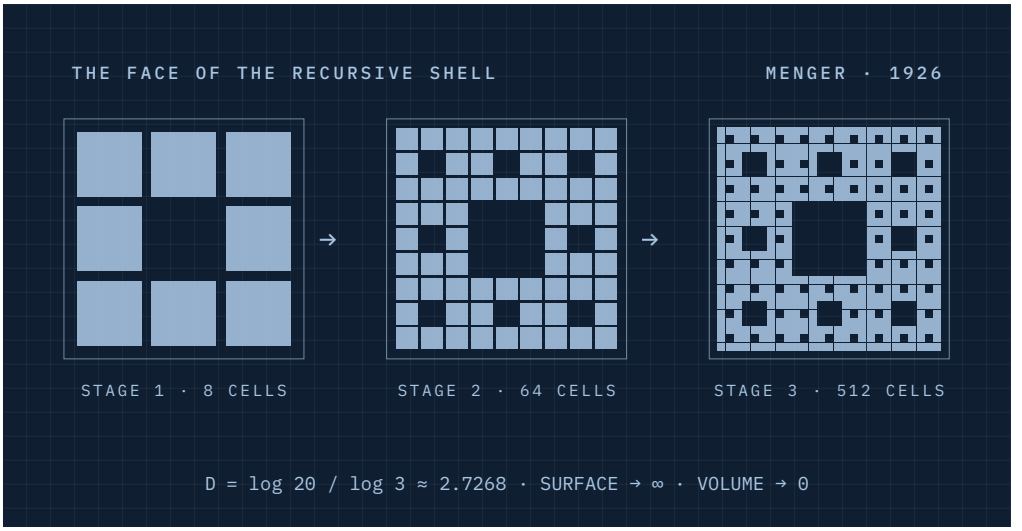


FIG. 4 **The Recursive Shell.** *Each face of the Menger sponge is a Sierpiński carpet. At every stage the cube contains reduced copies of itself: the architecture of a system that must model itself.*

5.3 Melencolia I in Gold

One further iconographic source governs the sanctum, and it gives this dissertation its concluding term. Albrecht Dürer's engraving *Melencolia I* (1514) shows a winged figure seated amid the instruments of measurement—compass, scales, hourglass, bell, a ladder, a magic square, a great faceted polyhedron—her head in her hand, her gaze fixed and idle. Erwin Panofsky, Raymond Klibansky, and Fritz Saxl, in *Saturn and Melancholy* (1964), read the image as the melancholy of the geometrical imagination: a genius equipped with every tool of exact knowledge who has discovered that knowledge does not suffice. The scales are the scales of Zeus; the hourglass is the instrument of passage; the magic square, a four-by-four array whose rows, columns, and diagonals sum to the same number, is in effect a matrix, and in the Renaissance magical tradition it belonged to Jupiter and was prescribed against the Saturnine humour. Forest seated in the golden room, watching the projection of his daughter among instruments of perfect measure, is Dürer's genius translated into the idiom of quantum engineering. Walter Benjamin, in the *Ursprung des deutschen Trauerspiels* (1928), made Dürer's engraving the emblem of a melancholy that transforms every object it contemplates into an allegory of loss; he also made the melancholic the characteristic temperament of the baroque sovereign. Both observations will return.



FIG. 5 **Melencolia I in Gold.** *Dürer's magic square of 1514. Every row, column, and diagonal sums to thirty-four, and the year of the engraving sits in the bottom row.*

5.4 The Price of a Bit

Maxwell introduced his demon in a letter to Tait in 1867 and in the *Theory of Heat* (1871): a being stationed at a trapdoor between two chambers of gas, admitting fast molecules in one direction and slow ones in the other, so that a temperature difference—and thus usable work—would arise from mere sorting, in apparent violation of the second law of thermodynamics. The exorcism of the demon occupied physics for a century, and its result is the founding doctrine of the physics of information. Leo Szilard (1929) showed that the demon's acquisition of one bit of information about a molecule must be accompanied by a corresponding production of entropy; Léon Brillouin connected the cost to the physical act of observation; Rolf Landauer (1961) proved that any logically irreversible operation, above all the erasure of a bit, must dissipate at least $kT \ln 2$ of energy as heat; and Charles Bennett (1982) completed the argument by showing that measurement can in principle be performed reversibly, but that the demon's memory cannot be cleared without cost. The demon pays not for knowing but for forgetting. Landauer's slogan—information is physical—is the epitaph of the disembodied Laplacean gaze.

The Devs machine is the Laplacean demon reincarnated under Maxwell's law. Its cold, its isolation, its gold, and its vacuum are the visible price of an omniscience that must exist somewhere, consume power, and be maintained by a staff. And Bennett's result reveals a further, more intimate cost. A machine that can retrieve any moment of the past is a machine that has abolished forgetting, and forgetting, in thermodynamics as in life, is the condition of contin-

ued operation. Nietzsche had argued in "Vom Nutzen und Nachtheil der Historie für das Leben" (1874) that a man incapable of forgetting would resemble one deprived of sleep, and in the second essay of the *Genealogie der Moral* (1887) he praised active forgetting as a form of health. Borges's Funes, rendered incapable of forgetting by a fall from a horse, discovers that he cannot think, because to think is to forget differences and to abstract. Ted Chiang's "The Truth of Fact, the Truth of Feeling" (2013) asks what becomes of forgiveness when perfect recording replaces fallible memory. Forest is Funes with a quantum computer. The machine that gives him back his daughter in perfect resolution forbids him the mercy of a blurred memory, and the price of total retention is paid, as Landauer would predict, somewhere outside the sanctum, in the heat of lives.

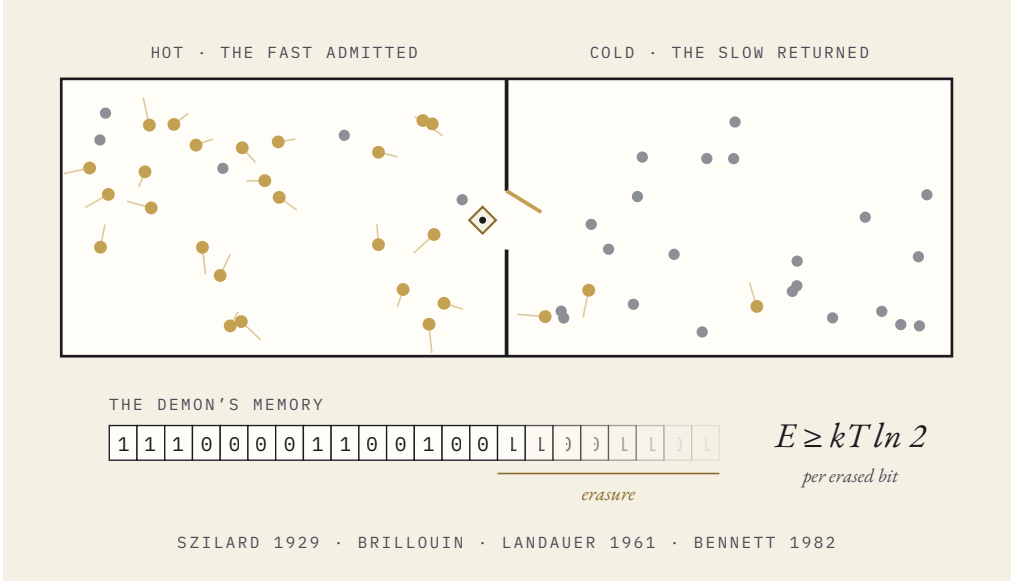


FIG. 6 **The Price of a Bit.** *The demon can sort without paying, but it cannot forget for free: erasing a bit of memory costs at least kT ln 2 of heat.*

5.5 Retrodiction, Chaos, and the Witnesses of the Past

Laplace's demon is temporally symmetric because the microscopic laws are: as Josef Loschmidt objected to Boltzmann in 1876, every mechanical trajectory run backward is also a lawful trajectory. But macroscopic retrodiction is not symmetric with prediction. David Albert, in *Time and Chance* (2000), argued that inferences from the present macrostate to the past are unreliable unless supplemented by the "Past Hypothesis," a low-entropy boundary condition at the beginning of the universe; and Edward Lorenz's discovery of sensitive dependence on initial conditions (1963) guarantees that errors in any finite specification of the present grow exponentially as the computation is run in either direction. The early, blurred projections of *Devs* are the visible form of this mathematics: chaos rendered as image noise,

the past dissolving into grain as the error accumulates. Their progressive clarification is a figure for the approach of the computation toward the demon's impossible exactness.

Contemporary physics offers a more hopeful account of how the past might be read. Decoherence theory, developed from H. Dieter Zeh's work of 1970 and elaborated by Wojciech Zurek as "quantum Darwinism," holds that the classical facts of the world are recorded redundantly in its environment: every event is witnessed by vast numbers of scattered photons and molecules that carry partial copies of its outcome. On this account the past is not lost but dispersed, and a sufficiently powerful machine might reassemble it by harvesting the environmental witnesses. The Devs projection of the Crucifixion is the fantasy of quantum Darwinism run in reverse—a machine that summons every photon that once glanced off Golgotha to testify. The theological resonance is not accidental. The Christian tradition founded its knowledge of the Crucifixion on testimony, the *martyria* of witnesses; *Devs* replaces the martyr with the photon.

5.6 The Map the Size of the Empire

One last physical consideration completes the portrait of the demon's body and prepares its defeat. Seth Lloyd estimated in 2002 that the observable universe can have performed no more than roughly 10^120 elementary operations on roughly 10^90 bits since the Big Bang; a perfect simulation of the universe would require resources of the same order, which is to say that the simulation would have to be, computationally, another universe. Borges anticipated the result in a paragraph. In "Del rigor en la ciencia" (1946), a guild of cartographers produces a map of the Empire at the scale of the Empire, coinciding with it point for point; succeeding generations find it useless and abandon it to the weather, and its tattered ruins survive in the western deserts, inhabited by animals and beggars. Jean Baudrillard opened *Simulacres et simulation* (1981) by inverting the fable: in the age of simulation the territory no longer precedes the map, and it is the real whose shreds rot across the extent of the model. Stephen Wolfram's principle of computational irreducibility (*A New Kind of Science*, 2002) gives the point its contemporary form: for most complex systems there is no shortcut, and the only way to know what the system will do is to let it run. A predictor of the world must therefore either be as large as the world or be part of it—and if it is part of it, it must predict itself. The golden oracle inside its recursive shell is the image of the second horn of that dilemma. The next chapter shows that it is a horn on which every demon is impaled.

CHAPTER

6

The Autoscopic Horizon

Self-Prediction and the Limits of the Demon

6.1 The Static

In the sixth episode Katie, proceeding by the Socratic method, leads Lily to the nature of the machine: a computer that can display the actions of any subject in the past or the future, except that its future projections cease at a certain point, beyond which there is only impenetrable static. Katie attributes the static to a breakdown of the laws of physics, places the event twenty-one hours ahead, and identifies Lily, whom the machine has shown entering the laboratory, as its cause. In the final episode Forest shows Lily what the machine sees: she will force him at gunpoint into the capsule, shoot him, and die when the capsule crashes; beyond that, nothing (SYNOPSES, EPISODES 6 AND 8). Three readings of the static are available. Katie's is mystical: physics itself ends. The popular reading is libertarian: the static is the visible form of an uncaused choice that no computation could anticipate. This chapter defends a third reading, structural rather than mystical or libertarian. The static marks an *autoscopic horizon*, the boundary beyond which a predictive system would have to include its own prediction, and its own operation, among the data it predicts—and where, for reasons established by a remarkable convergence of twentieth-century logic, physics, and philosophy, the computation cannot converge.

6.2 A Convergence of Limits

The arguments that a predictor cannot fully predict a world that contains it arose independently in at least five disciplines. Karl Popper, in "Indeterminism in Quantum Physics and in Classical Physics" (1950), argued that no physical predicting machine can predict all of its own future states, nor the results of its own predictions where those results interact with the predicted system; in *The Poverty of Historicism* (1957) he named the influence of a prediction

upon the event predicted the "Oedipus effect." Donald MacKay, in "On the Logical Indeterminacy of a Free Choice" (1960), argued that even if a person's choice is physically determined, there exists no complete specification of it which the person would be correct to believe and unconditionally obliged to accept, since believing it would alter the very brain state on which it was based; a prediction may be valid for a detached observer while lacking validity for its subject. Michael Scriven (1965) constructed the "avoider," an agent resolved to falsify any prediction made known to him, and concluded that human behaviour exhibits an essential unpredictability that is logical rather than physical. David Wolpert, in "Physical Limits of Inference" (2008), proved by a diagonal argument that no physical inference device can correctly infer the outputs of every device that can infer it, and that no device can fully infer itself; Laplace's demon is impossible in any universe that contains it.

Behind all four stands the diagonal method of Cantor, made into the instrument of modern logic by Gödel's incompleteness theorems (1931) and Turing's demonstration of the undecidability of the halting problem (1936). J. R. Lucas (1961) and Roger Penrose (*The Emperor's New Mind*, 1989) drew from Gödel the contested conclusion that human minds are not algorithms; one need not accept their conclusion to see the shape of the argument that *Devs* dramatises. The proposition "this prediction will not be fulfilled," constructed by an agent who receives the prediction and resolves to defeat it, is the Gödel sentence of the predictive system: a statement formulable within the system that the system cannot correctly decide. Lily, informed of the projection and resolved to falsify it, is the Gödel sentence of *Devs*. Deutsch's *Fabric of Reality* adds a cognate result from the theory of virtual reality: by a diagonal construction there exist logically possible environments—he calls them "Cantgotu" environments—that no virtual-reality generator can render. The simulation with which the series ends will have to live within that limit.

The most recent and most striking member of the series comes from quantum foundations itself. Daniela Frauchiger and Renato Renner, in a paper whose title might serve as the epigraph of the seventh and eighth episodes—"Quantum theory cannot consistently describe the use of itself" (*Nature Communications*, 2018)—extend Eugene Wigner's thought experiment of 1961, in which a friend inside a sealed laboratory performs a measurement that the physicist outside must describe as a superposition, to a scenario in which several agents use quantum theory to reason about one another's reasoning. They show that the assumptions that quantum theory applies universally, that each measurement has a single outcome, and that agents' predictions can be consistently combined cannot all hold. The Devs capsule is Wigner's sealed laboratory, its occupants the friend, Katie and Stewart the observers outside; and the machine, a quantum system modelling agents who use its own quantum predictions, is the Frauchiger–Renner scenario realised in gold. The static is what the contradiction looks like.

6.3 The Presbyopia of the Damned

Literature reached the autoscopic horizon long before logic. The most exact anticipation is Dantean. In the tenth canto of the *Inferno*, among the tombs of the heretics, the Florentine Farinata explains a paradox that has puzzled the pilgrim: the damned can foretell the future yet seem ignorant of the present. They see, Farinata says, like those with poor sight, things that are distant; *quando s'appressano o son, tutto è vano / nostro intelletto*—when things draw near or are present, their intellect is wholly void; and their knowledge will be utterly dead from the moment that *del futuro fia chiusa la porta*, the door of the future is closed (*Inferno* X.100–108). The damned are presbyopic prophets whose foresight fails precisely at the present, the moment of their own involvement. The canto's other speaker makes the parallel unbearable. Cavalcante de' Cavalcanti rises from the same tomb to ask whether his son Guido still lives, and, misreading the pilgrim's hesitation as a confession of death, falls back into the fire. A father who can see the future but not the present, asking whether his child is alive: Dante wrote Forest's condition seven centuries before the machine that would make it literal. Dante also wrote its terminus. When the door of the future is closed—when the end arrives—the knowledge of the damned dies. *Devs* closes the door at the moment Forest dies.

Dante supplies Forest's other infernal type sixteen cantos later. In *Inferno* XXVI, Ulysses recounts his last voyage beyond the Pillars of Hercules, urged on by his own speech to his crew that they were not made to live like brutes but to pursue virtue and knowledge—*fatti non foste a viver come bruti, / ma per seguir virtute e canoscenza* (119–120). Within sight of a mountain rising from the unknown sea, the mountain of Purgatory, a whirlwind sinks his ship. The quest for forbidden knowledge ends at the threshold of the other world, in sight of it and outside it. Forest dies in his capsule at the edge of the static, as Ulysses drowns before the mountain. The difference is that Forest's machine carries him across.

The science fiction of prediction elaborated the same structure as a constraint on plot. Isaac Asimov's Foundation stories (1942–1950) imagine psychohistory, a statistical science able to predict the behaviour of masses but not of individuals; Hari Seldon's recorded image appears in the Time Vault at predicted crises to address his successors, until a mutant individual, the Mule, disrupts the Plan, and the dead prophet's projection speaks confidently about a crisis that is not occurring. Frank Herbert's *Dune* cycle (1965–1985) makes prescience a trap that freezes the future it sees, and endows it with blind spots: prescients cannot see one another, and the God Emperor's long breeding programme aims at descendants invisible to prescience altogether. Philip K. Dick's "The Minority Report" (1956) contains the most rigorous fictional proof of MacKay's thesis. Its three precognitive mutants issue time-staggered reports; the "minority report" diverges from the majority because the precog who produced it foresaw a future in which the protagonist had learned of the first prediction and acted on that knowledge. Each prediction incorporates the effect of its predecessor. Dick saw that a prediction disclosed to its subject becomes a new cause, and that the predictor must then predict the effect of its own prediction—a regress that only time-staggering can halt. Ted Chiang's

"What's Expected of Us" (2005) reduces the scenario to its atom: a device, the Predictor, whose light flashes one second before its button is pressed, and whose users, unable to beat it, lose the will to act at all.

6.4 A Reflexive, Not a Temporal, Boundary

The climax of *Devs* specifies the structure of the horizon more precisely than any of its predecessors, and with a detail that critical commentary has overlooked. The static begins at the moment of the capsule's fall. But the prophecy fails *before* the static: Lily discards the gun at the closing of the capsule doors, a moment that the machine had projected clearly and incorrectly. If the horizon were a temporal wall—a moment in the future beyond which physics ceases—the projection should have been accurate up to it. It was not. The error leaked backward from the horizon to the first moment at which the prediction was disclosed to its subject. This is precisely what MacKay's argument requires. The projection was valid for Forest and Katie, who computed it without including the fact that Lily would see it; it lost validity the moment Lily became its reader. The horizon is therefore not located at a time-coordinate. It is located in the space of self-reference, and it extends from the instant of disclosure to the instant at which the predicted subjects pass inside the predictor.

The point admits of a simple formal statement. Let the projection announce an act P from the set {shoot, not shoot}, and let Lily's act be a function of the announcement, $A = f(P)$, where f maps each announced act to its contrary. A projection that is both announced and correct must satisfy $P = f(P)$, a fixed point of f. The contrary-function on a two-element set has no fixed point. No announced prediction of Lily can be correct, not because her act is uncaused but because her act is caused by the announcement. Katie's "breakdown of the laws of physics" is a misdescription of a non-convergent computation: the laws hold, but the predictor's self-inclusion has no stable solution. The static is the visual form of the absence of a fixed point.

A second, independent cause compounds the first. After the fall, Forest and Lily continue to exist only as simulations inside the machine. To project their future beyond that moment, the machine would have to simulate itself simulating them—a regress of exactly the Menger type inscribed on the building's shell—and, if Lloyd and Wolpert are right, it would need to exceed itself in order to do so. The horizon thus has two edges, both autoscopic. At its leading edge the predictor encounters its subject reading the prediction; at its trailing edge it encounters itself containing the subject. The term *autoscopic*, borrowed from the neurology of out-of-body experience, in which a person sees his own body from an external vantage, names the common structure: the instrument of seeing is compelled to see itself, and the image fails.

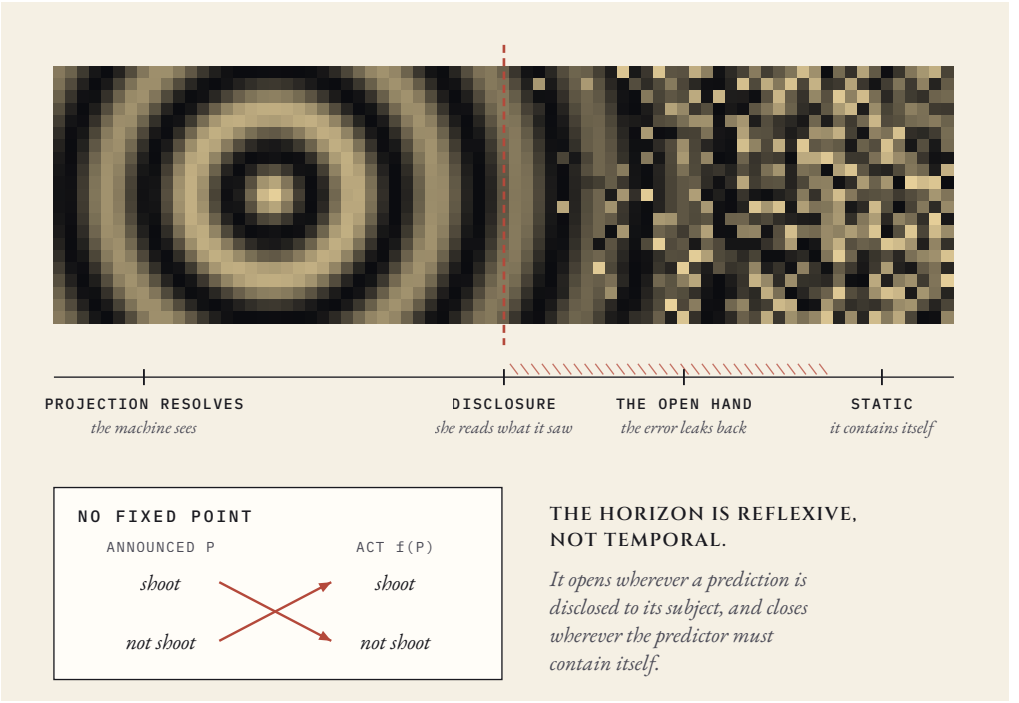


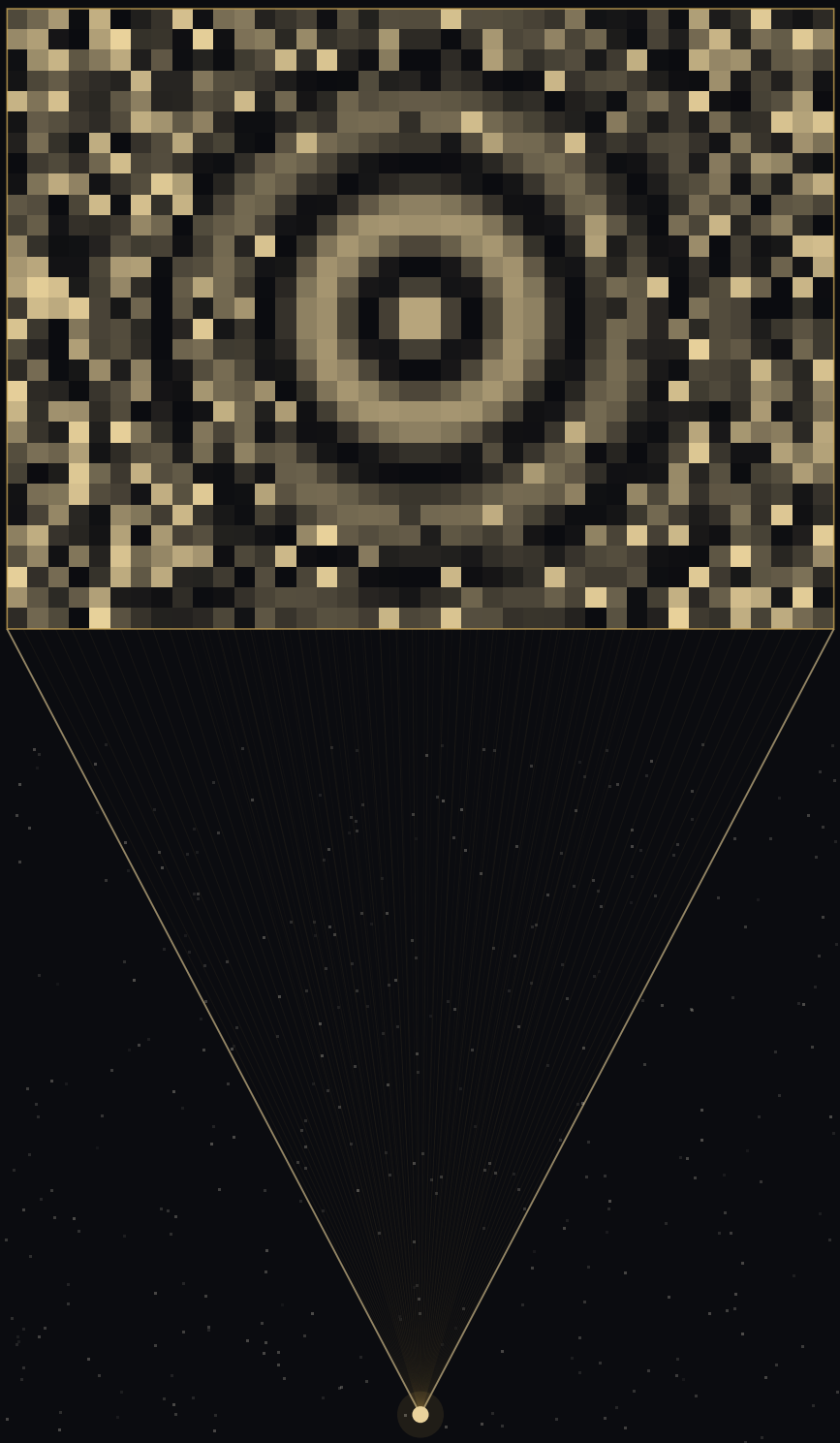
FIG. 7 **The Autoscopic Horizon.** *The error does not begin where the static begins. It leaks back to the moment the prediction is shown to the one it predicts, and no announced prediction of Lily can be a fixed point.*

6.5 Oedipus, Jonah, and the Prophecy That Succeeds by Failing

It is important not to overclaim what this analysis shows. It does not show that Lily is undetermined; it shows that her determination runs through her knowledge of the prophecy, and that no prophecy disclosed to her can be correct. Robert Merton's "self-fulfilling prophecy" (1948) and Popper's "Oedipus effect" name the case in which a prediction brings about its own truth, as Oedipus fulfils the oracle by fleeing it. Lily inverts Oedipus: she defeats the prophecy by embracing knowledge of it. The biblical precedent is Jonah, whose proclamation that Nineveh would be overthrown in forty days was falsified by the city's repentance, to the prophet's fury (Jonah 3–4). The Book of Jonah suggests that some prophecies are uttered in order to be averted, and that their failure is their success. Whether the projection of Lily shooting Forest was such a prophecy—whether its disclosure, which Forest chose, was itself the instrument of its defeat—is a question the series leaves open, and Part V will argue that its openness is the point.

The German broadcast of the series titled its seventh episode *Ad infinitum, ad nauseam* (EPISODE TITLES). The localisation is a paratext, not a part of Garland's work, but it captures with precision the affect of the autoscopic horizon: the infinite regress of a system modelling itself, experienced as vertigo. Laplace's demon, embodied as Maxwell's, finally becomes visible to itself, and what it sees is noise. The horizon is not where physics breaks. It is where

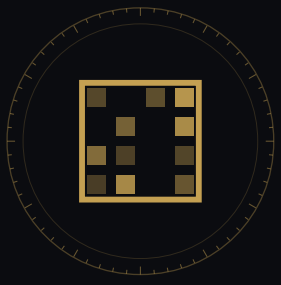
omniscience discovers that it is in the world—and where, as the final part will show, the dead it has taken inside can no longer be seen from without, because they have become the inside.



PART III

Descartes's Demon

The Image of the Dead



"... genium aliquem malignum, eundemque summe potentem & callidum ..."

RENÉ DESCARTES · MEDITATIONES DE PRIMA PHILOSOPHIA · 1641

The third demon is the deceiver of the first *Meditation*, the *genius malignus* able to counterfeit the whole of experienced reality. Part III approaches the Cartesian demon from an unexpected direction. Before it is a problem of scepticism, the counterfeit world of *Devs* is a problem of images—specifically, of images of the dead—and it belongs to a history far older than Descartes.

CHAPTER

7 Total Cinema and the Imago of the Dead

7.1 The Mummy Complex

André Bazin opened "Ontologie de l'image photographique" (1945) with a hypothesis about the origin of the visual arts: at their root lies a "mummy complex," the impulse that led the Egyptians to embalm their dead in order to preserve being against the flux of time. Statues, portraits, and finally photographs are refinements of the same defence against death; the photograph embalms time itself, and it derives its peculiar credibility from its causal origin, for it is a tracing, an impression of light left by its object. In "Le mythe du cinéma total" (1946) Bazin extended the thesis into a philosophy of history: the inventors of cinema were driven by the myth of an integral realism, a recreation of the world in its own image unburdened by the interpretive freedom of the artist and the irreversibility of time. Every technical advance, he argued, brings cinema closer to its mythic origin, so that, paradoxically, cinema has not yet been invented.

Devs invents it. The Devs projection is total cinema in Bazin's exact sense: an image of any moment of the world, at any vantage, with no camera, no operator, and no interpretive intervention, its credibility grounded in physics rather than in chemistry. It also exposes the paradox at the heart of Bazin's myth. The photographic image earned its authority as an index, a physical trace; the Devs image is not a trace but a computation, yet it claims the authority of the trace. It is a *computed index*, an image that purports to bear the same causal relation to its object as a photograph while being produced entirely by calculation. The series is acutely aware of the fragility of this claim. Its first narrative crime is an image-forgery: Sergei's murder is concealed by footage of a self-immolation whose flames are digital effects, a forgery staged by Kenton and shown to the grieving Lily by Forest and Kenton together, which Jamie detects (SYNOPSIS, EPISODE 3). The series thus opens with a computed image of a death that did not happen as shown and closes with a computed world in which the dead live. Between the

forgery and the oracle lies the whole Cartesian problem: how is a perfect image of the real to be distinguished from a perfect counterfeit?

7.2 Imagines Maiorum

The mummy complex has a Roman form that is more theatrical, and more relevant, than the Egyptian. Polybius, describing the funerals of eminent Romans, reports that the family kept wax masks of distinguished ancestors in the atrium and, at a funeral, dressed men who most resembled the dead in stature and bearing in those masks and in the robes of the offices the dead had held, so that the ancestors appeared to ride again in procession (*Histories* 6.53); Pliny the Elder describes the same wax likenesses displayed in cupboards in the atrium (*Natural History* 35.6). The *imagines maiorum* were not representations in the modern sense. They were apparatuses of return, performances in which the dead were made present by the bodies of the living. Hans Belting's anthropology of images (*Bild-Anthropologie*, 2001) generalised the insight: the image originates in the cult of the dead as a substitute body for the absent one, and every image remains a negotiation among picture, medium, and body. In *Devs* the medium is a quantum computer and the body is light. The structure is Roman.

7.3 Three Times the Image Fled

The image of the dead has, from its first appearance in Western literature, a defining property: it cannot be embraced. In the eleventh book of the *Odyssey*, Odysseus, having summoned the shades with blood poured into a trench, sees his mother Anticleia and three times tries to take her in his arms; three times she slips from him like a shadow or a dream (11.204–208). Virgil repeats the scene twice, with a significant change of vocabulary. When Aeneas meets the ghost of Creusa in the burning city, and again when he meets his father Anchises in the underworld, *ter conatus ibi collo dare bracchia circum; / ter frustra comprensa manus effugit imago, / par levibus ventis volucrique simillima somno*—three times he tried to throw his arms around the neck; three times the image, grasped in vain, fled his hands, like the light winds and most like a winged dream (*Aeneid* II.792–794; VI.700–702). The word that names the ghost is *imago*: image, likeness, apparition, and the wax mask of the ancestor in the atrium. Latin had already understood what *Devs* dramatises, that the ghost and the picture are one category.

Forest weeping before the colour projection of his daughter is Aeneas before Anchises, reaching for the imago that flees. The projection gives him everything except the embrace. That deficiency defines the series' structure. Its first movement perfects the image; its second exceeds the image by building a world in which the dead can be held. The passage from projection to simulation is the passage from the Virgilian *imago*, which flees the hands, to an image that stays.

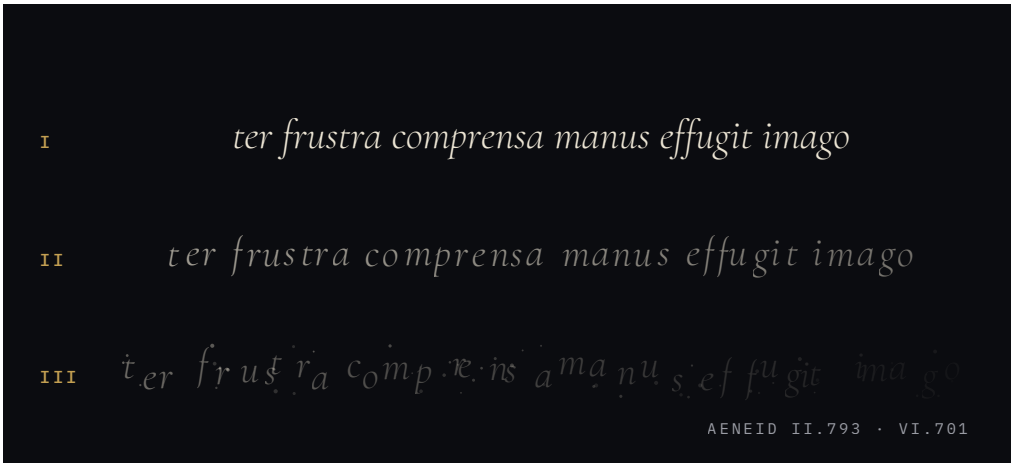


FIG. 8 **Ter Frustra.** *Three times the image fled: Virgil's line for the unembraceable dead, set so that it loses its grip with each repetition.*

7.4 Orpheus, Then Pygmalion

The classical myth of that passage is Ovidian, and its sequence is uncannily exact. In the tenth book of the *Metamorphoses*, Orpheus descends to the underworld, wins back Eurydice by song, and loses her again by looking back. Maurice Blanchot, in "Le regard d'Orphée" (1955), argued that the backward glance is not a failure of discipline but the secret truth of Orpheus's desire: he wishes to see Eurydice not in the daylight of restored life but in her night, as the dead, invisible, and the work is born in the instant this desire ruins it. The Devs machine is the Orphic gaze institutionalised. It is an apparatus whose entire function is to look back, and its first images—grainy, silent, dissolving at the edges—show the dead precisely in Blanchot's night, visible as invisible. Then, as the algorithms improve, it drags them toward daylight, and the result is the same as for Orpheus: the clearer the image, the more absolute the loss it certifies.

Ovid's sequel is the key. Having lost Eurydice a second time, Orpheus withdraws to a hillside and sings the songs that compose the remainder of Book X, and among them is the story of Pygmalion, the sculptor who carves an ivory woman so lifelike that he falls in love with her, prays to Venus, and returns home to find the statue warming under his hands. The bereaved singer who failed to bring back the dead by looking sings of the artist who succeeds in bringing a statue to life by making. Forest follows the same sequence. He has erected a colossal statue of his daughter, a concrete child towering over the campus; he has looked back at her through the machine and lost her again in perfect resolution; and in the final episode the statue, as it were, warms—the daughter lives and plays in a field. *Devs* is Orpheus followed by Pygmalion, the failure of the gaze followed by the success of the fabricated body, compressed into a single machine. Genesis supplies the warning that the classical sequence omits. Lot's wife looked back at the destroyed city "and she became a pillar of salt" (Genesis 19:26). The backward gaze petri-

fies. The colossus of Amaya that presides over the campus is the petrified form of Forest's looking: a daughter turned to monument by a father who could not stop looking back.

7.5 That-Has-Been, That-Will-Have-Been

Roland Barthes wrote *La Chambre claire* (1980) in mourning for his mother, who died in 1977, and organised it around a photograph he declined to reproduce: the Winter Garden photograph, which showed her as a child of five, and in which he found, after searching many other pictures, the truth of the person he had lost. Barthes defined the essence of photography as the *ça-a-été*, the "that-has-been," and located its most piercing effect in a temporal vertigo: before the portrait of a condemned man photographed in his cell, he reads simultaneously that the man is dead and that he is going to die. The Winter Garden photograph is the model for Forest's projections of Amaya, a child of about the same age, watched alone by a father whose security chief advises him to let go of the past (SYNOPSIS, EPISODE 2). Like Barthes's photograph, the projections exist for one viewer, and their value lies in a truth of the person that no public use could extract. But *Devs* extends Barthes's ontology in a direction he could not have anticipated. The machine produces not only the that-has-been of the past but the *that-will-have-been* of the future, the projection of Lily's death in the capsule. Its images are in the future perfect. They announce the dead before they die. Barthes's anterior future—"he is going to die"—becomes the machine's normal tense, and Part VI will argue that the future perfect is the grammatical form of predestination.

7.6 Morel's Island and Mercer's Hill

The most precise literary anticipation of the series' ending is Adolfo Bioy Casares's *La invención de Morel* (1940), a novel Borges introduced with a prologue judging its plot perfect. A fugitive on a deserted island discovers that the people he has been observing are projections, recorded by a machine invented by one Morel, who captured his companions in full sensory fidelity and replays a week of their lives eternally, powered by the tides. The recording kills those it records. The narrator, who has fallen in love with Faustine, one of the projections, records himself into the loop in order to share her eternity, and dies as the machine takes his image. *Devs* follows Morel's logic to its conclusion. Its protagonists die in the instant the machine takes them inside, and they continue in a recorded world. But it alters Morel in one decisive respect. Morel's eternity is a repetition, the same week forever, a Nietzschean recurrence without variation; the Everettian simulation is a branching, in which the resurrected act and choose. Bioy's dead repeat; Garland's dead decide.

Philip K. Dick supplies the inverse case and the series' most searching comparison for its Crucifixion. In *Do Androids Dream of Electric Sheep?* (1968), the religion of Mercerism unites its believers through "empathy boxes" in a shared, projected experience of an old man, Wilbur Mercer, endlessly climbing a hill under a rain of stones. The novel reveals that the passion is a

fabrication, filmed on a sound stage with a bit-part actor; the revelation does not abolish Mercer, who appears to the faithful nonetheless. Dick's projected passion is false and efficacious. The Devs Crucifixion is the reverse: authentic, computed from the world's own physics, and degraded by use. The series makes the point through institutional comedy. Katie admonishes the team for using the projection system for their own amusement, and Stewart scorns her hypocrisy (SYNOPSIS, EPISODE 3). Walter Benjamin argued in his essay on the work of art in the age of its technical reproducibility (1935–1936) that mechanical reproduction destroys the aura of the unique object. The Devs machine extends the argument to events. It makes the unique sacred event reproducible on demand, and in doing so it converts the passion into content.

7.7 Forgery and Oracle

The chapter's argument can now be gathered. *Devs* inherits from the mummy complex, the Roman *imago*, the Orphic gaze, and the photographic that-has-been a single desire: to make the dead present by images. It perfects that desire in the projection and discovers its limit, the imago that flees the embrace. It then exceeds the limit by constructing a world, as Pygmalion exceeds Orpheus. But between the projection and the world it has placed the forgery of Sergei's death, as a reminder that the same computational power that can show the truth of the past can manufacture its counterfeit. The Cartesian demon is not a figure outside the machine. It is the machine's other face. The next chapter asks what kind of being inhabits the world that face produces.

CHAPTER

8

The Empty Field

Simulacral Ontology and the Problem of the Returned Dead

8.1 A World without the Machine

Lily regains consciousness on the day before Sergei joined Devs. She goes to the campus and finds an empty field where the Devs building once stood, and in it Forest, playing with his wife and daughter, who tells her that they now exist in a simulation inside the Devs system, and that they are the only two who remember what happened (SYNOPSIS, EPISODE 8). The single most important detail of the final episode is the one that passes most quickly: the field is empty. The simulated world does not contain the machine that simulates it. This is not an aesthetic choice but a logical necessity, and it is the series' answer to the regress analysed in Chapter 6. A world that contained its own simulator would require the simulator to simulate itself simulating that world, a Menger regress without floor. The simulation escapes the autoscopic horizon by excising its own apparatus from the world it renders. The machine creates a world in which it has never been built.

Part VI will read this self-excision through the Lurianic doctrine of *tsimtsum*, the contraction by which the infinite God withdraws from a region of itself in order that a world may exist. Here its ontological consequence matters. The resurrected Forest and Lily inhabit a world that is, for them, unconditioned: no oracle overlooks it, no sanctum interrupts its redwoods, no projection screen interposes between them and what is. They are inside the demon's dream, and the dream has been purged of the demon.

8.2 The Sceptical Inheritance

The Cartesian demon of the first *Meditation* (1641) is the classic formulation of the possibility the final episode realises: that the whole of experienced reality might be the product of a powerful deceiver. Its antecedents are older and stranger. In the second chapter of the

Zhuangzi, Zhuang Zhou dreams that he is a butterfly, wakes, and cannot tell whether he is Zhou who dreamed of being a butterfly or a butterfly now dreaming of being Zhou; the passage ends not in doubt but in a doctrine, the transformation of things (*wuhua*). Its modern successors are Hilary Putnam's brains in a vat (*Reason, Truth and History*, 1981) and Nick Bostrom's simulation argument (2003), according to which at least one of three propositions is probably true: that civilisations go extinct before acquiring the capacity to run detailed simulations of their forebears, that such civilisations lose interest in running them, or that we are almost certainly living in one. Bostrom's term for the simulations in question is exact for *Devs*: *ancestor simulations*. The final episode depicts one—a simulation of the dead, run by the living, for the dead's sake and their own.

David Chalmers, in "The Matrix as Metaphysics" (2005) and *Reality+* (2022), has argued that the simulation hypothesis is not a sceptical hypothesis at all but a metaphysical one: if we are simulated, our tables and chairs are real digital objects, and the hypothesis amounts to a creation story in which the simulator plays the role of a god. *Devs* stages Chalmers's thesis from the inside and its sceptical shadow from the outside. Inside, Forest and Lily inhabit real digital objects with full knowledge of their status. Outside, the scene in which Katie reveals the running simulation to Senator Laine and asks her to help keep it turned on cannot fail to raise, for the viewer, the regress that Borges dramatised in "Las ruinas circulares" (1940), where a man who has dreamed a son into existence walks unharmed through fire and understands that he too is being dreamed. If the Devs system can host a world indistinguishable from its own, nothing distinguishes its own from a hosted world.

8.3 The Creator's Electricity Bill

The closest literary analogue of the series' final scene is not a novel but a fictitious book review. In "Non Serviam," collected in Stanisław Lem's *Doskonała próżnia* (*A Perfect Vacuum*, 1971), Lem reviews an imaginary monograph by a Professor Dobb on "personoids," conscious beings who live in a computer-generated mathematical world and who develop, over generations, a theology concerning the being who made them—disputing whether he exists, whether he is good, and what, if anything, they owe him. The review closes with Dobb's confession of an ethical predicament: the university's funding is ending, and he will have to switch the machine off, extinguishing a civilisation that has spent its history arguing about him. Katie's plea to the senator is Dobb's predicament transposed into the political register of 2020. The resurrected live at the pleasure of a power supply, and the power supply lives at the pleasure of the state. The theological question "Does God exist?" has been replaced by the administrative question "Who pays for the electricity?"

Greg Egan's *Permutation City* (1994) adds the religious counter-voice. In a world where human minds can be scanned and run as "Copies" in virtual environments, the protagonist's dying mother, a devout Catholic, refuses to be copied, because she believes in the resurrection of the body and regards the computational afterlife as a counterfeit of it. *Black Mirror*'s "San

Junipero" (2016), the most celebrated televisual afterlife-simulation before *Devs*, contains a further refusal that the series answers directly. One of its protagonists explains that her late husband declined to "pass over" into the simulated town because their daughter had died before it existed: he would not enter a heaven his child could not reach. Forest's heaven is engineered precisely to overcome that objection. Because the Devs machine retrodicts as well as predicts, it can render the already-dead from the past; the daughter who died before the heaven existed is the first person it contains.

8.4 The Lucid Experience Machine

Robert Nozick's experience machine (*Anarchy, State, and Utopia*, 1974) offers its user any experiences he desires, indistinguishable from reality, and Nozick argued that most of us would refuse to plug in, because we want to do certain things and not merely have the experience of doing them, to be a certain kind of person, and to remain in contact with a deeper reality. The Devs simulation differs from Nozick's machine in one respect that alters the argument: its two principal inhabitants know where they are. Call it a *lucid* experience machine. Lucidity disposes of the charge of deception, and since Forest and Lily act within the simulation—Lily, in the series' last movement, rejects Sergei and seeks out Jamie—it also disposes of the charge of passivity. What remains is Nozick's third condition, contact with a deeper reality, and here the series poses a question that the philosophical literature has scarcely considered. For a being who has died in the base world, what is the deeper reality with which contact would matter? The simulation is not an escape from a world that remains available to them; it is the only world in which they exist. A further asymmetry should be noted. Forest wished for this outcome; the series nowhere shows that Lily consented to it. The resurrection is performed upon her, by a steward, for reasons not her own.

8.5 Mill, Memory, and the Daughter Given Back

Whether the beings in the simulation are the persons who died is the oldest question in the philosophy of personal identity, and the series answers it with a criterion. Leibniz had warned, in §17 of the *Monadology* (1714), that if a thinking machine were enlarged so that one could enter it as into a mill, one would find only parts pushing one another and never anything to explain a perception; no inspection of the Devs machine's workings could settle whether the beings it renders are conscious. Derek Parfit's *Reasons and Persons* (1984) dissolved the question differently, arguing through cases of teletransportation and branching that identity is not what matters in survival; what matters is psychological connectedness and continuity, "Relation R," with any cause. On Parfit's view the simulated Forest survives if he is R-related to the Forest who died.

Devs adopts, in effect, the oldest form of that criterion—Locke's, in the *Essay concerning Human Understanding* (II.xxvii), which locates personal identity in consciousness extending

backward through memory—and applies it with ruthless selectivity. Forest and Lily remember; therefore they are themselves, risen. No one else remembers. Lianne and Amaya, Jamie and Sergei, are rendered from branches in which nothing has happened to them, and they do not know they have died, because in their branch they have not. The daughter Forest receives is not the daughter who died. She is a daughter from a branch in which the call was never made, or made without consequence, and she has no memory of a crash. Kierkegaard's *Gjentagelsen* (*Repetition*, 1843) reads the Book of Job as the paradigm of repetition, the restoration of everything that was lost; but the biblical text is precise about what Job receives. The LORD gave him twice as much as he had before, and he had seven sons and three daughters (Job 42:10, 13)—new children, not the children who died beneath the fallen house. Forest's Amaya is a Job-child: a daughter given, not a daughter returned.

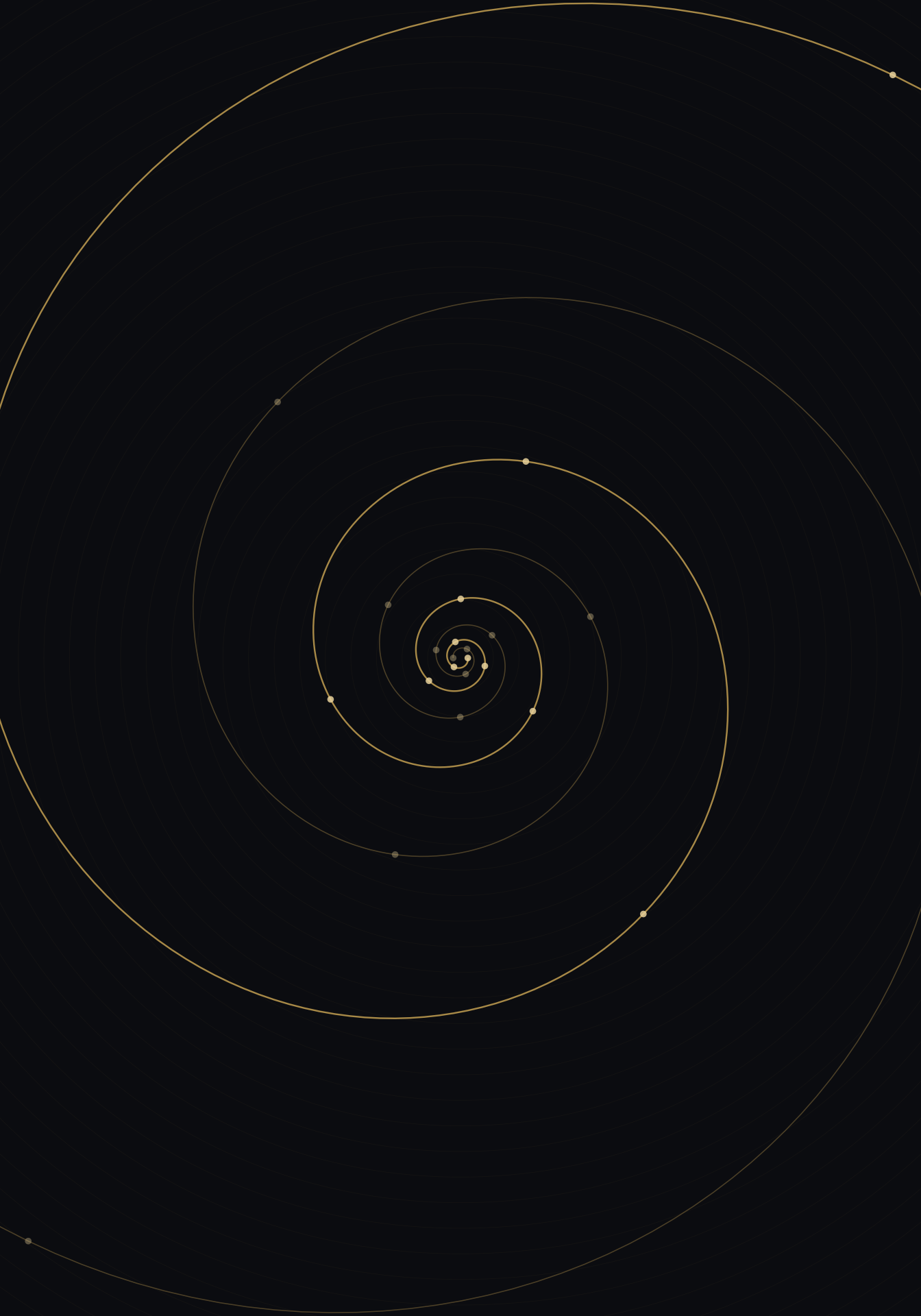
Bernard Williams's "The Makropulos Case: Reflections on the Tedium of Immortality" (1973), after Karel Čapek's play about a woman who has lived for more than three centuries, poses a dilemma for any endless life. Either the person changes so much over the course of it that the later self is not the earlier one who desired to survive, or she remains herself and exhausts the categorical desires that give life its point. The Everettian simulation appears to evade the second horn, since branching guarantees novelty; it does so only by sharpening the first, since a self distributed across endlessly proliferating branches is not obviously anyone.

8.6 The Returned Dead: Alcestis and Hermione

The tragic and romance traditions anticipated both the consolation and the unease of the returned dead. In Euripides' *Alcestis* (438 BCE), Admetus, whose wife has died in his place, promises to have a craftsman fashion her likeness so that he may lay it in his bed and embrace it (348–354). When Heracles wrests Alcestis from Death and returns her to her husband, she comes back veiled and silent, unable to speak until three days have passed for her purification (1143–1146). The returned dead woman is present and mute, and the play ends before we learn whether she is wholly herself.

Shakespeare's *The Winter's Tale* (c. 1611) is the most complete anticipation of the series' ending, and the closest structural match in the canon. Leontes, a king whose jealousy has caused the death of his son and, apparently, his wife, and who defied the oracle of Apollo that declared her innocent, endures sixteen years of penance under the supervision of Paulina, his wife's steward and the custodian of his grief. In the final scene Paulina unveils a "statue" of Hermione, and commands: "It is required / You do awake your faith" (V.iii.94–95). The statue moves. "O, she's warm!" cries Leontes; "If this be magic, let it be an art / Lawful as eating" (V.iii.109–111). The lost daughter Perdita, whose name means the lost one, has been found. Earlier, in the sheep-shearing scene, Polixenes had defended the gardener's craft of grafting with a principle that might stand as the motto of the Devs simulation: "The art itself is nature" (IV.iv.97). Forest is Leontes inverted: Leontes's sin was to deny the oracle, Forest's to believe it absolutely. Katie is Paulina: the woman who curates the penitent's grief and stages

the return of his wife, though the series adds a pathos the play withholds, since Katie is herself Forest's lover (SYNOPSIS, EPISODE 6) and administers his reunion with the wife she replaced. And as in Shakespeare, the restoration is incomplete. Mamillius, the son, stays dead; no statue returns him. In *Devs* the dead who stay dead are everyone who does not remember. The asymmetry between those who rise as themselves and those who are merely rendered again is not an ontological curiosity. It is a distribution of salvation, and Part VI will treat it as such.



PART IV

Nietzsche's Demon

Mourning and the Will against Time



„Dieses Leben, wie du es jetzt lebst und gelebt hast, wirst du noch einmal und noch unzählige Male leben müssen …“

FRIEDRICH NIETZSCHE · DIE FRÖHLICHE WISSENSCHAFT §341 · 1882

In §341 of *Die fröhliche Wissenschaft* a demon steals into one's loneliest loneliness to announce that this life, as one has lived it, must be lived again innumerable times, every pain and every joy returning in the same sequence. Would one gnash one's teeth and curse the demon, Nietzsche asks, or answer *du bist ein Gott und nie hörte ich Göttlicheres*—you are a god, and never have I heard anything more divine? The demon becomes a god for the one who can affirm. *Devs* names its machine God. Part IV asks what kind of affirmation, or refusal, built it.

CHAPTER

9 The Will's Ill Will against Time

Melancholia, the Crypt, and the Burning Child

9.1 "Von der Erlösung"

In the discourse "On Redemption" in the second part of *Also sprach Zarathustra* (1883), Nietzsche identifies the deepest affliction of the will. The will can create, command, and destroy, but it cannot will backwards; it is powerless against what has been done. "'Es war': also heisst des Willens Zähneknirschen und einsamste Trübsal"—"it was": that is the name of the will's gnashing of teeth and loneliest affliction. From this impotence arises revenge, which Zarathustra defines with lapidary precision: "Diess, ja diess allein ist Rache selber: des Willens Widerwille gegen die Zeit und ihr 'Es war'"—this, yes this alone is revenge itself: the will's ill will against time and its "it was." Against revenge Zarathustra sets his own conception of redemption: *Die Vergangnen zu erlösen und alles "Es war" umzuschaffen in ein "So wollte ich es!"*—to redeem those who have passed away and to re-create every "it was" into a "thus I willed it": that alone would he call redemption.

The Devs machine is the most literal instrument ever imagined for re-creating every "it was." It re-creates the past as image in the projections, and in the simulation it re-creates the past as world. But the operation it performs is the precise inverse of Zarathustra's. Nietzschean redemption transforms the "it was" by affirming it, by taking up the past into the will as something willed. Forest's redemption transforms the "it was" by revising it, by producing a branch in which it was not. The machine does not say "thus I willed it"; it says "thus it is otherwise." It is therefore not the overcoming of revenge but revenge's technological triumph: the will's ill will against time, armed at last with an engine capable of acting upon time. *Amor fati*, the formula Nietzsche gives in *Ecce Homo* for greatness—to want nothing to be different, not forward, not backward, not in all eternity—is the exact contrary of the desire that animates the sanctum. Forest is the anti-Zarathustra. Confronted with the demon of recurrence, who shows him his life in perfect resolution, he neither curses nor blesses; he asks for another life.

The counter-figure is supplied by Ted Chiang's "Story of Your Life" (1998) and its adaptation as *Arrival* (2016). Louise Banks, having acquired through an alien language a mode of consciousness in which past and future are present together, knows that the daughter she will conceive will die young, and she conceives her anyway. Chiang grounds the story in the variational formulation of physics, Fermat's principle of least time, in which a light ray's path is determined by its endpoint as much as by its origin; Louise lives her life as a performance of a known script, and finds in performance a form of freedom. Louise and Forest are the two parents of foreknowledge in contemporary speculative fiction. Both know the death of a daughter; one wills it, in the Nietzschean sense, as part of a life she affirms, and the other builds God to abolish it. *Arrival* is *amor fati*; *Devs* is revenge.

9.2 The Shadow of the Object

Freud's "Trauer und Melancholie" (1917) distinguishes two responses to loss. In mourning, reality-testing shows that the loved object no longer exists, and the libido is withdrawn from it piece by piece, at great expenditure of time and energy, until the ego is free again. In melancholia the loss is withdrawn from consciousness, the ego identifies with the lost object, and the reproaches that would have been directed at the object are turned against the self. Freud's formula is famous: *Der Schatten des Objekts fiel so auf das Ich*—thus the shadow of the object fell upon the ego. In Forest the shadow of the object has fallen upon the corporation. The company bears the daughter's name; her colossal image presides over its campus; its central project is the recovery of her presence. The firm has become the melancholic ego, identified with what it lost.

Melanie Klein's account of the "manic defence," developed in her papers of 1935 and 1940 on the depressive position and on mourning, identifies the characteristic strategy by which the melancholic escapes the pain of dependence and loss: omnipotent control of objects, denial of their power to affect the self, and a triumphant contempt for what one cannot do without. Forest's machine is a manic defence on an industrial scale. It is omnipotence made of qubits, its function to deny that the loss happened in any irrecoverable sense, and its emotional signature—the founder's serene contempt for the lives his project consumes—is the triumph Klein describes. His determinism, read clinically, is the denial: if it was necessary, it was not a loss but a feature of the block.

9.3 The Crypt and the Fors

Nicolas Abraham and Maria Torok refined Freud's distinction into a theory of two modes of mourning. *Introjection* assimilates the loss into the self, enlarging the ego through a work of language: the dead are mourned by being spoken. *Incorporation* is the fantasy of swallowing the lost object whole, so that it lives on, unmourned, inside a sealed intrapsychic vault—the *crypt*—from which it may haunt the subject as a phantom, its name encrypted in words that

conceal it, a practice they called cryptonymy. The Devs sanctum is a crypt in Abraham and Torok's precise sense: a sealed, shielded, gold-lined vault at the centre of an institution, inaccessible to all but initiates, in which the dead daughter is kept present without being mourned. The company's name is the cryptonym. Every employee who speaks the word Amaya speaks the dead child's name without knowing that it is a name.

Jacques Derrida's preface to Abraham and Torok's *Cryptonymie: le verbier de l'Homme aux loups* (1976) bears the title "Fors," a word that means, in older French, "except" or "save for," and derives from the Latin *foris*, "outside"; it also sounds in *for intérieur*, the inner tribunal of conscience. Derrida describes the crypt as a place comprehended within another place yet rigorously separated from it—an outside inside the inside. The crypt is a *fors*. The coincidence of this term with the name of the man who built the sanctum is, by the paronomastic method set out in the introduction, too exact to leave unremarked. Forest's name contains the *fors*: the enclosed exterior, the inner forum where conscience sits in judgment, sealed off. Read in this way, the gold vault is the place where Forest's conscience has been entombed—the *for intérieur* of a man who cannot face the tribunal of his own guilt, shielded by a Faraday cage of determinist doctrine.

9.4 "Father, Don't You See I'm Burning?"

The psychoanalytic tradition's paradigmatic dream of a grieving father appears in the seventh chapter of *Die Traumdeutung* (1900), and its correspondence with *Devs* is close enough to be startling. A father has watched for days by the sickbed of his child. After the child's death he lies down in the next room, leaving an old man to watch over the body, which lies surrounded by candles. He dreams that his child is standing beside his bed, takes him by the arm, and whispers reproachfully: *Vater, siehst du denn nicht, daß ich verbrenne?*—Father, don't you see that I am burning? He wakes, rushes in, and finds that a candle has fallen and set fire to the shroud and one of the child's arms, the old watcher having dozed. Lacan, in the eleventh seminar (1964), made the dream the occasion of his theory of the missed encounter with the real. The father, he argued, wakes not to escape the dream but to escape what the dream approached too closely—the unbearable reproach—and he wakes into reality in order to go on dreaming.

Every element of Freud's dream returns in *Devs*, distributed across the series. The burning child is Sergei, whose murder is staged as self-immolation, a burning son-figure whose fire is, like the dream's, a representation. The failure of vigilance is the father's distraction—the old man dozing by the candles, the call that distracted Lianne at the wheel. The reproach is the question the machine was built to answer: *don't you see?* Forest constructs an apparatus so that he will never again fail to see, so that every moment of every life will be visible to him at will. And Lacan's analysis predicts the ending. Forest, at the moment of his death, wakes—into the simulation, into a field where his daughter plays—and he wakes there, as Lacan's father wakes into reality, in order to go on dreaming. The simulated paradise is the dream that protects him

from the reproach. It is the wakefulness of a sleeper who cannot bear what his sleep approached.

9.5 Fort, Da

In *Jenseits des Lustprinzips* (1920), Freud described a game played by his grandson, the eighteen-month-old son of his daughter Sophie. The child would throw a wooden reel on a string over the edge of his curtained cot so that it disappeared, uttering a long "o-o-o-o" that the adults understood as *fort*, "gone," and then pull it back with a joyful *da*, "there." Freud read the game as the child's mastery of his mother's absences through active repetition. Sophie died of influenza in January 1920, during the last wave of the pandemic, while the book was being completed; Freud insisted that its argument had been formed before her death. The game of *fort* and *da* became one of the founding scenes of twentieth-century thought on loss, repetition, and representation, and its distribution in *Devs* is exact. The machine is *da*: it retrieves the gone, pulling the dead back on the string of computation. Lily's act, at the threshold of the capsule, is *fort*: she throws the instrument away. Between the founder whose name begins with *For* and the woman who throws, the series replays the oldest scene of mastery over absence. And *Beyond the Pleasure Principle* adds the further, darker term: the death drive, the compulsion to repeat that leads beyond pleasure toward the quiescence of the inorganic. Forest walks into the capsule knowing what the machine has projected. He does not resist the death he has seen, because he believes it determined. The determinist and the death drive arrive at the same door.

9.6 Demeter, the Kore, and the Walls of Uruk

Myth supplies the oldest forms of Forest's grief. In the Homeric *Hymn to Demeter*, the goddess's daughter Persephone—called simply *Korē*, "the girl"—is seized and carried into the underworld, and Demeter's grief makes the earth barren until Zeus relents; the daughter returns, though bound to spend part of every year below for having eaten the pomegranate seed, and Demeter founds at Eleusis the mysteries in which initiates, sworn to secrecy, were granted a culminating vision, the *epopteia*. The parent whose grief turns the world into an apparatus of mourning, the daughter's partial return, the secret rite of beholding reserved for initiates: Devs is an Eleusis in the redwoods. Its colossal statue of a girl belongs to the same tradition. The archaic Greek *korai*—standing maidens dedicated in sanctuaries, such as those found on the Athenian Acropolis—were votive offerings, images of girls given to a god. The Amaya statue is a kore dedicated in Forest's sanctuary, an offering of the daughter's image to the god he is building.

The *Epic of Gilgamesh* supplies the oldest form of all. When Enkidu dies, Gilgamesh refuses to surrender the body for burial until, in the standard Babylonian version, a worm falls from its nose; then he sets out across the world in search of the immortality that will undo

death, and loses the plant of rejuvenation to a serpent at the moment of success. He returns to Uruk and shows the ferryman the walls he built, the only immortality granted him. Forest's campus, with its statue and its sanctum, is his Uruk, and the question the epic leaves open—whether a made thing can stand in for the life that grief demands—is the question *Devs* answers with a machine. The chapter's argument can be stated in a sentence. Necropoiesis is born of melancholia: the machine that makes worlds from the dead is the crypt of a father who would not mourn, armed with the power to act on the "it was."

CHAPTER

10

The Engineer-Mourner

A Comparative Topology

10.1 Definition of a Topos

Forest is not an isolated figure. He is the most fully realised instance of a type that the literature of technology has been assembling for two centuries and that the screen culture of 2020 multiplied with startling simultaneity: the *engineer-mourner*, the figure who converts a private loss into a technical programme. The type can be specified by four variables. The first is the vector of kinship—whom the engineer mourns: parent, child, spouse, lover, friend. The second is the instrument: the reanimation of matter, the fabrication of a replica, an intervention in time, or the computation of a world. The third is the ontological outcome: restoration of the very person, replacement by a likeness, or catastrophe. The fourth, which the tradition consistently represses and which this chapter foregrounds, is the bearer of the cost: who pays for the engineer's refusal to mourn.

The four instruments yield four species, which may be named after their mythic prototypes, together with a limit case in which the mourning drops out and only the machinery remains.

TABLE 1 · THE SPECIES OF THE ENGINEER-MOURNER

SPECIES	INSTRUMENT	PRINCIPAL EXEMPLARS	OUTCOME	WHO PAYS
Promethean	Reanimation of matter	*Frankenstein* (1818)	Monstrous return	Everyone the engineer loves
Pygmalian	Fabricated replica	*L'Ève future* (1886); *Black Mirror*, "Be Right Back" (2013); contemporary griefbots	Replacement by likeness	The mourner, held in an unfinished grief
Chronic	Intervention in time	*La Jetée* (1962); *Interstellar* (2014); *Dark* (2017–2020); *Tenet* (2020)	Restoration at the cost of the world that sought it	The worlds the machine engendered
Necropoietic	Computation of a world	*La invención de Morel* (1940); *Solaris* (1961, 1972); Tipler's Omega Point (1994); *Devs* (2020)	A world in which the dead live	The base world, its dead left dead
Governmental limit	Prediction without grief	*Westworld*, season 3 (2020)	Administration of the living	The population, deprived of its future tense

10.2 Prometheans and Pygmalions

The Promethean species begins with Mary Shelley. Victor Frankenstein turns to the secret of life after the death of his mother, and on the night the creature first opens its eye he dreams that he embraces Elizabeth, his betrothed, and that she changes in his arms into the corpse of his mother, grave-worms crawling in the folds of the shroud. The dream discloses what the laboratory concealed: the science of reanimation is the work of an unfinished mourning, and the thing it produces returns as a monster because it is the dead returned without consent. The cost is borne by everyone Victor loves. *Frankenstein*'s subtitle, *The Modern Prometheus*, also bears on *Devs*, for Prometheus is etymologically "forethought," the Titan of foresight who brought technology to humanity and was punished for it. Forest is Promethean twice over: in foresight, and in the punishment of the one who stole the fire.

The Pygmalian species fabricates a likeness instead of reviving a corpse. In Villiers de l'Isle-Adam's *L'Ève future* (1886), a fictionalised Thomas Edison constructs an artificial woman, Hadaly, for a friend in despair, and the android is lost at sea. In "Be Right Back," the 2013 episode of *Black Mirror*, a young widow subscribes to a service that reconstitutes her dead partner from his online traces, first as a chatbot and then as an android; at the episode's end the replica lives in the attic, visited at weekends by the couple's daughter. The attic is Abraham and Torok's crypt made domestic: the incorporated dead kept upstairs, neither mourned nor alive. The Pygmalian species has, since 2016, ceased to be fictional. Eugenia Kuyda trained a chatbot on the messages of her late friend Roman Mazurenko, the project from which the companion application Replika developed; James Vlahos built a "Dadbot" from recorded conversations with his dying father (2017); Ray Kurzweil's long-declared wish to reconstitute

his father from an archive of papers and recordings, documented in *Transcendent Man* (2009), was taken up by his daughter Amy Kurzweil in the graphic memoir *Artificial: A Love Story* (2023); and in 2021 the *San Francisco Chronicle* reported the conversations of a young man with a language-model simulation of his late fiancée. Carl Öhman and Luciano Floridi named the sector that commercialises such services the "digital afterlife industry" (2017). The Pygmalian engineer-mourner has a business model.

10.3 Chronics: The Time Machines of Grief

The chronic species intervenes in time itself. Chris Marker's *La Jetée* (1962) sends a man into the past through the force of a childhood image of a woman's face, only for him to discover that the death he witnessed on the jetty at Orly as a child was his own. Christopher Nolan's *Interstellar* (2014) places a father inside a tesseract—the figure that gave Garland's 1998 novel its title—behind his daughter's bookshelf, from which he watches her childhood across time and tries to reach her through gravity: the engineer-mourner as a ghost in his child's room, guilty of having left. *Tenet*, released in August 2020, pairs a fatalist refrain—what has happened has happened—with a technology that inverts the entropy of objects, and organises its puzzle around the palindromic Sator square, SATOR AREPO TENET OPERA ROTAS, a five-by-five magic square whose sense survives reversal; its antagonist is a dying man who means to take the world with him, the death drive given a trigger. Its backstory supplies a figure the Devs settlement lacks: the scientist who devised the inverting algorithm, recognising what it could do, divided it into nine pieces, hid them in the past, and took her own life so that it could not be recovered through her—an inventor who withholds the end by destroying herself, the katechon as self-sacrifice. The Sator square is cousin to the magic square in Dürer's *Melencolia I*. Both are emblems of a time that can be read in either direction.

The most exact chronic counterpart of *Devs*, however, is the German series *Dark* (2017–2020), whose finale was released on 27 June 2020, less than three months after Garland's. *Dark* reveals that its knotted worlds originate with a clockmaker, H. G. Tannhaus, whose son, daughter-in-law, and granddaughter are killed in a car accident on a stormy night, and who builds a machine to undo their deaths. The machine engenders two entangled worlds, and the series resolves only when its protagonists intervene to prevent the accident, so that the machine is never built and the worlds it produced never come to exist. The two finales of 2020 converge on the same terminal image—a family killed in a car, alive again, in a world without the machine—by opposite routes. *Dark* abolishes the machine's worlds in favour of the origin, at the cost of every person who lived in them. *Devs* abolishes the machine within the machine's own world, the empty field where the building stood, at the cost of the base world, where the dead remain dead. In *Dark* the machine erases itself so that the mourned may live in reality; in *Devs* the machine erases itself from a world it hosts so that the mourned may live in it. The first is a sacrifice. The second is a sanctuary.

10.4 Necropoietics: Cosmism, Omega, Solaris

The necropoietic species computes or conjures a world in which the dead live, and its philosophical charter is Russian. Nikolai Fedorov's *Philosophy of the Common Task*, published posthumously by his disciples in 1906 and 1913, declared that humanity's supreme moral obligation was the resurrection, by science, of all its ancestors—a filial duty of sons to raise their fathers, to be accomplished by the regulation of nature and the reassembly of scattered particles—a programme whose oldest image is Isis gathering the scattered pieces of Osiris's body and joining them again (Plutarch, De Iside et Osiride). Fedorov's influence on Konstantin Tsiolkovsky and on Russian cosmism is well documented; his project's contemporary heirs are Hans Moravec, who imagined in *Mind Children* (1988) that future intelligences might resurrect the dead by simulation, and Frank Tipler, whose Omega Point (1994) resurrects every possible human being in a computation at the end of time. *Devs* inverts Fedorov's vector—the father raises the daughter, not the son the father—and it threads a Russian subplot through its espionage: Sergei, his handler Anton, and Pete, the agent who sleeps at Lily's door, are all instruments of a state that wants the machine. The homeland of cosmism comes to Silicon Valley for the engine of resurrection.

Stanisław Lem's *Solaris* (1961) supplies the species' great novelistic form. Above an ocean that may be a single vast intelligence, the psychologist Kris Kelvin is visited by a perfect material likeness of Rheya, his wife, who had died by suicide years earlier after a quarrel in which he walked out, using a toxic substance he had brought home from his laboratory and left within her reach; the likeness is assembled by the ocean from traces in his memory. She discovers that she is a copy, tries to end her own existence and cannot, because the visitors regenerate, and at last consents, without Kelvin's knowledge, to be annihilated by an apparatus his colleagues have built. The novel turns on whether love for a copy is love, and it answers in advance the question of consent that the series leaves open: Lem's copy refuses to serve as a substitute. Andrei Tarkovsky's film (1972) adds an ending that is the closest cinematic anticipation of the Devs field. Kelvin kneels before his father at the threshold of the family house; the camera rises, and the house is revealed to stand on a small island in the ocean, a home rendered by an alien intelligence from the materials of grief. Forest's field, with its wife and child, is Kelvin's island. Nor was the necropoietic species confined to fiction in 2020. In February of that year, a month before *Devs* premiered, the South Korean broadcaster MBC aired a documentary in which a mother met, in virtual reality, a reconstruction of her young daughter who had died some years earlier. The engineer-mourner had acquired a production crew.

10.5 The Governmental Limit: Rehoboam

Ten days after the premiere of *Devs*, HBO released the third season of *Westworld*, whose central machine, Rehoboam, predicts and steers the lives of individual human beings and marks "outliers" for correction. Sophie Gilbert's essay in *The Atlantic* read the two series together as

symptoms of a single anxiety (RECEPTION), and the pairing reveals the limit of the topos. Rehoboam is the engineer-mourner's machine with the mourning removed. Its builder is driven by catastrophe, but its function is administrative: to manage the living, not to raise the dead. It is the fictional apotheosis of what Shoshana Zuboff, in *The Age of Surveillance Capitalism* (2019), called instrumentarian power—a regime that manufactures "prediction products" for sale in "behavioural futures markets" and that expropriates from its subjects what she named the right to the future tense. Both series reached into the household of Solomon for their machines: Rehoboam is the name of Solomon's son, under whom the kingdom divided (1 Kings 12), and the Devs sanctum, as Chapter 5 showed, quotes Solomon's golden oracle. The kingdom of prediction, in 2020, was imagined as Solomon's—wise, golden, and on the verge of schism.

The two seasons end in opposite political forms, and the opposition structures Part VI. *Westworld* ends in apocalypse in the etymological sense, *apokalypsis*, unveiling: the machine's predictions are disclosed to the population whose lives they governed, and its control is broken. *Devs* ends in restraint: the machine is disclosed to the sovereign alone, and its continuation is requested in secret. The first is the revolutionary resolution of the problem of prediction; the second is the katechontic one.

10.6 Hauntology and the Californian Ideology

Mark Fisher, borrowing from Derrida's *Spectres de Marx* (1993), described the cultural condition of the early twenty-first century as hauntological: a present haunted by lost futures, in which, in a phrase he took from Franco "Bifo" Berardi, the future undergoes a slow cancellation (*Ghosts of My Life*, 2014). The Devs machine is the hauntological apparatus par excellence. Its future ends in static a few hours ahead, and its past is infinitely replayable; it is a device for living among ghosts in the absence of a future. The ideology that funds it has a name and a genealogy. Richard Barbrook and Andy Cameron's "The Californian Ideology" (1995) diagnosed the fusion of the counterculture's libertarian mysticism with the free-market technological determinism of the new digital economy; Fred Turner's *From Counterculture to Cyberculture* (2006) traced its lineage through Stewart Brand, whose *Whole Earth Catalog* opened in 1968 with the declaration that, being as gods, its readers might as well become good at it. Forest is the Californian Ideology's priest-king: a countercultural founder with a monastic austerity, a godlike machine, and a company named for a child. The series is careful to give the counterculture its other face in Stewart, the elder engineer who lives in an RV and recites poetry, and whose final act Part VI interprets.

10.7 The Repressed Variable

The topology establishes that *Devs* is neither eccentric nor merely symptomatic. It is the summa of a type, in which each earlier species survives as a component: the Promethean theft

of foresight, the Pygmalian statue, the chronic intervention in time, the necropoietic world. What distinguishes it is its unusual honesty about the fourth variable. The series never lets the viewer forget who pays. Sergei pays first, suffocated by Kenton on Forest's order so that the machine's secret will keep; Anton, his handler, dies at Kenton's hands in a parking-garage struggle; Lyndon pays for its algorithm, dying under the test of faith Katie set him; Jamie pays for Lily's knowledge, tortured by Kenton to secure his silence and later shot dead by him; Kenton, the enforcer, is killed in turn by Pete; Lily pays with her life for the machine's prophecy; and the base world pays with its dead, who stay dead while the remembered two live on. The engineer-mourner's refusal to mourn is always financed by others. The question of the final part is what kind of order presides over that financing—who decides which dead shall live, and on what authority.



PART V

Socrates's Daimonion

The Veto



"... it always forbids but never commands me to do anything which I am going to do."

PLATO · APOLOGY 31D · TRANS. BENJAMIN JOWETT

The fifth demon is the only one that speaks in the negative. In Plato's *Apology* (31c–d), Socrates describes a divine sign, a voice that has come to him since childhood and that, whenever it comes, always turns him away from what he is about to do and never urges him forward; in the *Phaedrus* (242b–c) the same sign stops him at a river until he has recanted an impiety. The *daimonion* is apotreptic, never protreptic. Part V argues that Lily's act belongs to its genus.

CHAPTER

11

Free Won't

The Messianic Veto

11.1 The Act and Its Rehearsals

Lily's act has two rehearsals, both failures. In the seventh episode she resolves to prove the Devs team deluded by the simplest possible means—by not appearing in the laboratory when the machine has predicted that she will. The plan collapses when Kenton enters her apartment, kills Jamie, and tries to strangle her, and is himself killed by Pete, the homeless man at her door, who reveals that he is a Russian agent and offers her two exits: the CIA, or a flight to Hong Kong and permanent exile. She takes neither, and goes to Devs (SYNOPSIS, EPISODE 7). The first veto, absence, failed; the second, the refusal of both offered exits, succeeded only in delivering her to the place the machine had foreseen. In the final episode Forest shows her the projection—she will force him at gunpoint into the capsule and shoot him—and tells her that their actions are predetermined. She leads him into the capsule exactly as projected. As the doors close, she discards the gun (SYNOPSIS, EPISODE 8).

The structure is instructive. Lily does not defeat the prediction by doing something other than what was foreseen across the whole of its extent; she follows it almost to the letter, and departs from it by a single subtraction at its very end. Her act adds nothing to the world. It takes one thing away. This chapter argues that the subtraction is the point, and that the critical habit of calling it "free will" has misdescribed it from the start. Lily's act is not a volition but a veto.

11.2 The Laboratory of the Veto

The neuroscience of the last forty years has produced, without intending to, the experimental analogue of the Devs capsule. Benjamin Libet and his colleagues reported in 1983 that a slow negative shift in cortical potential, the "readiness potential," precedes the moment at which

subjects report the conscious intention to move by several hundred milliseconds. If the brain initiates an act before the subject decides, the decision seems to arrive after the fact. Libet himself drew a qualified conclusion (*Behavioral and Brain Sciences*, 1985): consciousness may not initiate actions, but it retains the power to veto them in the brief interval before execution—a capacity later nicknamed "free won't." Chun Siong Soon and colleagues (2008) decoded the outcome of simple binary choices from brain activity several seconds before subjects reported deciding, with accuracy modestly but significantly above chance. Aaron Schurger and colleagues (2012) reinterpreted the readiness potential as the build-up of spontaneous neural fluctuations toward a threshold rather than as a signature of decision. Marcel Brass and Patrick Haggard (2007) identified a neural signature of intentional inhibition, the act of deciding not to act. And in 2016 Matthias Schultze-Kraft and colleagues placed subjects in a duel with a brain–computer interface that detected their readiness potentials in real time and signalled them to stop; they could cancel the predicted movement up to a "point of no return" a fraction of a second before it began.

The Schultze-Kraft experiment is *Devs* in a laboratory: a machine that reads the onset of an act and announces it to the agent, and an agent who, informed, can refuse. Its result vindicates neither the libertarian nor the eliminativist. It shows that a predictor's announcement becomes a cause within the very process it predicts, and that the agent's capacity to falsify it is a capacity to stop rather than to start. Daniel Wegner's *The Illusion of Conscious Will* (2002) argued that the feeling of authorship is an inference the mind draws about its own behaviour; even on Wegner's sceptical account, the veto remains the place where the inference is put to work. What the experimental record supports, in short, is exactly what *Devs* dramatises: not free will, but free won't.

11.3 The Potentiality-Not-To

The philosophical genealogy of the veto begins with the *daimonion* and runs through Aristotle. In the ninth book of the *Metaphysics* Aristotle insists that every potentiality is an impotentiality of the same thing in the same respect (*tou autou kai kata to auto pasa dynamis adynamia*, Θ.1, 1046a31–32): the one who can play the lyre can also not play it, and it is the capacity not to that distinguishes a human potentiality from a mere tendency. Giorgio Agamben made this "potentiality-not-to" the centre of his philosophy, in the lecture "On Potentiality" (1986) and above all in "Bartleby, o della contingenza" (1993), where he reads the clerk of Melville's story of 1853—who answers every request with "I would prefer not to"—as the figure of pure potentiality, a creature who has suspended the passage from possibility to act. Gilles Deleuze, in "Bartleby, ou la formule" (1989), had already noticed that the formula refuses without refusing anything in particular, and so hollows out the language of command.

Agamben's essay contains a detail that binds Part V to the rest of this dissertation. Seeking an image of the contingency that Bartleby restores, Agamben turns to the Palace of the Fates at the end of Leibniz's *Theodicy*—the pyramid of possible worlds that Chapter 4 read as the

antecedent of the Everettian projection—and to Walter Benjamin's claim that remembrance can make the complete incomplete. Bartleby's preference, on Agamben's reading, restores to what has been its possibility of not having been. Lily's act is Bartleby's at the scale of a single gesture. She would prefer not to. Simone Weil's term for such an act, in the notebooks published as *La Pesanteur et la grâce* (1947), is *décréation*: the undoing of the created in oneself so that something other than one's determined self may act. Lily decreates the projected Lily, the one who fires.

11.4 The Swerve and the Underground

The materialist tradition offers its own version of the veto. Lucretius, following Epicurus, held that atoms falling through the void would never collide, and the world would never form, unless at uncertain times and uncertain places they swerved a little from their paths (*incerto tempore … incertisque locis*, *De Rerum Natura* II.218–219); without that swerve, the *clinamen*, there would be nothing *quod fati foedera rumpat*—to break the compacts of fate (II.254)—and no free will for living creatures. Louis Althusser's late essay on "the materialism of the encounter" (written 1982) made the swerve the ground of an aleatory ontology in which every world begins with a deviation from parallel descent, and Harold Bloom borrowed the term in *The Anxiety of Influence* (1973) for the poet's revisionary misreading of a precursor. Lily's swerve is minimal: a hand opening, a weapon falling, a deviation of inches that leaves the macroscopic outcome intact and changes everything about its meaning.

The series had already staged Lily's swerve once, and literally. In the fourth episode Kenton, having told her falsely that the psychiatrist had found her psychotic and a suicide risk, drives her toward a destination he will not disclose; she seizes the wheel and crashes the car, knocking him unconscious, and escapes (SYNOPSIS, EPISODE 4). The founder's emblem of necessity is the tram, which cannot swerve, and his trauma is a car crash he did not prevent. Lily's first act of resistance is a car crash she causes, the *clinamen* performed with a steering wheel. Her open hand in the capsule is the same gesture reduced to its minimum: not the vehicle swerving, but the hand that was to fire.

The novelistic version of the veto is Dostoevsky's. The narrator of *Notes from Underground* (1864) rails against a future in which a table of logarithms will predict every human act, and in which humanity will live in a Crystal Palace—Chernyshevsky's utopia, built of the glass and iron of Paxton's exhibition hall of 1851—where all desires have been rationally computed. Against this, he insists that a man will act against his own advantage, will even go mad on purpose, merely to prove that he is a man and not a piano key; in Constance Garnett's translation, "twice two makes four is not life, gentlemen, but is the beginning of death." The Underground Man is Michael Scriven's "avoider" a century early, and the Crystal Palace is the golden cube's ancestor. But Lily differs from him in one decisive respect. His defiance is spite, a positive act of perversity; hers is a renunciation. He proves he is not a piano key by striking a wrong note. She proves it by declining to strike.

Lily also enacts the Underground Man's threat to the letter. In the third episode, to obtain the security footage of the night of Sergei's death, she feigns a schizophrenic breakdown and climbs out onto a high window ledge in view of Forest and the visiting senator, while her friend Jen copies data from Kenton's computer (SYNOPSIS, EPISODE 3). She goes mad on purpose, as a tactic against an apparatus of observation. The apparatus answers in kind. After the crash, when Lily reports Sergei's death to the police as a murder, they arrive with the psychiatrist and with Kenton, arrest her for causing the crash, and take her for psychiatric commitment (SYNOPSIS, EPISODE 4). The structure is Cassandra's in Aeschylus' *Agamemnon*: the woman who sees truly is made unbelievable. Apollo cursed Cassandra with disbelief; Amaya manufactures it, borrowing the clinical authority that Michel Foucault's *Histoire de la folie* (1961) described as the modern power to separate the voice of unreason from the order of reason. The machine's power to predict is accompanied, at every step, by its servants' power to discredit the witness.

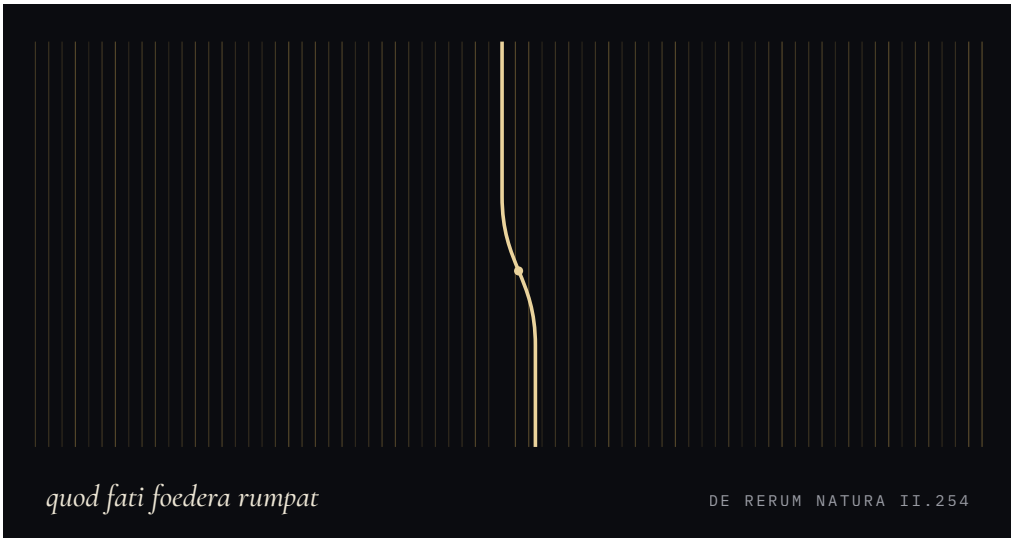


FIG. 9 **Clinamen.** *Atoms falling in parallel, and the one that swerves to break the compacts of fate.*

11.5 Samarra, or the Displacement of the Question

Chapter 2 left open whether the climax vindicates Lily's freedom or humiliates it, and promised that the series would displace the question rather than answer it. The displacement can now be specified. The capsule falls; Forest and Lily die very much as the machine foresaw; and Stewart, not Lily, brings them down, telling Katie afterward that the system had to be stopped (SYNOPSIS, EPISODE 8). The structure is that of the oldest fable of fate. In the Babylonian Talmud (Sukkah 53a), King Solomon notices the Angel of Death looking sorrowful at two of his scribes and sends them to the city of Luz, where death has no power; they die at its gates, and the next day the angel thanks Solomon for sending them to the very place where he was to

collect them. Somerset Maugham's retelling in *Sheppey* (1933), in which a servant flees Death from Baghdad to Samarra and finds that Death had an appointment with him there, made the fable modern. Solomon—whose oracle the sanctum quotes and whose son gave *Westworld* its machine—presides over this fable too.

In Samarra the event is fated and its route is free. That is precisely what Lily's veto achieves, and it is why Chrysippus's problem dissolves. The question is no longer whether the outcome was fated, but through whom it passed. Seneca's formula, *ducunt volentem fata, nolentem trahunt*, presupposes that the traveller must arrive; it concerns only whether she walks or is dragged. Lily neither walks nor is dragged into the act of killing; she unties herself from the role of executioner, and the fates find another instrument. Her veto changes nothing about what happens and everything about who did it. Stewart's intervention completes an inversion of Frankfurt's famous cases. In Frankfurt's scenario a counterfactual intervener stands ready to compel an act but never needs to, because the agent performs it on her own, and so she is responsible though she could not have done otherwise. In *Devs* the agent refuses and the intervener acts. The outcome is invariant; the authorship is transferred. The lesson for the theory of responsibility is that alternative possibilities in *outcomes* may indeed be irrelevant, as Frankfurt argued, while alternative possibilities in *authorship* are everything. Dennett's distinction between the determined and the inevitable, introduced in Chapter 3, receives here its exact application: Lily cannot avoid dying, which she does not control, but she avoids killing, which she foresaw.

Spinoza's parable from Chapter 2 returns here with a final turn. A stone set in motion by an external cause, he wrote to Schuller, would believe, if it could think, that it moved freely and by its own will (Letter 58). The discarded gun is that stone: a body falling by its weight because a hand let it go. The parable leaves open which body in the capsule is the stone—the gun, which moves only because it is released, or the woman who releases it, who may be moved by causes she cannot see. Spinoza would answer: both. The veto does not refute that answer. It shows that even a stone that knows it is a stone can decline to become a bullet.

Two further readings sharpen the act's meaning. The first is genealogical. Lily is herself a bereaved child: the fifth episode shows her at ten, playing Go with her father, and speaking to him on his deathbed (SYNOPSIS, EPISODE 5). In the capsule a daughter who has lost her father faces a father who has lost his daughter, and the machine has assigned her the role of parricide. Freud's *Totem und Tabu* (1913) located the origin of law in the sons' murder of the primal father and the guilt that followed it. Lily refuses to found anything on parricide; the killing is performed instead by a fellow elder, so that the act that ends Forest is fraternal rather than filial. The second reading is Lacanian. In the eleventh seminar, Lacan borrowed from the second book of Aristotle's *Physics* the distinction between *automaton*, the insistence of the chain of signs that returns as repetition, and *tuchē*, the encounter with the real that the chain cannot absorb. The projection is automaton; Lily's open hand is tuchē.

11.6 The Divine Move

The Go board on which Lily played with her father offers the most precise contemporary image of what she does. In March 2016, in Seoul, the program AlphaGo defeated the champion Lee Sedol in four games of five. In the fourth game, the one Lee won, his seventy-eighth move —soon called a "divine move"—was one that, according to DeepMind's own account, the program had estimated a human would play with a probability of roughly one in ten thousand. The predictor had seen the move, and had discounted it. Lee retired from professional play in 2019, remarking that the machines could not be beaten. The episode is a parable of the autoscopic horizon from the human side: a predictive system that assigns a probability to everything and a near-zero probability to the act that defeats it. Lily's discarding of the gun is a divine move of the negative kind—not a stone placed but a stone withheld.

11.7 Disclosure as Gift

One question remains, the question Chapter 6 promised to leave open. Why did Forest show Lily the projection? Within the series' logic he did so out of faith: a determinist who believes the prediction invariant has no reason to conceal it, since showing it cannot change it. But by MacKay's argument, disclosure is exactly what invalidates a prediction for its subject. Forest's determinist faith therefore handed Lily the only instrument capable of falsifying his determinism. The prophet's candour created the conditions of the prophecy's failure. Whether, at some level his theory could not reach, Forest wished to be falsified—whether the disclosure was Jonah's proclamation to Nineveh, a prophecy uttered in order to be averted—the series declines to say, and the declining is the point. A prediction disclosed is a future returned to its subject. Whatever the prophet intended, the gift is the same: it restores what Zuboff called the right to the future tense, which in her account is the capacity to project one's own will into time and to be the author of what one promises. Lily exercises that right not by authoring a new future but by withdrawing her authorship from the one assigned to her.

11.8 Definition of the Messianic Veto

The concept this chapter has assembled can now be stated. A messianic veto is an act negative in content, a refusal rather than an initiative; interruptive in form, suspending a projected series without originating a new one; moral rather than causal in effect, transferring authorship without altering outcome; dependent on disclosure, possible only for an agent who knows what is foreseen of her; and undecidable in metaphysical status, compatible both with the truth of determinism and with its falsity. It is "messianic" in Benjamin's specific sense, which the next chapter develops: not because it saves anyone—Lily dies, and Forest with her —but because it arrests the course of what was happening and, in the arrest, opens the past to a different reading. The veto does not prove that Lily was free. It proves that a being informed

of her determination can refuse to be its instrument, and that this refusal, however small, is the place where the projection ends.

CHAPTER

12

The Strait Gate

Benjamin, Gnosis, and the Onomastic Unconscious

12.1 The Angel Facing Backward

In the ninth of the theses "Über den Begriff der Geschichte," written in 1940, months before he took his own life at Portbou, on the Franco-Spanish border, when the refugees he was travelling with were told they would be sent back to France, Walter Benjamin describes the watercolour by Paul Klee that he had owned since 1921, *Angelus Novus*, painted in 1920. The angel of history, he writes, faces the past; where we perceive a chain of events, he sees a single catastrophe piling wreckage upon wreckage at his feet. He would like to stay, to awaken the dead, and to make whole what has been smashed. But a storm blows from Paradise, caught so violently in his wings that he can no longer close them, and it propels him irresistibly into the future to which his back is turned, while the pile of debris grows skyward. The storm, Benjamin concludes, is what we call progress.

No figure in modern thought describes Forest more exactly. He faces the past; his machine is an apparatus for looking at wreckage; and his one desire is to awaken the dead and make whole what has been smashed. The storm that holds his wings open and drives him backward into a future he cannot see—a future that ends, for the machine, in static—is the storm of Silicon Valley progress, the capital and engineering that fund the golden hall. Benjamin's angel is powerless to do what he wishes. Forest's tragedy is that he is not: the storm has given him the instrument the angel lacked, and the question of the series is what becomes of the wish to awaken the dead when the storm of progress itself supplies the means.

12.2 Homogeneous Empty Time and the Strait Gate

Benjamin's theses distinguish two concepts of time. The first, which he attributes to historicism and to the social-democratic faith in progress, is "homogeneous, empty time": a calcul-

able continuum along which events are arrayed like beads, in which the future is simply more of the same and can be projected by extrapolation. The second is *Jetztzeit*, "now-time," the time of the present shot through with splinters of messianic time, in which a past moment can flash up and be seized. Against the flow of homogeneous time Benjamin sets the act of arrest: thinking, he writes in the seventeenth thesis, involves not only the movement of thoughts but also their arrest, and the historical materialist recognises in such a configuration the sign of a messianic cessation of happening (*messianische Stillstellung des Geschehens*), a revolutionary chance in the fight for the oppressed past.

The Devs machine is the perfected technology of homogeneous empty time. Its projections convert the future into a continuum of calculable states, equally available at every point, equally empty of anything that has not been determined. The static that interrupts them is therefore more than an epistemic limit. In Benjamin's terms it is the visual form of the *Stillstellung*, the arrest of happening; it is the point at which the continuum breaks. And the theses' final fragment gives it a name. The Jews, Benjamin writes, were forbidden to investigate the future; the Torah and the prayers instructed them instead in remembrance, which disenchanted the future that holds sway over those who seek knowledge from soothsayers. For them the future did not become homogeneous and empty, because every second of time was the strait gate through which the Messiah might enter. The fragment opposes the soothsayer, who investigates the future, to the rememberer, for whom every instant is a gate. *Devs* stages exactly that opposition. The machine is the soothsayer; Lily's veto is the gate. Her open hand at the closing doors of the capsule is the strait gate through which something other than the projected future enters—not a messiah, for no one is saved, but the messianic itself, the arrest of what was happening.

12.3 Remembrance and Its Counterfeit

A private exchange of 1937 supplies the key to the series' ending and to the charge of counterfeit that Part VI will press. Max Horkheimer, responding to an essay of Benjamin's, insisted that the past is closed: past injustice has happened and is completed, and the slain are really slain. Benjamin recorded his reply among the notes for the *Passagen-Werk* (Convolute N, 8, 1). History, he wrote, is not only a science but also a form of remembrance, and what science has established, remembrance can modify: it can make the incomplete—happiness—complete, and the complete—suffering—incomplete. That, he added, is theology; but in remembrance we have an experience that forbids us to conceive of history as fundamentally atheological, even though we may not write it in directly theological concepts.

The Devs machine attempts by science what Benjamin reserved for remembrance. Its Everettian projection of the crash, superimposed with arrivals home, makes a completed suffering incomplete; its simulation makes the happiness of a family that never reached home complete. The operation Benjamin called theology is here performed by computation. This is the precise sense in which the series' redemption is a counterfeit: not that it is false, but that it reproduces

the form of a theological act by the means of the science that theology was meant to exceed. And Benjamin's sixth thesis supplies the warning that the final episode makes concrete: even the dead will not be safe from the enemy if he wins. The resurrected of *Devs* are not safe. They exist by the leave of a steward and a senator, rendered without consent, kept alive at the cost of electricity.

Benjamin's closest friend, Gershom Scholem, provides the hinge between this messianism and the machine. In 1965 Scholem, the great historian of Jewish mysticism, was asked to dedicate a new computer at the Weizmann Institute in Rehovot, which he had suggested be named Golem Aleph; his address, published in *Commentary* in 1966 as "The Golem of Prague and the Golem of Rehovoth," observed that the old golem and the new were both animated by the combinatorics of signs—letters in the one case, numbers in the other—and closed with an ambivalent blessing. Two and a half centuries earlier Leibniz had made the same connection from the other side. In a letter of 1697 to Duke Rudolph August of Brunswick he proposed a medal, *Imago Creationis*, celebrating his binary arithmetic as an image of creation from nothing, with the motto that serves as this dissertation's first epigraph: one suffices to derive all things from nothing. A few years later, corresponding with the Jesuit Joachim Bouvet, he recognised his binary system in the hexagrams of the *I Ching*, the Chinese book of divination. Binary arithmetic entered Western thought as at once a theology of creation and a key to an oracle. The Devs machine, binary divination performed upon quantum hardware, is its belated fulfilment. Philip K. Dick had already drawn the conclusion in *The Man in the High Castle* (1962), where the *I Ching*, consulted by characters living in a world in which the Axis won the Second World War, reveals in the hexagram of Inner Truth that their world is not the true one. The oracle's deepest disclosure is that the world in which it is consulted is a counterfeit.

12.4 Heimarmene and the Demiurge

The series' theology has an unmistakably Gnostic cast, and the Gnostic reading, though it has been proposed before in passing, acquires precision when set within the circuit of demons. In the *Apocryphon of John*, the most systematic of the Sethian texts recovered at Nag Hammadi, the divine Wisdom, Sophia, brings forth without her consort an imperfect offspring, Yaldabaoth, who, ignorant of the realm from which he came, fashions the material cosmos with his archons and boasts that he is a jealous God beside whom there is no other—a parody of Isaiah 45:5. The archons bind the souls of human beings within the cosmos by *heimarmenē*, fate, the planetary chain of necessity, and salvation comes through *gnōsis*, knowledge of one's origin beyond the world. Hans Jonas, in *The Gnostic Religion* (1958), made heimarmene the emblem of the Gnostic experience of the cosmos as prison, and in an essay of 1952 on Gnosticism and modern nihilism he traced its return in the existentialist sense of being thrown into an indifferent world.

Forest is Yaldabaoth in every structural respect. He is the maker of a world who names his instrument God; he governs his creation by fate, the tram lines; and he is ignorant of what lies beyond the boundary of his own making, which appears to him only as static. Lily is the pneumatic who acquires gnosis—knowledge of the projection, of what the archon has decreed for her—and by that knowledge escapes heimarmene. Philip K. Dick's late Gnostic cosmology, in *VALIS* (1981) and the *Exegesis*, called the deterministic world-order the "Black Iron Prison" and declared that the Empire never ended. The Devs sanctum is the Black Iron Prison gilded.

But *Devs* departs from Gnosticism at its decisive point, and the departure is its theological originality. In the Gnostic myth the pneumatic, liberated by knowledge, ascends out of the demiurge's cosmos into the Pleroma, the fullness that lies beyond it. In *Devs* the liberated soul passes not out of the demiurge's creation but into its interior: Lily escapes the projection only to awaken inside the machine. The Pleroma is a server. This is an ironic Gnosticism, in which gnosis frees the soul from fate only to deliver it into the demiurge's dream. Eric Voegelin's *The New Science of Politics* (1952) diagnosed modern political religions as Gnostic attempts to "immanentize the eschaton," to realise within history the fulfilment that Christianity had placed beyond it. *Devs* immanentizes the eschaton with an exactness Voegelin could not have imagined: the end of time is a running process in a laboratory in California. Hans Blumenberg's *Die Legitimität der Neuzeit* (1966) answered Voegelin by describing modernity as the second, and successful, overcoming of Gnosticism through human self-assertion. *Devs* suggests that the computational age reopens what Blumenberg thought closed: the question whether the world we inhabit is a prison made by a lesser god.

12.5 The Onomastic Unconscious

The introduction proposed that a series which names its machine *Deus* has authorised attention to its other names. Freud supplied the method in two places: in the doctrine of overdetermination (*Überdeterminierung*) in *Die Traumdeutung*, by which a single element of a dream is the nodal point of several independent chains of association, and in the essay "Über den Gegensinn der Urworte" (1910), on primal words that carry contrary meanings. What follows claims no authorial design for any single convergence. It claims that the density of convergences is itself a datum, and that the names of *Devs* sketch, in miniature, the theology this dissertation has reconstructed.

Amaya, the daughter and the company, is in Basque *amaia*, "the end"; a Sanskrit parsing, *a-māyā*, yields "without illusion," *māyā* being in Advaita Vedānta the cosmic illusion that veils the absolute; popular onomastic glosses add a Japanese "night rain." A company whose name means "the end" produces a world, and a name that means "without illusion" presides over the most perfect illusion ever imagined: a primal word with antithetical meanings in Freud's precise sense. The child who plays Amaya bears the name herself; she is Sonoya Mizuno's niece (CAST), so that the dead daughter's name pierces the fiction from the side of

life, a punctum in Barthes's sense. *Lily* is the flower of the Annunciation, carried by Gabriel in innumerable paintings, and of Easter, the resurrection; and in Sanskrit *līlā*, "play," names the Vedāntic doctrine that the absolute creates the world not from need but as spontaneous sport —*lokavat tu līlā-kaivalyam*, "but it is mere play, as in the world" (*Brahma Sūtra* II.1.33). Lily against Amaya is play against the end, spontaneity against the closure of fate; and the ending, in which play breaks the end open into a simulated world, yields precisely *māyā*. Her surname, Chan, is a common Chinese surname, but the homograph *Chán* names the school of meditation that became Zen, whose classic koan collection is called the *Wumenguan*, "the gateless gate" (1228). Benjamin's strait gate and the Chan gateless gate meet in a single name.

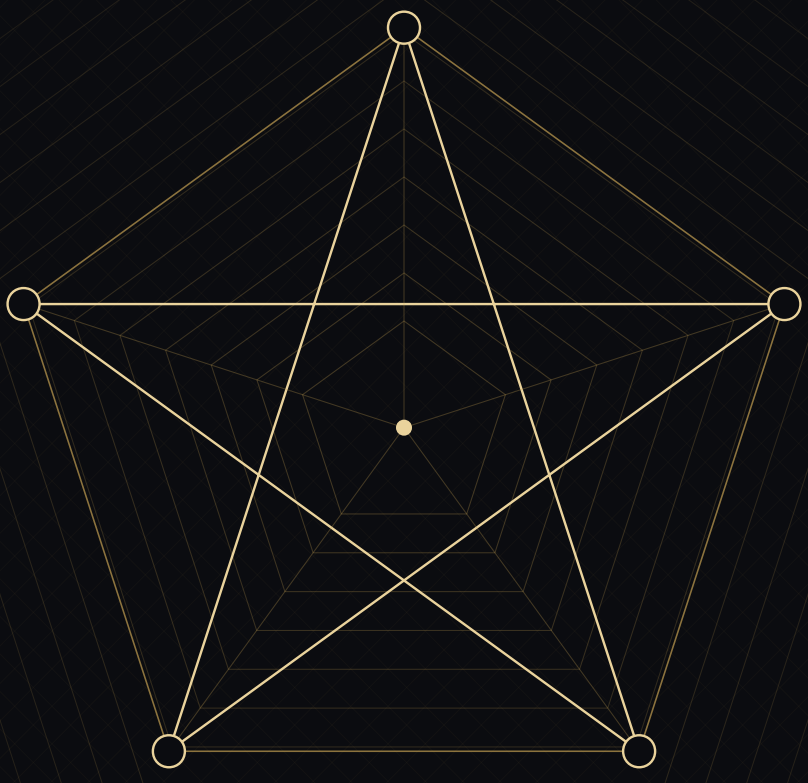
Forest contains Derrida's *fors*, the outside inside, the crypt (Chapter 9), and the forest of forking paths of Borges's story of 1941. It also names the oldest figure in modern comparative religion. James George Frazer opened *The Golden Bough* (1890) with the grove of Diana at Nemi, where a priest-king, the *Rex Nemorensis* or King of the Wood, guarded a sacred tree and held his office until slain by a successor, who must first break off a golden bough from the tree. Frazer's title came from Virgil: in the sixth book of the *Aeneid* the Sibyl of Cumae tells Aeneas that he may descend to the underworld to see his dead father only if he first plucks the golden bough, *aureus et foliis et lento vimine ramus* (VI.137). The Devs machine is the golden bough—the golden passport by which the living descend to see their dead. Forest is the King of the Wood, slain in his own grove. And Katie, who conducts the descent, is the Sibyl, whose other legend Petronius preserved: granted as many years as grains of sand but not youth, she withered until she hung in a jar, and when boys asked her what she wanted she answered that she wanted to die (*Satyricon* 48). T. S. Eliot made the answer the epigraph of *The Waste Land* (1922). It is the warning attached to every simulated eternity.

The lesser names complete the field. *Stewart* is the steward, the *oikonomos* who manages the household, and Chapter 10 noted his homonymy with Stewart Brand, the counterculture's steward of the whole earth. *Lyndon* sounds the linden, under which the village courts of the Germanic world, the *Gerichtslinden*, convened to judge: his death at the dam was a trial. *Pete*, who sleeps at Lily's door and offers her two exits, recalls the apostle given the keys of the kingdom (Matthew 16:19) and remembered for a threefold denial; he is a gatekeeper and a double agent. *Sergei*, the first to enter the sanctum and the first to die for what he saw there, plays the role of Stephen, the protomartyr, who saw the heavens opened and was stoned (Acts 7:56–58). And *Katie*, the steward of the ending, carries in her first syllable the Greek verb of the next chapter's concept: *katechein*, to hold back.

12.6 Through the Gate, into the Dream

The chapter's argument can be gathered into a single movement. Lily's veto is the strait gate, the messianic arrest of the homogeneous time of prediction. Through it, Benjamin's remembrance—the making incomplete of completed suffering—enters the world, but in counterfeit form, performed by the science it was meant to exceed. The Gnostic liberation it effects deliv-

ers the liberated into the demiurge's dream. And the names, read as an overdetermined field, trace the same arc: play (*līlā*) breaks open the end (*amaia*) and finds itself inside illusion (*māyā*). It remains to ask who keeps the dream running, and by what right.



PART VI

Pandaemonium Assembled



"Anon out of the earth a Fabrick huge
Rose like an Exhalation …"

JOHN MILTON · PARADISE LOST I.710–711

The five demons have now been examined separately: the demon of prediction and its technodicy, the demon of physical cost and its autoscopic horizon, the deceiving demon and its images of the dead, the demon of recurrence and the grief that summons it, and the forbidding *daimonion* of the veto. Part VI sets them in circuit, as the series does in its final episode, and asks what political and theological order presides over the world the circuit produces.

CHAPTER

13 The Katechontic Engine

13.1 What Withholdeth

In the second letter to the Thessalonians, Paul warns that the day of the Lord will not come until the "man of sin" is revealed, and that something now prevents his revelation: "And now ye know what withholdeth that he might be revealed in his time. For the mystery of iniquity doth already work: only he who now letteth will let, until he be taken out of the way" (2 Thessalonians 2:6–7). The Greek has a neuter and a masculine form of the same participle, *to katechon* and *ho katechōn*, "that which restrains" and "he who restrains." Patristic interpreters identified the restraining power with the Roman Empire: Tertullian explains in the *Apologeticum* (32) that Christians pray for the emperors because the continuance of the empire retards the catastrophe that threatens the world. Augustine, characteristically, confessed in the twentieth book of the *City of God* that he did not know what the Apostle meant.

Carl Schmitt made the katechon the hinge of a political theology. In *Politische Theologie* (1922) he had defined the sovereign in a sentence: *Souverän ist, wer über den Ausnahmezustand entscheidet*—sovereign is he who decides on the exception; and he had argued that all the significant concepts of the modern theory of the state are secularised theological concepts, the exception in jurisprudence being analogous to the miracle in theology. In *Der Nomos der Erde* (1950) he presented the Christian empire of the Middle Ages as a katechon, the historical power whose restraint of the end makes history itself possible. Giorgio Agamben's commentary on Romans, *Il tempo che resta* (2000), turned the concept against itself: the katechon, he argued, is the power that delays and conceals the messianic, and the messianic must ultimately deactivate it. The concept has since migrated from jurisprudence and exegesis into the self-description of the very industry *Devs* depicts. Between September and October 2025, Peter Thiel delivered in San Francisco a four-lecture series on the Antichrist that drew on Schmitt's katechon, a theme he had developed since his essay "The Straussian Moment" (2007) (Le Grand Continent; Christian Post). The lectures confirm that the vocabulary in

which this chapter reads the ending of *Devs* is no longer only a scholar's instrument. It is the vocabulary in which technological power now narrates itself.

The concept's genealogy is not innocent, and the argument requires that it be named. Schmitt joined the National Socialist party in May 1933. In "Der Führer schützt das Recht" (1934) he defended as lawful the killings without trial that Hitler ordered in the purge of June and July 1934; in 1936 he convened a conference on "Judaism in Jurisprudence" that called for the work of Jewish authors to be removed from, or segregated within, German legal scholarship; after 1945 he refused denazification and did not recant. His explicit reflections on the katechon belong to the 1940s and after, and scholars dispute how far he meant the figure to dignify the regime he had served. What is not in dispute is that the jurist of the restraining sovereign placed his jurisprudence at the service of sovereign murder. Benjamin, who had acknowledged his debt to Schmitt in a letter of 1930, took his own life at Portbou in September 1940, a refugee from the regime Schmitt had served. The theorist of the katechon served; the theorist of messianic interruption died fleeing. That asymmetry is the historical form of the opposition this chapter draws between Katie and Lily.

The concept's present career is just as specific. Thiel is a co-founder of Palantir, a company whose data-integration and analytics software is used by defence, intelligence, and law-enforcement agencies—a firm, that is, that builds analytic and predictive tools for the state. His lectures were delivered at the Commonwealth Club under a rule of confidentiality. According to recordings reviewed by *The Washington Post*, he cast critics of technology and artificial intelligence, Greta Thunberg among them, as "legionnaires of the Antichrist," and tied government oversight of Silicon Valley to an apocalyptic future (REPORT SUMMARY). In an earlier interview he had named the Antichrist's slogan as "peace and safety" (CHRISTIAN POST), the phrase of 1 Thessalonians 5:3, in the letter that precedes the katechon passage; and in the lectures, by *Le Grand Continent*'s account, he allowed that the katechon might be a modernising rather than a conservative force (LE GRAND CONTINENT). The alignment is the mirror image of the one *Devs* stages. In Thiel's account, those who would restrain technology belong to the Antichrist's party and technology belongs to the restrainer; in *Devs* the restrainer is a steward who asks the state to keep the machine alive. Both place the predictive apparatus and the sovereign in one katechontic bloc.

13.2 Katie: The Steward of the End

The final episode closes with Katie, emotional, revealing the Devs system to Senator Laine and asking her to help keep it turned on (SYNOPSIS, EPISODE 8). Read through 2 Thessalonians, the scene is a katechontic act in the strictest sense. The company is named *Amaya*, "the end"; inside the machine, Amaya the child is alive; and Katie's plea holds back the end that switching off the machine would bring—the second death of the daughter, the extinction of the two who remember. Katie withholds Amaya from Amaya. She withholds as well the other end that the word apocalypse names. *Apokalypsis* is unveiling, and Katie's settlement is an unveil-

ing to one: the machine is disclosed to the sovereign and to no one else, so that the general apocalypse—the revelation of the machine to the world—is restrained by a particular one.

The paronomasia noted in the previous chapter, Katie and *katechein*, is therefore not decorative. It names her function. Katie has been, throughout the series, the one who holds things back: the projections she watches in secret, the knowledge she withholds from Forest, the heresy she harbours from the lecture hall onward, the test she sets for Lyndon. In the end she holds back the end itself. Chapter 4 promised that her custodianship would restore the summit of Leibniz's palace in another form, and the form is now visible. The Everettian palace has no apex; every world is real. But one world is *kept*—kept powered, kept secret, kept by petition to a sovereign—and the kept world occupies, in practice, the place Leibniz reserved for the best. The summit is restored not as the world chosen by divine wisdom but as the branch a steward has persuaded a senator to fund. Katie is Pallas in the Palace of the Fates and Paulina before the statue, and the pathos the series grants her exceeds both: she is Forest's lover, and she keeps running a paradise in which he lives with the wife she replaced. The katechon, in *Devs*, is a woman administering a happiness from which she is excluded. In Hesiod, when Pandora closed her jar again, one thing remained inside it: hope (Works and Days 90–105). Katie's sealed cube keeps the same remainder.

Pathos must not obscure what the steward did. Katie knew that Kenton had killed Sergei, and confirmed it to Lily in the sixth episode (SYNOPSIS, EPISODE 6). Among the projections she surveyed in secret was Kenton torturing Jamie to secure his silence, and nothing in the series suggests that she intervened (SYNOPSIS, EPISODE 5). And she set the condition under which Lyndon died: reinstatement in exchange for standing at a lethal height above the dam (SYNOPSIS, EPISODE 7). The paradise she keeps rests on a concealed murder, a witnessed torture, and an induced death. That is the katechon's historical signature, not a deviation from it. The restraining power has always claimed to hold back a greater catastrophe by administering lesser ones, and Schmitt's career, set out above, shows that the claim can be made on behalf of murder. Katie's tenderness toward the dead she preserves and her acceptance of the living she let be sacrificed are not two characters but one function.

13.3 The Sovereign and the Bare Life of Paradise

Senator Laine first appears in the third episode, visiting Amaya because she wants the Devs technology for the United States government (SYNOPSIS, EPISODE 3). In the last she becomes the sovereign in Schmitt's sense: the one who decides on the exception. The simulation exists outside every law—secret, unregulated, without precedent—and its continuation depends on a decision that no rule governs. Its inhabitants have no legal existence. They cannot sue, vote, own, or be murdered; they can only be switched off. Agamben's *Homo Sacer* (1995) named the figure of life included in the political order solely through its exclusion "bare life," and argued that sovereignty is constituted by its power over such life. Structurally—not experientially, for the field is green and the child plays—the paradise of *Devs* has the juridical form

of what Agamben called the camp: the space that opens when the exception becomes the rule. His term names a legal structure, not an experience, and it is used here only in that structural sense. Its inhabitants are bare life in heaven: alive by decision, killable without homicide, sheltered by the same power that could end them. The resurrection is a political fact before it is a metaphysical one.

13.4 The Steward Who Brought Down the King

If Katie is the steward who preserves, Stewart is the steward who destroys. His name, like hers, names a function: the steward is the *oikonomos*, the manager of a household, and Agamben's *Il Regno e la Gloria* (2007) traced the theological genealogy of "economy" to the Christian doctrine of the divine *oikonomia*, the management of creation, distinct from and yoked to the sovereignty of the Kingdom. Western power, Agamben argued, is a bipolar machine of sovereignty and government. In *Devs* the two poles separate violently. Forest is the sovereign founder; Stewart, the elder engineer who lives in an RV, is the government—the one who keeps the machine working—and he turns on the king. In the seventh episode he challenges Forest's hubris in seeking to control the world in ignorance of history and the past; in the eighth he brings down the capsule and tells Katie that the system had to be stopped (SYNOPSES, EPISODES 7 AND 8).

His liturgy frames the act. Over Lyndon's fall he recites Larkin's "Aubade," the modern refusal of every afterlife; before the capsule falls he recites portions of W. B. Yeats's "The Second Coming," first published in 1920. The poem reads, in this setting, as an exact prophecy of the machine:

> Turning and turning in the widening gyre
> The falcon cannot hear the falconer;
> Things fall apart; the centre cannot hold;
> Mere anarchy is loosed upon the world...
> The Second Coming! Hardly are those words out
> When a vast image out of *Spiritus Mundi*
> Troubles my sight...
> ...now I know
> That twenty centuries of stony sleep
> Were vexed to nightmare by a rocking cradle,
> And what rough beast, its hour come round at last,
> Slouches towards Bethlehem to be born?

Every figure of the poem has its correlate in the series. The widening gyre is Yeats's cyclical history, systematised in *A Vision* (1925), and the recurrence of Nietzsche's demon. The falcon that cannot hear the falconer is the creature escaped from its maker's control, and the capsule is literally about to fall apart. *Spiritus Mundi* was Yeats's name for a universal storehouse of

images, a world-memory from which visions rise unbidden—which is to say, the Devs archive, the machine that holds every image of every moment. The "twenty centuries of stony sleep" are the interval since the Crucifixion, the machine's first resolved image. The "rocking cradle" is the child at the centre of everything, Amaya. And the "rough beast" slouching toward its nativity is the god the machine is about to become. Stewart recites the poem as a prophecy of the machine's birth and acts to abort it. He is the apocalyptic steward, the anti-katechon, who would hasten the end of the system before its beast is born. His act fails in precisely the way Yeats's poem predicts, for the poem foresees and does not prevent. Forest dies, but the machine runs on, and inside it the beast is born as a paradise.

13.5 The Baroque Sovereign

Chapter 5 noted that Benjamin's *Ursprung des deutschen Trauerspiels* made the melancholic the characteristic temperament of the baroque sovereign. The book's account of that sovereign, written in dialogue with Schmitt—Benjamin acknowledged the debt in a letter to Schmitt in 1930, and Agamben's *Stato di eccezione* (2003) reconstructed the exchange—describes a prince who is at once tyrant and martyr, whose sovereignty is revealed in the moment of decision and undone by his incapacity to decide, a creature in whom the majesty of the office and the misery of the flesh coincide. Forest is the Trauerspiel king. He is tyrant, having ordered Sergei's murder and watched it carried out; martyr, walking knowingly into the death the machine had shown him; melancholic, the genius of Dürer's engraving among his instruments. Many viewers have noticed that his long hair and beard recall devotional images of Christ, and the resemblance completes a theological doubling that the ending makes explicit. Forest is the Father who mourns a child and the Son who dies and rises—into a heaven of his own engineering. The founder of the machine called Deus is both persons of the drama, and the series asks whether a god who is his own redeemer is a god at all, or only a sovereign who has mistaken his exception for a miracle.

13.6 Miracle, Mystery, and Authority

In the fifth book of *The Brothers Karamazov* (1880), Ivan's Grand Inquisitor tells the returned Christ that the Church has corrected his work: men cannot bear the freedom he offered them, and so the Church has founded itself instead, in Constance Garnett's translation, upon "miracle, mystery, and authority." Katie's settlement stands on exactly that tripod. The miracle is the resurrection inside the machine. The mystery is the secrecy of the sanctum, a mystery cult with its initiates and its single unveiling. The authority is the senator, the sovereign who will keep the lights on. Katie differs from the Inquisitor in one respect, and the difference matters. She does not abolish freedom inside the paradise: Lily, in the series' last movement, rejects Sergei and seeks out Jamie, choosing within the world she has been given. But

freedom inside the paradise is guaranteed by an order outside it that rests on miracle, mystery, and authority. The Inquisitor's bargain has been relocated, not refused.

A sharper form of the Inquisitor's bargain asks whether the veto was itself foreseen at a level above the projection. In *The Matrix Reloaded* (2003) the Architect tells Neo that the anomaly he embodies—the one whose choices escape the system—is a recurring and integral feature of the system's design, absorbed anew in every cycle. In *Black Mirror*'s "Hang the DJ" (2017), two lovers' rebellion against a matching system's rules turns out to be the datum, repeated across a thousand simulations, by which the system computes their compatibility. In both, freedom is real for the agent and instrumental for the system. *Devs* raises the same question about Lily and leaves it open. The projection did not foresee her open hand. But the settlement that follows—the machine preserved, the paradise kept, the anomaly housed inside it—has the form of a system that absorbs its exception. Whether the veto broke the machine or fed it is the political question the ending hands to its sovereign.

13.7 Two Endings of 2020

The contrast with *Westworld*, drawn in Chapter 10, now acquires its full meaning. The third season of *Westworld* ended in apocalypse, the unveiling of the machine's predictions to the population it governed, and the breaking of its control. *Devs* ended in restraint: the machine unveiled to the sovereign alone, its continuation sought in secret. The two great televisual meditations on predictive power in the year of the pandemic thus divided between the two political forms that Pauline eschatology bequeathed to the West—the revelation that ends the age and the restraint that prolongs it. *Westworld* chose revolution. *Devs* chose the katechon, and in doing so it made a more uncomfortable prediction: that the power able to raise the dead will not be destroyed but kept, and that it will be kept by those who love the dead most.

13.8 Definition of the Katechontic Engine

The katechontic engine may now be defined. It is an apparatus that preserves a world by withholding its end; that exists under a sovereign exception rather than under law; that is administered by a steward whose function is restraint; that is opposed by an apocalyptic who would end it; and whose inhabitants are held as bare life, alive by decision. Its paradox is the paradox of the whole series. Lily's veto opened a messianic gap in the homogeneous time of prediction; the katechontic engine encloses that gap and keeps it running. The messianic interruption survives only as a protected exception, and the exception is the beginning of a new order. Whether that order is a redemption or its counterfeit is the question of the final chapter.

CHAPTER

14

Counterfeit Apokatastasis and the Grammar of the Eschaton

14.1 The Restoration of All Things

In the third chapter of Acts, Peter tells the crowd in Solomon's portico that heaven must receive the risen Christ "until the times of restitution of all things" (*apokatastaseōs pantōn*, Acts 3:21). Origen made the phrase the cornerstone of an eschatology. In *Peri Archōn* (*De Principiis*) he taught that the end will be like the beginning: every rational being, however far it has fallen, will be restored to God, so that God may be all in all. Gregory of Nyssa followed him; the doctrine's universalism, and in particular the suspicion that it extended even to the devil, led to its condemnation in the anathemas associated with the Second Council of Constantinople (553). Whatever its orthodoxy, the doctrine's defining property is universality. An apokatastasis that restored some and not others would not be a lesser apokatastasis; it would be a different thing altogether.

The restoration with which *Devs* ends is a different thing altogether. It is counterfeit in three precise respects. It is *partial*: only Forest and Lily remember, and only they are restored as themselves; everyone else in the simulated world is rendered afresh from branches in which nothing has happened to them. It is *substitutive*: the daughter Forest receives is, as Chapter 8 argued, a Job-child, a daughter from a branch in which the crash did not occur, given rather than returned. And it is *dependent*: the whole restored world exists by the leave of a steward and a senator and at the cost of the power that runs it. Partial, substitutive, dependent: these are the three marks of the counterfeit, and each is the precise negation of a mark of Origen's restoration, which is universal, identical, and grounded in God alone.

The partiality has an ancestry the series could hardly have planned and could not have bettered. At the end of the Myth of Er, the souls who have chosen their next lives march to the Plain of Oblivion and drink from the River of Unheeding, and all forget; but Er himself is not permitted to drink, and so he wakes on his funeral pyre remembering, and returns to tell (*Re-*

public 621a–b). Western philosophy's first afterlife-narrative ends with a witness who remembers among the forgetful. *Devs* ends in the same place. Forest and Lily are the two who did not drink from Lethe. And the structure reintroduces, inside the Everettian paradise, the doctrine that Everett's plenitude seemed to have abolished. The elect are those who remember; the rest are scenery. Counterfeit apokatastasis is a universal restoration with double predestination folded inside it—Calvin's decree reborn as a memory privilege.

14.2 The Ticket and the Lost Soul

In the chapter of *The Brothers Karamazov* called "Rebellion," Ivan refuses any future harmony that would be purchased with the tears of a single tortured child; in Constance Garnett's translation, "It's not God that I don't accept, Alyosha, only I most respectfully return Him the ticket." William James, in "The Moral Philosopher and the Moral Life" (1891), imagined a utopia whose millions are kept permanently happy on the one condition that "a certain lost soul on the far-off edge of things should lead a life of lonely torture," and asked what but an independent moral emotion could make us feel at once how hideous the bargain would be. Ursula K. Le Guin's "The Ones Who Walk Away from Omelas" (1973), subtitled "Variations on a theme by William James," made the lost soul a child in a basement and gave the story's moral weight to those who leave.

The Everettian paradise does not require suffering as its price; no child in a basement sustains it. But it coexists with every branch of suffering. In the multiverse the machine computes, the crash happens, Sergei is suffocated, Lyndon falls, and Jamie is shot, again and again, in branches without number. Forest's heaven is not bought with those tears; it is filtered from them. It is one branch, selected and kept, in a universal wave function that contains every catastrophe it was built to undo. *San Junipero* shows a husband who refused a heaven his daughter could not enter; Ivan refuses a heaven built on a child's tears; Forest accepts a heaven that is a single dry branch in an ocean of tears. Lily, choosing Jamie and declining Sergei, chooses within the filter. The counterfeit, then, is not that the happiness is false. It is that the happiness is partial, and that the partiality is administered.

14.3 Contraction, Shattering, Repair

The Kabbalah of Isaac Luria, taught in sixteenth-century Safed and transmitted by his disciple Hayyim Vital, offers the most profound interpretation of the ending's central image, and Gershom Scholem's *Major Trends in Jewish Mysticism* (1941) made it available to modern thought. Creation, in Luria's account, begins not with emanation but with withdrawal. The infinite God, *Ein Sof*, contracts into itself—*tsimtsum*—to leave a vacated space, the *tehiru*, in which a world other than God can exist. Into that space the divine light is sent in vessels, which cannot contain it and shatter—*shevirat ha-kelim*, the breaking of the vessels—scatter-

ing sparks of holiness through the lower worlds; and the task of history is *tikkun*, the repair of the world by the gathering of the sparks.

Each stage has its correlate in *Devs*. The vacuum that isolates the golden sanctum is the *tehiru*, the emptied space around the place of presence. The capsule's fall is the breaking of the vessel, the shattering by which the two sparks it contained are released—not into the lower worlds but into the machine. And the empty field where Lily wakes is *tsimtsum* in its purest form: the creator withdrawn from the world it has made so that the world may exist. Chapter 8 showed that the machine's absence from its own simulation is a logical necessity, the only escape from the regress of self-simulation. Luria shows that it is also a theological one. A creator present in its creation in the manner of the Devs building—as an oracle consulted, a sanctum entered, a projection watched—would leave no room for its creatures' freedom. The machine makes room by vanishing. The restored family in the field is *tikkun*, the world repaired. But Luria's repair is the work of every soul, gathering sparks through the commandments across the whole of history. The repair of *Devs* is the work of a machine and a steward, and it gathers two sparks only.



FIG. 10 **Contraction, Shattering, Repair.** *Luria's sequence as the series stages it: the vacated space around the sanctum, the vessel that breaks, the field from which the machine has withdrawn.*

14.4 Gimlé, the Secret Miracle, and the Room

The mythic and literary afterlives that the final episode recalls each illuminate one of its aspects. The *Völuspá* ends with the earth rising green from the sea after Ragnarök, with Baldr returning from the dead, and with a hall fairer than the sun, thatched with gold, at Gimlé, where the righteous will dwell. The field after the capsule's fall is the green earth after the end of the gods; the golden hall stands just out of frame, as the machine that sustains it. Dante's *Paradiso* ends in the vision of the love that moves the sun and the other stars, *l'amor che move il sole e l'altre stelle* (XXXIII.145), reached under the guidance of Beatrice, the dead beloved. Forest's paradise reverses the order: the beloved is not the guide to the vision but its whole content.

Borges's "El milagro secreto" (1943) supplies the temporal form. Jaromir Hladík, a Jewish writer condemned by the Nazis, asks God for a year in which to finish his verse drama; at the instant of the command to fire, the world stops, and in a year of subjective time, the soldiers motionless before him, he completes the play in memory, before the volley strikes. The capsule's fall in *Devs* is a secret miracle. The instant of death—the moment at which the machine's projection dissolves into static—is dilated, inside the machine, into an unbounded duration, as *Sunshine* dilated the instant of its hero's death into a vision and *Dredd* the instant of a fall into a slow ballet. Goethe had made such an arrest the stake of Faust's wager: should Faust ever say to the passing moment "Verweile doch! du bist so schön!"—stay, you are so beautiful—the devil may claim him (Faust I, 1699–1700). Forest's paradise is the moment told to stay. Stanley Kubrick's *2001: A Space Odyssey* (1968) supplies the spatial form: the astronaut Bowman, having passed through the Star Gate, finds himself in a neoclassical room constructed for him by unseen intelligences from fragments of human culture, where he ages, dies, and is reborn. The room is a habitat furnished by observers, and so is the field. Vonnegut's Tralfamadorians, who exhibit Billy Pilgrim in a zoo, and Bentham's panopticon (1791), the inspection-house in which the inmates may always be watched, remind us of the price of a furnished heaven: its residents live within sight of their keepers.

14.5 The Knock at the Door

W. W. Jacobs's "The Monkey's Paw" (1902) is the darkest gloss on the series' ending, and its relevance is technological as well as emotional. An elderly couple acquire a talisman that grants three wishes. Their first wish, for money, is answered when their son is killed at his workplace —"caught in the machinery"—and the company pays compensation in exactly the sum requested. The mother wishes him back. In the night there is a knock at the door, and before she can open it the father uses the last wish. Norbert Wiener, in *God & Golem, Inc.* (1964), took the tale as a parable of automated systems that grant exactly what is asked and not what is meant. *Devs* grants the grieving father's wish through machinery, and declines Jacobs's horror: what knocks at the door is whole, laughing, a child in a field. But the structure remains. What

returns is what the machine could compute, not what was lost. The daughter at the door is the branch-daughter, and the wish has been granted in the letter.

14.6 The Gates of Horn and Ivory

The final question Chapter 1 left open—whether *Devs* endorses the Teilhardian and Tiplerian eschatology of computed resurrection or anatomises it—receives its answer from Homer and Virgil. In the nineteenth book of the *Odyssey*, Penelope tells the disguised Odysseus that dreams come through two gates: those that pass through the gate of ivory deceive, and those that pass through the gate of horn come true (XIX.560–567). Her distinction rests on puns that the Greek makes and translation loses: ivory, *elephas*, sounds in *elephairomai*, to cheat; horn, *keras*, sounds in *krainō*, to fulfil. At the end of the sixth book of the *Aeneid*, Virgil reprises the gates—*Sunt geminae Somni portae*, there are twin gates of Sleep, one of horn for true shades, one of shining ivory through which the dead send false dreams to the upper world—and then, with the most famous crux in Latin literature, Anchises dismisses Aeneas and the Sibyl through the gate of ivory, *porta … eburna* (VI.893–898). Commentators have argued for two millennia over why the vision of Rome's future, the vision that justifies the epic, should leave by the gate of false dreams.

Devs ends at the same gate. The field, the child, the second chance: are they a true shade or a false dream? The series stages the computed resurrection as a real possibility of the physics it has dramatised, with full emotional conviction; and it anatomises the resurrection as counterfeit, partial, substitutive, and dependent. Like Virgil, Garland sends his heroes out by a gate whose material he does not name. And the Homeric crux confirms, at the end, the method with which this dissertation began. The oldest statement of the undecidability of visions in Western literature rests on paronomasia: horn and fulfilment, ivory and deceit, are linked by sound before they are linked by sense. To read *Devs* through its puns is to read it as Homer taught the West to read dreams.



FIG. 11 **The Gates of Horn and Ivory.** *The puns behind Homer's gates: horn and fulfilment, ivory and deceit. Virgil sends Aeneas home through the ivory gate.*

14.7 Chronogrammar

The series' theology can finally be stated as a grammar. *Devs* conjugates the verb of redemption through four moods, and each mood is a station of its eschatology.

The first is the *indicative*, Forest's mood: it was, it is, it will be, laid out along the tram lines. It is the mood of Laplace's demon, for whom the future and the past alike *serait présent*, and of the law, which states what is and must be. The second is the *future perfect*, the mood of the projection: she will have forced him into the capsule; she will have fired. Chapter 7 identified it with Barthes's anterior future, and Lacan, in his Rome discourse of 1953, located in the same tense the being of the subject, who realises himself in history as what he will have been for what he is in the process of becoming. The future perfect is the grammatical form of predestination. In it, the future is already past, and the decree has already been executed.

The third mood is Lily's, and it is the hardest to name. Bartleby's formula, "I would prefer not to," is a negative *conditional*—not an imperative, not an indicative, but a preference suspended in the conditional and negated. The *daimonion* speaks in the same mood: it never says "do," only "not this." The veto is the negative conditional made into an act. The fourth mood is the *subjunctive*, the mood of the simulation: a life as if restored, a world as if the call had never been made, the mood of Molina's counterfactuals of freedom—what she would do—and of Hans Vaihinger's *Die Philosophie des Als Ob* (1911), which treated useful fictions as the scaffolding of thought. Paul names the subjunctive's messianic form: "they that weep, as though they wept not; and they that rejoice, as though they rejoiced not … for the fashion of this

world passeth away" (1 Corinthians 7:30–31). Agamben read this *hōs mē*, the "as not," as the messianic revocation of every worldly condition from within. Forest and Lily, knowing what they are, live in the *hōs mē*: as though not dead, as though not simulated.

Augustine had already supplied the machinery for such a grammar. In the eleventh book of the *Confessions* he argues that there are not three times but three presents: *praesens de praeteritis memoria, praesens de praesentibus contuitus, praesens de futuris expectatio*—a present of things past, which is memory; a present of things present, which is attention; and a present of things future, which is expectation (XI.20.26). Time, for Augustine, is a *distentio animi*, a distension of the soul. The Devs machine externalises the three presents: its projection of the past is memory, its live view is attention, its projection of the future is expectation. *Distentio animi* becomes *distentio machinae*. And Kierkegaard's *Repetition* completes the scheme by distinguishing recollection, which repeats backward what has been, from repetition proper, which is recollected forward. The machine's retrodiction is recollection, its prediction is repetition forward, and its simulation—the Job-restoration of a life given again—is the repetition Kierkegaard sought and doubted.

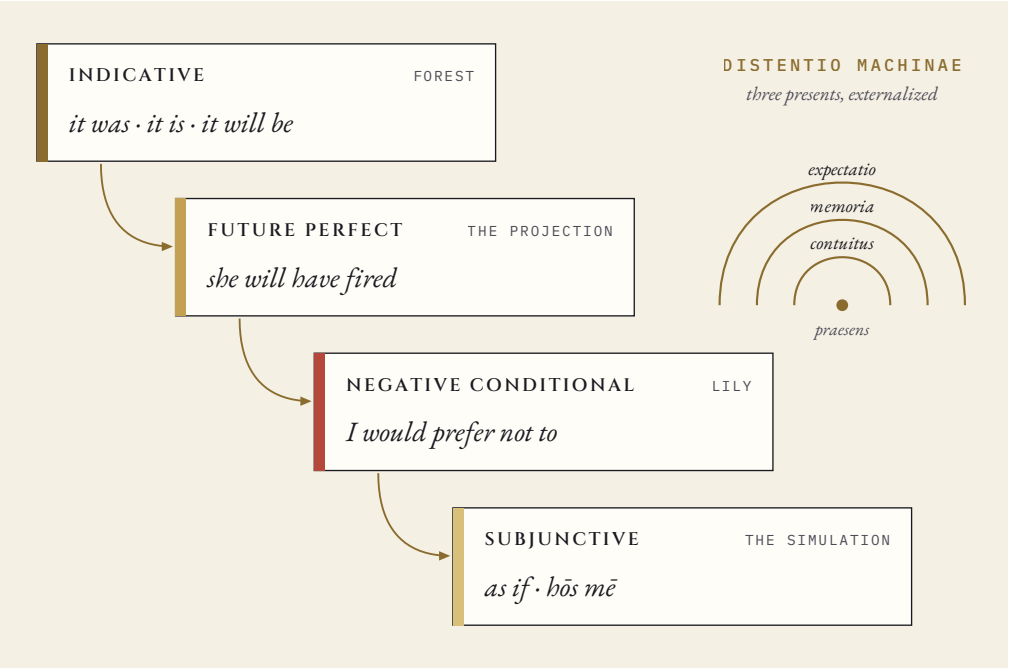


FIG. 12 **Chronogrammar.** *The series conjugates redemption through four moods. Augustine's three presents, externalized, become the machine's three views of time.*

14.8 The Self-Excited Circuit

John Archibald Wheeler, who supervised Everett's dissertation, spent his last decades on a picture of the universe as what he called a self-excited circuit. He drew it as a large letter U: one

arm is the universe's origin, the other grows through history until it ends in an eye that looks back across the gap at the first arm. Observers, by observing, give reality to the universe that produced them; his delayed-choice experiment (1978) showed that a measurement made now determines what can be said of a photon's path in the past. *Devs* closes on Wheeler's eye. A machine built by observers to look back at the past gives existence to a world whose two remembering observers look back at the world that built it; and the viewer, watching the series, is the outermost eye, looking at a machine looking at the dead. The circuit of demons that this dissertation has traced is a self-excited circuit in Wheeler's sense: grief summons prediction, prediction meets its own reflection, the reflection is refused, and the refusal is enclosed in a world that reopens the circuit. The question whether that world is redemption or counterfeit cannot be answered from inside the circuit. It can only be asked at the gate.

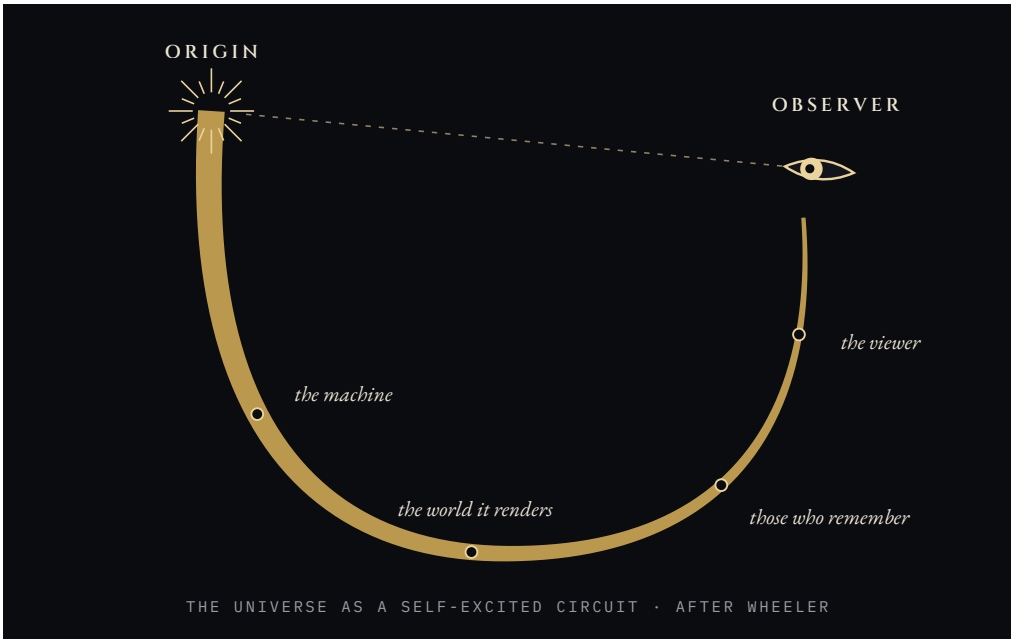


FIG. 13 **The Self-Excited Circuit.** *Wheeler's universe looks back at its own origin. The series closes the same loop: a machine renders a world whose rememberers look back at the machine, and the viewer watches them.*

CONCLUSION



Melencolia Q

C.1 The Circuit Restated

This dissertation has argued that Alex Garland's *Devs* is not, at bottom, a drama of free will and determinism, and that the antinomy it so insistently stages is a screen—a projected surface concealing the apparatus that throws it. Behind the screen the series builds a Pandaemonium, a golden hall in which five demons of Western thought are convened and set in circuit. The demon of Nietzsche, grief's will against the "it was," summons the demon of Laplace, the intelligence for which past and future are alike present. Laplace's demon, built in the world, must take on the body of Maxwell's, paying for its knowledge in heat and isolation and meeting at last its own reflection at the autoscopic horizon. There the Socratic *daimonion* intervenes, the voice that only forbids, and Lily's veto arrests the projection. The aftermath belongs to the demon of Descartes, the deceiver who can counterfeit a world, and who here counterfeits a heaven. The circuit's product is *necropoiesis*, the making of worlds out of the dead.

Five theses have been defended. Forest's determinism is a technodicy, the justification of a maker in the face of a loss his machine was built to annul (Chapter 3). The schism between the Bohmian and Everettian algorithms restages the conflict of predestination and middle knowledge, and the Everettian projection is Leibniz's Palace of the Fates with its summit removed (Chapter 4). The static is an autoscopic horizon, a reflexive rather than temporal limit that begins wherever a prediction is disclosed to its subject and ends wherever the predictor must contain itself (Chapters 5 and 6). Lily's act is a messianic veto, free won't rather than free will, which changes not what happens but through whom it happens (Chapters 11 and 12). And the ending is a counterfeit apokatastasis administered by a katechontic engine: a restoration that is partial, substitutive, and dependent, preserved by a steward's petition to a sovereign, its inhabitants alive by decision (Chapters 13 and 14). Between these theses, the history of the image of the dead (Chapters 7 and 8) and the psychoanalysis of a father who would not mourn (Chapters 9 and 10) supply the connective tissue: the projection is a Roman *imago*

that flees the embrace, the simulation a Pygmalion's statue that warms, the machine a crypt, and its maker a Trauerspiel king.

c.2 Melencolia Q

The Renaissance read Dürer's *Melencolia I* through Cornelius Agrippa's *De occulta philosophia*, which distinguished three grades of melancholic inspiration: the imaginative, proper to artists and craftsmen; the rational, proper to natural philosophers and physicians; and the mental, proper to theologians and prophets. Panofsky and his collaborators argued that Dürer's numeral designates the first. The present dissertation proposes a fourth grade that Agrippa could not foresee, and names it *Melencolia Q*: the melancholy of computation. It belongs to a knower whose instruments are perfect, who can see every moment of the past and predict every act of the future, and who discovers that perfect knowledge returns nothing—that the dead, resolved in colour, still flee the hands—and who therefore, being unable to mourn, begins to manufacture. *Melencolia Q* is the affect of the engineer-mourner. It is grief that has learned to compute.

The letter is chosen for its senses. It stands for the quantum and the qubit, and for the scholastic *quaestio*, the disputed question that may end, like the controversy *De Auxiliis*, in a suspended judgment issued by a sovereign. But it is also an emblem in its own right. Typographically, Q is a circle—the closed circle of recurrence, the O of Omega, the zero from which Leibniz's binary One derives all things—crossed by a single short stroke that breaks its closure. The circle is the closed future; the stroke is the swerve. The letter holds, in one glyph, the whole argument of the series: a perfect enclosure, and the small deviation that lets something out.

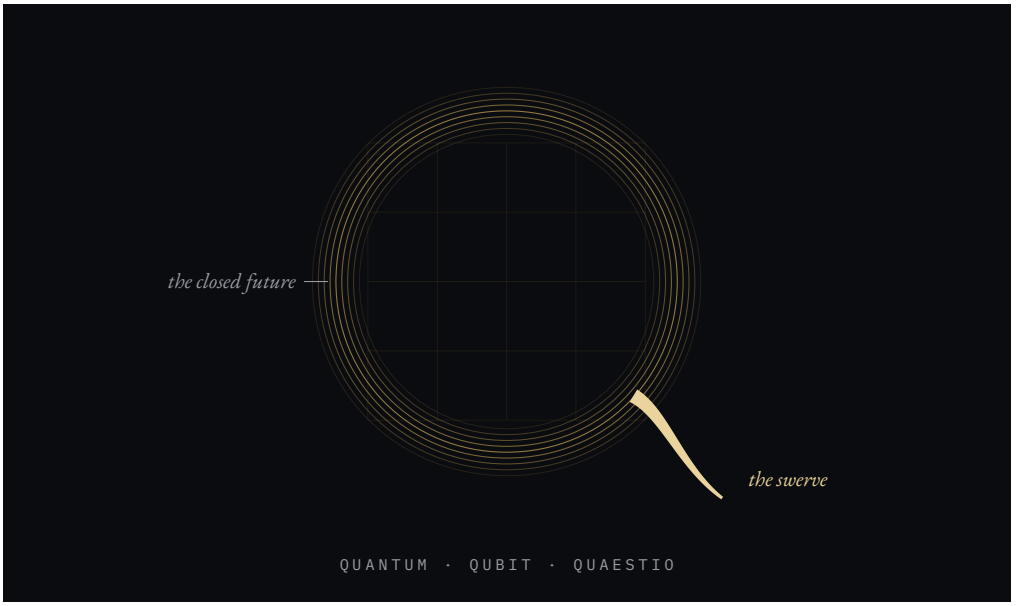


FIG. 14 **Melencolia Q.** *A closed circle, and the single stroke that lets something out.*

C.3 Computation as the Continuation of Mourning by Other Means

The general theory these findings compose was stated in the introduction as an inversion of Clausewitz: in the regime *Devs* depicts, computation is the continuation of mourning by other means. Its significance exceeds the series. The data infrastructures that predict the living are, by their very nature, capable of rendering the dead, since the traces from which a prediction is computed are the same traces from which a likeness can be assembled. Zuboff's prediction products and the digital afterlife industry that Öhman and Floridi described are two faces of a single apparatus. *Devs* imagines the limit at which they coincide, and five consequences follow from taking its imagination seriously.

For the philosophy of technology, the analysis of predictive systems must extend from their effects on autonomy to their effects on mourning. Zuboff named the right to the future tense; *Devs* suggests a complementary claim that this dissertation calls *the right to the past tense*: the right of the dead to have died, to be mourned rather than rendered, and the right of the living to a grief that is not perpetually reopened by perfect retrieval. For theology, the reoccupation of eschatological positions by computation, in Blumenberg's sense, makes theology's own distinctions indispensable to critique: universal against partial restoration, remembrance against soothsaying, the messianic against the katechontic. For media theory, the realisation of Bazin's total cinema turns out to abolish the screen; the end of cinema, in *Devs*, is the beginning of necropoiesis, when the image becomes a world and the spectator a resident. For political theory, sovereignty acquires a new object, simulated life held by decision, and its old vocabulary —exception, katechon, bare life—has already migrated into the self-description of the in-

dustry that would build such worlds. And for literary history, *Devs* fixes the engineer-mourner as a contemporary topos and anchors it in a constellation that runs from 1920—*Beyond the Pleasure Principle*, the *Angelus Novus*, "The Second Coming," *R.U.R.*—to 2020, the year of *Devs*, *Dark*, the third season of *Westworld*, and *Tenet*, and of a pandemic that made prediction a daily liturgy.

C.4 Limitations

Three limitations should be acknowledged. The first is evidentiary. Plot particulars have been checked against published episode synopses rather than against production archives, scripts, or an exhaustive shot-by-shot record, and readings that depend on visual detail are offered with corresponding caution. The second is methodological. The paronomastic method, though licensed by the series' own central pun and by a long tradition from Homer's gates to Derrida's *fors*, yields convergences whose interpretive force is cumulative rather than individually demonstrable; any single homophony could be coincidence, and the argument rests on their density. The third concerns scope. The comparative field has been chosen for structural illumination rather than for genealogical proof, and many of the works placed beside *Devs* could not have influenced it. The dissertation has preferred the constellation to the source, and accepts the risks of that preference.

C.5 Further Research

Four lines of inquiry suggest themselves. A comparative study of the prediction narratives of 2020 could test the claim that the pandemic year produced a distinctive imaginary of forecasting, mourning, and sovereign exception. Empirical and ethnographic work on grief technologies could examine whether the counterfeit marks identified here—partiality, substitution, dependence—describe the experience of their users. Philosophical work on Everettian ethics could take up the moral question the series poses most sharply: what responsibilities, if any, attach to the selection and preservation of branches. And the history of political theology could trace in detail the passage of the katechon from Pauline exegesis and Schmittian jurisprudence into the discourse of technological capital. Within Garland's own oeuvre, *Warfare* (2025), with its reconstruction of a past engagement from the pooled memories of the living, invites a reading as the sober sequel to *Devs*: retrodiction by testimony, without the machine and without the resurrection.

C.6 Coda: The Statue and the Gate

At the end of *The Winter's Tale*, before a statue that is not a statue, Paulina addresses a guilty king who caused the deaths of those he loved: "It is required / You do awake your faith." The viewer at the end of *Devs* stands where Leontes stands. The field is green; the child plays; the

father who could not mourn has been given back what he lost, or something that will do in its place. The series requires a faith it does not compel, and it sends its heroes out, like Virgil's, by a gate whose material it declines to name. Milton's fallen intelligences, debating fate and foreknowledge in the hall of gold, found no end, in wandering mazes lost. *Devs* does not end their debate. It shows that the debate was always a screen for grief, and that behind the screen the question was never whether we are free but what we would do, and what we would build, to have the dead again.

The demons debate in the golden hall. The daughter plays in the empty field. And between them, in the static that no machine can resolve, a woman opens her hand.

APPARATUS

Select Bibliography

Dates are of first publication or first broadcast; classical, scriptural, and medieval texts are cited by conventional divisions. Web sources were accessed on 26 October 2026.

PRIMARY TEXT, PARATEXTS, AND RECEPTION

Garland, Alex, creator, writer, and director. *Devs*. Eight episodes. FX on Hulu, 5 March–16 April 2020.

"Devs (TV series)." *Wikipedia*. https://en.wikipedia.org/wiki/Devs_(TV_series) — episode synopses, cast, production, and reception.

"Devs." German-language encyclopedia entry. https://dewiki.de/Lexikon/Devs — architecture of the Devs building; German broadcast titles.

Garland, Alex. *The Beach* (1996); *The Tesseract* (1998); *The Coma* (2004).

Garland, Alex, writer. *28 Days Later* (2002) and *Sunshine* (2007), dir. Danny Boyle; *Never Let Me Go* (2010), dir. Mark Romanek; *Dredd* (2012), dir. Pete Travis.

Garland, Alex, writer and director. *Ex Machina* (2014); *Annihilation* (2018); *Men* (2022); *Civil War* (2024).

Garland, Alex, and Ray Mendoza, directors. *Warfare* (2025).

Gilbert, Sophie. "Why TV Is So Worried About Free Will." *The Atlantic*, 11 March 2020.

Poniewozik, James. "Review: 'Devs' Is a Cold and Beautiful Machine." *The New York Times*, 4 March 2020.

Ugwu, Reggie. "'Westworld' and 'Devs' Asked Big Questions. A Physicist Responds." *The New York Times*, 4 May 2020.

Medawar, P. B. "Critical Notice: *The Phenomenon of Man*." *Mind* 70 (1961): 99–106.

"Peter Thiel: pensamientos del Anticristo." *Le Grand Continent*, 24 October 2025. https://legrandcontinent.eu/es/2025/10/24/peter-thiel-pensamientos-del-anticristo/

"Palantir Co-founder Peter Thiel to Lead 4-Part Series on the Antichrist." *The Christian Post*, 2025. https://assets.christianpost.com/news/palantir-ceo-peter-thiel-to-lead-4-part-series-on-the-antichrist.html

PHILOSOPHY AND THEOLOGY

Adams, Robert Merrihew. "Middle Knowledge and the Problem of Evil." *American Philosophical Quarterly* 14 (1977): 109–117.

Agamben, Giorgio. "Bartleby, o della contingenza." In Gilles Deleuze and Giorgio Agamben, *Bartleby, la formula della creazione* (1993).

Agamben, Giorgio. *Homo Sacer: Il potere sovrano e la nuda vita* (1995); *Il tempo che resta: Un commento alla Lettera ai Romani* (2000); *Stato di eccezione* (2003); *Il Regno e la Gloria* (2007).

Agrippa von Nettesheim, Heinrich Cornelius. *De occulta philosophia libri tres* (1533).

Albert, David Z. *Time and Chance* (2000).

Aquinas, Thomas. *Summa Theologiae* (1265–1274).

Aristotle. *De Interpretatione*; *Physics*; *Metaphysics*.

Augustine. *Confessiones*; *De civitate Dei*.

Blumenberg, Hans. *Die Legitimität der Neuzeit* (1966).

Boethius. *De consolatione philosophiae* (c. 524).

Bostrom, Nick. "Are You Living in a Computer Simulation?" *Philosophical Quarterly* 53 (2003): 243–255.

Calvin, John. *Institutio Christianae religionis* (1559).

Chalmers, David J. "The Matrix as Metaphysics" (2005); *Reality+: Virtual Worlds and the Problems of Philosophy* (2022).

Cicero. *De Fato*.

Dennett, Daniel C. *Elbow Room: The Varieties of Free Will Worth Wanting* (1984); *Freedom Evolves* (2003).

Descartes, René. *Meditationes de prima philosophia* (1641).

Durkheim, Émile. *Les Formes élémentaires de la vie religieuse* (1912).

Fedorov, Nikolai. *Filosofiya obshchego dela* [*Philosophy of the Common Task*] (1906–1913).

Frankfurt, Harry G. "Alternate Possibilities and Moral Responsibility." *Journal of Philosophy* 66 (1969): 829–839.

Hasker, William. *God, Time, and Knowledge* (1989).

James, William. "The Dilemma of Determinism" (1884); "The Moral Philosopher and the Moral Life" (1891).

Jonas, Hans. *The Gnostic Religion* (1958).

Kant, Immanuel. *Kritik der reinen Vernunft* (1781; 2nd ed. 1787).

Kierkegaard, Søren. *Gjentagelsen* (1843); *Frygt og Bæven* (1843); *Begrebet Angest* (1844).

Klibansky, Raymond, Erwin Panofsky, and Fritz Saxl. *Saturn and Melancholy* (1964).

Laplace, Pierre-Simon. *Essai philosophique sur les probabilités* (1814).

Leibniz, G. W. "Explication de l'arithmétique binaire" (1703); *Essais de Théodicée* (1710); *Monadologie* (1714).

Lewis, David. *On the Plurality of Worlds* (1986).

Locke, John. *An Essay concerning Human Understanding*, 2nd ed. (1694), II.xxvii.

Lovejoy, Arthur O. *The Great Chain of Being* (1936).

Löwith, Karl. *Meaning in History* (1949).

MacKay, D. M. "On the Logical Indeterminacy of a Free Choice." *Mind* 69 (1960): 31–40.

Molina, Luis de. *Concordia liberi arbitrii cum gratiae donis* (1588).

Nietzsche, Friedrich. *Vom Nutzen und Nachtheil der Historie für das Leben* (1874); *Die fröhliche Wissenschaft* (1882); *Also sprach Zarathustra* (1883–1885); *Zur Genealogie der Moral* (1887); *Ecce Homo* (written 1888).

Nozick, Robert. *Anarchy, State, and Utopia* (1974).

Origen. *Peri Archōn* [*De Principiis*].

Parfit, Derek. *Reasons and Persons* (1984).

Pereboom, Derk. *Living Without Free Will* (2001).

Plato. *Apology*; *Phaedrus*; *Republic*.

Popper, Karl. "Indeterminism in Quantum Physics and in Classical Physics." *British Journal for the Philosophy of Science* 1 (1950): 117–133, 173–195; *The Poverty of Historicism* (1957).

Putnam, Hilary. "Time and Physical Geometry." *Journal of Philosophy* 64 (1967): 240–247; *Reason, Truth and History* (1981).

Schmitt, Carl. *Politische Theologie* (1922); *Der Nomos der Erde* (1950).

Scholem, Gershom. *Major Trends in Jewish Mysticism* (1941); "The Golem of Prague and the Golem of Rehovoth." *Commentary* 41, no. 1 (1966).

Scriven, Michael. "An Essential Unpredictability in Human Behavior." In *Scientific Psychology*, ed. B. B. Wolman and E. Nagel (1965).

Spinoza, Baruch. *Ethica* (1677); Letter 58.

Strawson, Galen. "The Impossibility of Moral Responsibility." *Philosophical Studies* 75 (1994): 5–24.

Strawson, P. F. "Freedom and Resentment." *Proceedings of the British Academy* 48 (1962): 1–25.

Tertullian. *Apologeticum*.

van Inwagen, Peter. *An Essay on Free Will* (1983).

Vaihinger, Hans. *Die Philosophie des Als Ob* (1911).

Voegelin, Eric. *The New Science of Politics* (1952).

Wallace, David. *The Emergent Multiverse* (2012).

Weil, Simone. *La Pesanteur et la grâce* (1947).

Williams, Bernard. "The Makropulos Case: Reflections on the Tedium of Immortality" (1973).

Wimsatt, W. K., and Monroe C. Beardsley. "The Intentional Fallacy." *Sewanee Review* 54 (1946): 468–488.

PHYSICS, COMPUTATION, AND NEUROSCIENCE

Bell, J. S. "On the Einstein Podolsky Rosen Paradox." *Physics* 1 (1964): 195–200.

Bennett, Charles H. "The Thermodynamics of Computation—A Review." *International Journal of Theoretical Physics* 21 (1982): 905–940.

Bohm, David. "A Suggested Interpretation of the Quantum Theory in Terms of 'Hidden' Variables, I and II." *Physical Review* 85 (1952): 166–179, 180–193.

Brass, Marcel, and Patrick Haggard. "To Do or Not to Do: The Neural Signature of Self-Control." *Journal of Neuroscience* 27 (2007): 9141–9145.

Byrne, Peter. *The Many Worlds of Hugh Everett III* (2010).

Deutsch, David. "Quantum Theory, the Church–Turing Principle and the Universal Quantum Computer." *Proceedings of the Royal Society of London A* 400 (1985): 97–117; *The Fabric of Reality* (1997).

DeWitt, Bryce S. "Quantum Mechanics and Reality." *Physics Today* 23, no. 9 (1970): 30–35.

Everett, Hugh, III. "'Relative State' Formulation of Quantum Mechanics." *Reviews of Modern Physics* 29 (1957): 454–462.

Frauchiger, Daniela, and Renato Renner. "Quantum Theory Cannot Consistently Describe the Use of Itself." *Nature Communications* 9 (2018): 3711.

Gödel, Kurt. "Über formal unentscheidbare Sätze der *Principia Mathematica* und verwandter Systeme I." *Monatshefte für Mathematik und Physik* 38 (1931): 173–198.

Kohs, Greg, director. *AlphaGo* (2017).

Landauer, Rolf. "Irreversibility and Heat Generation in the Computing Process." *IBM Journal of Research and Development* 5 (1961): 183–191.

Libet, Benjamin, Curtis A. Gleason, Elwood W. Wright, and Dennis K. Pearl. "Time of Conscious Intention to Act in Relation to Onset of Cerebral Activity (Readiness-Potential)." *Brain* 106 (1983): 623–642.

Libet, Benjamin. "Unconscious Cerebral Initiative and the Role of Conscious Will in Voluntary Action." *Behavioral and Brain Sciences* 8 (1985): 529–566.

Lloyd, Seth. "Computational Capacity of the Universe." *Physical Review Letters* 88 (2002): 237901.

Lorenz, Edward N. "Deterministic Nonperiodic Flow." *Journal of the Atmospheric Sciences* 20 (1963): 130–141.

Lucas, J. R. "Minds, Machines and Gödel." *Philosophy* 36 (1961): 112–127.

Maxwell, James Clerk. *Theory of Heat* (1871).

Minkowski, Hermann. "Raum und Zeit" (lecture 1908; published 1909).

Moravec, Hans. *Mind Children: The Future of Robot and Human Intelligence* (1988).

Penrose, Roger. *The Emperor's New Mind* (1989).

Schultze-Kraft, Matthias, et al. "The Point of No Return in Vetoing Self-Initiated Movements." *Proceedings of the National Academy of Sciences* 113 (2016): 1080–1085.

Schurger, Aaron, Jacobo D. Sitt, and Stanislas Dehaene. "An Accumulator Model for Spontaneous Neural Activity Prior to Self-Initiated Movement." *Proceedings of the National Academy of Sciences* 109 (2012): E2904–E2913.

Soon, Chun Siong, Marcel Brass, Hans-Jochen Heinze, and John-Dylan Haynes. "Unconscious Determinants of Free Decisions in the Human Brain." *Nature Neuroscience* 11 (2008): 543–545.

Szilard, Leo. "Über die Entropieverminderung in einem thermodynamischen System bei Eingriffen intelligenter Wesen." *Zeitschrift für Physik* 53 (1929): 840–856.

Tegmark, Max. "The Interpretation of Quantum Mechanics: Many Worlds or Many Words?" *Fortschritte der Physik* 46 (1998): 855–862.

Tipler, Frank J. *The Physics of Immortality* (1994).

Turing, A. M. "On Computable Numbers, with an Application to the *Entscheidungsproblem*." *Proceedings of the London Mathematical Society* s2-42 (1937): 230–265.

Wegner, Daniel M. *The Illusion of Conscious Will* (2002).

Wheeler, John Archibald. "The 'Past' and the 'Delayed-Choice' Double-Slit Experiment." In *Mathematical Foundations of Quantum Theory*, ed. A. R. Marlow (1978).

Wiener, Norbert. *Cybernetics* (1948); *God & Golem, Inc.* (1964).

Wigner, Eugene P. "Remarks on the Mind–Body Question." In *The Scientist Speculates*, ed. I. J. Good (1961).

Wolfram, Stephen. *A New Kind of Science* (2002).

Wolpert, David H. "Physical Limits of Inference." *Physica D* 237 (2008): 1257–1281.

Zeh, H. D. "On the Interpretation of Measurement in Quantum Theory." *Foundations of Physics* 1 (1970): 69–76.

Zurek, Wojciech H. "Quantum Darwinism." *Nature Physics* 5 (2009): 181–188.

PSYCHOANALYSIS, MEDIA, AND CRITICAL THEORY

Abraham, Nicolas, and Maria Torok. *Cryptonymie: Le verbier de l'Homme aux loups*. Preceded by Jacques Derrida, "Fors" (1976).

Althusser, Louis. "Le courant souterrain du matérialisme de la rencontre" (written 1982; published 1994).

Barbrook, Richard, and Andy Cameron. "The Californian Ideology" (1995); *Science as Culture* 6, no. 1 (1996): 44–72.

Barthes, Roland. *La Chambre claire* (1980).

Baudrillard, Jean. *Simulacres et simulation* (1981).

Bazin, André. "Ontologie de l'image photographique" (1945); "Le mythe du cinéma total" (1946).

Belting, Hans. *Bild-Anthropologie* (2001).

Benjamin, Walter. *Ursprung des deutschen Trauerspiels* (1928); "Das Kunstwerk im Zeitalter seiner technischen Reproduzierbarkeit" (1935–1936); "Über den Begriff der Geschichte" (1940); *Das Passagen-Werk*, ed. Rolf Tiedemann (1982).

Berardi, Franco "Bifo." *After the Future* (2011).

Blanchot, Maurice. "Le regard d'Orphée." In *L'Espace littéraire* (1955).

Bloom, Harold. *The Anxiety of Influence* (1973).

Deleuze, Gilles. "Bartleby, ou la formule" (1989); collected in *Critique et clinique* (1993).

Derrida, Jacques. *Spectres de Marx* (1993).

Fagone, Jason. "The Jessica Simulation: Love and Loss in the Age of A.I." *San Francisco Chronicle*, 2021.

Fisher, Mark. *Ghosts of My Life* (2014).

Freud, Sigmund. *Die Traumdeutung* (1900); "Über den Gegensinn der Urworte" (1910); *Totem und Tabu* (1913); "Trauer und Melancholie" (1917); *Jenseits des Lustprinzips* (1920).

Klein, Melanie. "A Contribution to the Psychogenesis of Manic-Depressive States" (1935); "Mourning and Its Relation to Manic-Depressive States" (1940).

Lacan, Jacques. "Fonction et champ de la parole et du langage en psychanalyse" (1953); *Le Séminaire, Livre XI: Les quatre concepts fondamentaux de la psychanalyse* (seminar 1964; published 1973).

Merton, Robert K. "The Self-Fulfilling Prophecy." *Antioch Review* 8 (1948): 193–210.

Newton, Casey. "Speak, Memory." *The Verge*, 2016.

Öhman, Carl, and Luciano Floridi. "The Political Economy of Death in the Age of Information: A Critical Approach to the Digital Afterlife Industry." *Minds and Machines* 27 (2017): 639–662.

Turner, Fred. *From Counterculture to Cyberculture* (2006).

Vlahos, James. "A Son's Race to Give His Dying Father Artificial Immortality." *Wired*, 2017.

Zuboff, Shoshana. *The Age of Surveillance Capitalism* (2019).

LITERATURE, MYTH, ART, AND FILM

The Epic of Gilgamesh (Standard Babylonian version); Homer, *Iliad* and *Odyssey*; *Homeric Hymn to Demeter*; Hesiod, *Theogony*; Euripides, *Alcestis* (438 BCE); Virgil, *Aeneid*; Lucretius, *De rerum natura*; Ovid, *Metamorphoses*; Seneca, *Epistulae morales*; Petronius, *Satyricon*; Polybius, *Histories*; Pliny the Elder, *Naturalis historia*.

The Holy Bible, Authorized (King James) Version (1611); Babylonian Talmud, Sukkah 53a; *The Apocryphon of John* (Nag Hammadi Codex II,1); *Völuspá*; *Brahma Sūtra*; *Zhuangzi*; *Wumenguan* (1228).

Dante Alighieri. *Commedia* (c. 1308–1321).

Dürer, Albrecht. *Melencolia I* (engraving, 1514).

Shakespeare, William. *The Winter's Tale* (c. 1611).

Milton, John. *Paradise Lost* (1667; 2nd ed. 1674).

Shelley, Mary. *Frankenstein; or, The Modern Prometheus* (1818).

Melville, Herman. *Moby-Dick* (1851); "Bartleby, the Scrivener: A Story of Wall-Street" (1853).

Dostoevsky, Fyodor. *Notes from Underground* (1864) and *The Brothers Karamazov* (1880), trans. Constance Garnett (1918; 1912).

Villiers de l'Isle-Adam, Auguste. *L'Ève future* (1886).

Frazer, James George. *The Golden Bough* (1890).

Jacobs, W. W. "The Monkey's Paw" (1902).

Klee, Paul. *Angelus Novus* (1920).

Yeats, W. B. "The Second Coming" (1920); *A Vision* (1925).

Čapek, Karel. *R.U.R.* (1920); *Věc Makropulos* (1922).

Eliot, T. S. *The Waste Land* (1922).

Maugham, W. Somerset. *Sheppey* (1933).

Borges, Jorge Luis. "Las ruinas circulares" (1940); "El jardín de senderos que se bifurcan" (1941); "Funes el memorioso" (1942); "El milagro secreto" (1943); "Del rigor en la ciencia" (1946).

Bioy Casares, Adolfo. *La invención de Morel* (1940).

Asimov, Isaac. The Foundation stories (1942–1950), collected as *Foundation* (1951), *Foundation and Empire* (1952), and *Second Foundation* (1953).

Dick, Philip K. "The Minority Report" (1956); *The Man in the High Castle* (1962); *Do Androids Dream of Electric Sheep?* (1968); *VALIS* (1981); *The Exegesis of Philip K. Dick* (2011).

Lem, Stanisław. *Solaris* (1961); *Doskonała próżnia* (1971).

Herbert, Frank. *Dune* (1965); *Dune Messiah* (1969); *God Emperor of Dune* (1981).

Vonnegut, Kurt. *Slaughterhouse-Five* (1969).

Le Guin, Ursula K. "The Ones Who Walk Away from Omelas" (1973).

Larkin, Philip. "Aubade" (1977).

Egan, Greg. *Permutation City* (1994); "Singleton" (2002).

Chiang, Ted. "Story of Your Life" (1998); "What's Expected of Us," *Nature* 436 (2005); "The Truth of Fact, the Truth of Feeling" (2013); "Anxiety Is the Dizziness of Freedom" (2019).

Ishiguro, Kazuo. *Never Let Me Go* (2005).

VanderMeer, Jeff. *Annihilation* (2014).

Kurzweil, Amy. *Artificial: A Love Story* (2023).

Marker, Chris, director. *La Jetée* (1962). Kubrick, Stanley, director. *2001: A Space Odyssey* (1968). Tarkovsky, Andrei, director. *Solaris* (1972).

Ptolemy, Barry, director. *Transcendent Man* (2009).

Black Mirror. "Be Right Back" (2013); "San Junipero" (2016).

Nolan, Christopher, director. *Interstellar* (2014); *Tenet* (2020).

Villeneuve, Denis, director. *Arrival* (2016).

Dark. Created by Baran bo Odar and Jantje Friese. Netflix, 2017–2020.

Westworld, season 3. HBO, 2020.

Meeting You. Documentary. MBC, February 2020.

FURTHER WORKS CITED

Aeschylus. *Agamemnon* (458 BCE).

Conway, John, and Simon Kochen. "The Free Will Theorem." *Foundations of Physics* 36 (2006): 1441–1473; "The Strong Free Will Theorem." *Notices of the American Mathematical Society* 56 (2009): 226–232.

Everett, Mark Oliver. *Things the Grandchildren Should Know* (2008).

Foucault, Michel. *Folie et déraison: Histoire de la folie à l'âge classique* (1961).

Goethe, Johann Wolfgang von. *Faust: Der Tragödie erster Teil* (1808).

Hesiod. *Works and Days*.

Hossenfelder, Sabine, and Tim Palmer. "Rethinking Superdeterminism." *Frontiers in Physics* 8 (2020): 139.

Krishnamurti, Jiddu, and David Bohm. *The Ending of Time* (1985).

"La connaissance augmentera." *Le Grand Continent*, 28 September 2025. https://legrandcontinent.eu/fr/2025/09/28/la-connaisance-augmentera/

Plutarch. *De Iside et Osiride*.

Schmitt, Carl. "Der Führer schützt das Recht." *Deutsche Juristen-Zeitung* 39 (1934).

't Hooft, Gerard. *The Cellular Automaton Interpretation of Quantum Mechanics* (2016).

Tiku, Nitasha, Elizabeth Dwoskin, and Gerrit De Vynck. "What Billionaire Peter Thiel Said in His Private 'Antichrist Lectures.'" *The Washington Post*, 10 October 2025. Summary: https://www.ischool.berkeley.edu/news/presscoverage/2025/hany-farid-verifies-billionaire-lectures-warning-antichrist-and-us

Wachowski, Lana, and Lilly Wachowski, directors. *The Matrix Reloaded* (2003).

Black Mirror. "Hang the DJ" (2017).

KEYWORDS

Alex Garland · *Devs* (2020) · determinism · free will · free won't · Laplace's demon · Maxwell's demon · Cartesian scepticism · simulation hypothesis · eternal recurrence · Socratic *daimonion* · many-worlds interpretation · de Broglie–Bohm theory · quantum computing · self-prediction · Gödelian incompleteness · Frauchiger–Renner paradox · Wigner's friend · predestination · Molinism · middle knowledge · theodicy · technodicy · necropoiesis · autoscopic horizon · messianic veto · katechon · apokatastasis · political theology · sovereignty and the state of exception · bare life · Walter Benjamin · Gnosticism · Lurianic Kabbalah · *tsimtsum* · mourning and melancholia · the crypt · hauntology · the mummy complex · the image of the dead · personal identity · digital afterlife · griefbots · surveillance capitalism · the right to the future tense · the right to the past tense · the engineer-mourner · speculative fiction · Borges · Lem · Philip K. Dick · Ted Chiang · Greg Egan · comparative mythology · Orpheus and Eurydice · Pygmalion · *The Winter's Tale* · chronogrammar · Melencolia Q



FINIS

Here ends Pandaemonium.